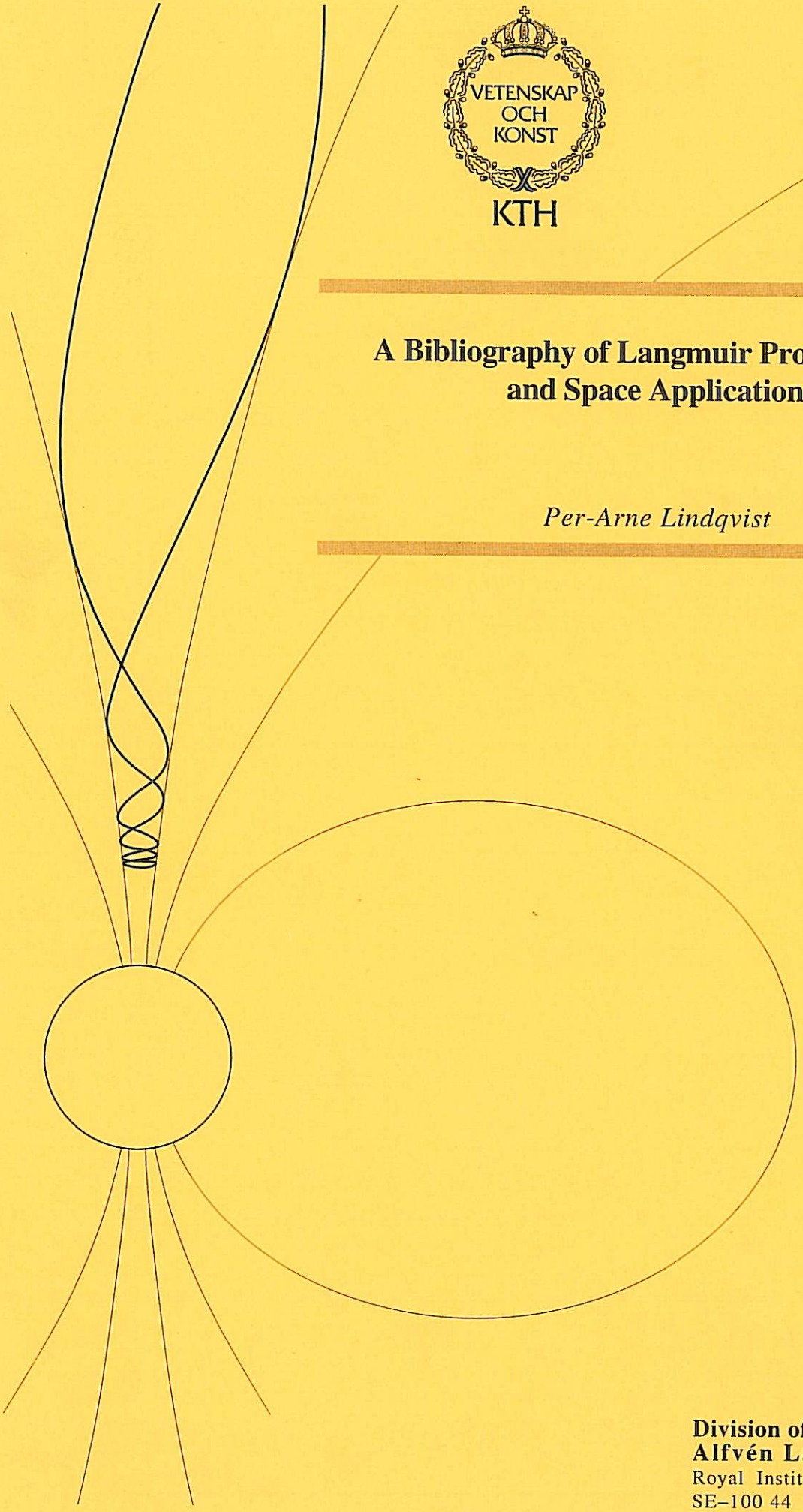
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A Bibliography of Langmuir Probe Theory and Space Applications

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A bibliography of Langmuir probe theory and applications to the measurements of electric fields, plasma density and temperature, and to satellite charging effects in space plasmas

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Introduction

Langmuir probe theory has been studied extensively since the pioneering results of Irving Langmuir and his colleagues in the 1920's. Initially the theory was developed for use in laboratory plasmas such as gas discharges. As the space era started around 1960, the theory was expanded to cover also such plasma regimes as encountered in space. A particular topic at the heart of the interest of the Space plasma group at the Alfvén Laboratory is the use of probes in space to measure electric fields and plasma characteristics. In this context, a need was felt to compile a bibliography of papers relevant to the understanding of probe behaviour in a plasma. The result is this report, where references to papers in the literature have been collected. The papers cover a variety of topics, but the main emphasis is on Langmuir probe theory and applications to the measurements of electric fields, plasma density and temperature in space plasmas, and to satellite charging effects. An effort has been made to make the list fairly comprehensive, but omissions are unavoidable, and should not always be taken to be intentional.

For ease of use, the list is repeated, sorted in different ways, as indicated in the list of contents. The first section contains the references sorted in alphabetical order by author. The second section contains the same material, sorted chronologically by publication year. The third section is again sorted by author, and includes abstracts of each publication, whenever available. In some cases it has not been possible to get hold of the abstract; this is indicated by an ellipsis ("..."). The fourth section contains the references sorted alphabetically by journal or publication.

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3. Sorted alphabetically by author, including abstracts

Adriaens, M. R. and B. Feuerbacher, Photoelectric uniformity of materials for satellite electric-field probes, Eldo-Cecles/Esro-Cers Sci. and Tech. Rev., 6, 161-167, 1974. (6 refs)

Abstract: The uniformity of photoelectric emission near the emission threshold has been investigated for vitreous carbon, graphite and Aquadag surfaces. Scans have been made over lengths of 8 to 18 cm using a mercury arc or a frequency-doubled argon laser beam. Vitreous-carbon spheres show photoelectric-yield variations that cannot be reduced below +-7 % by cleaning, etching, or heating. Graphite allows surfaces with +-2 % photoelectric-yield variation to be prepared consistently by abrasive cleaning. Surfaces coated with Aquadag may show variations of +-1.5 % if care is taken to ensure a uniform coating.

Ahedo, E. and J. R. Sanmartin, Current-voltage response of a spherical plasma contactor, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 1-8, Editrice Compositori, Bologna, 1992. (6 refs)

Abstract: A theoretical model for a contactor, collecting electrons from an ambient, unmagnetized plasma and emitting a current Ii is discussed. The relation between Ii and the potential bias of the contactor is found to be crucial for the formation of a quasineutral core around the anode and, consequently, for the current collected. Approximate analytical laws and charts for the current-voltage response are provided.

Ahedo, E., M. Martinez-Sanchez, and J. R. Sanmartin, Current collection by an active spherical electrode in an unmagnetized plasma, Phys. Fluids B (Plasma Phys.), 4, 3847-3855, 1992. (22 refs)

Abstract: A theoretical model for the steady-state response of anodic contactors that emit a plasma current I/sub i/ and collect electrons from a collisionless, unmagnetized plasma is presented. The use of a (kinetic) monoenergetic population for the attracted species, well known in passive probe theory, gives both accuracy and tractability to the theory. The monoenergetic population is proved to behave like an isentropic fluid with radial plus centripetal motion, allowing direct comparisons with ad hoc fluid models. Also, a modification of the original monoenergetic equations permits analysis of contactors operating in orbit-limited conditions. Besides that, the theory predicts that, only for plasma emissions above certain threshold current a presheath/double layer/core structure for the potential is formed (the core mode), while for emissions below that threshold, a plasma contactor behaves exactly as a positive-ion emitter with a presheath/sheath structure (the no-core mode). Ion emitters are studied as a particular case. Emphasis is placed on obtaining dimensionless charts and approximate asymptotic laws of the current-voltage characteristic.

Allred, D. B., J. D. Benson, H. A. Cohen, W. J. Raitt, D. A. Burt, I. Katz, G. A. Jongeward, J. Antoniades, M. Alport, D. Boyd, W. C. Nunnally, W. Dillon, J. Pickett, and R. B. Torbert, The SPEAR-1 experiment: high voltage effects on space charging in the ionosphere, IEEE Trans. Nucl. Sci., 35, 1386-1393, 1988. (3 refs)

Abstract: The SPEAR-1 rocket payload, flown on December 13, 1987, exposed biased conducting spheres at potentials of tens of kV to the ambient space environment at altitudes as high as 370 km (apogee) in the night-time ionosphere. Small chamber tests of the mock-up demonstrated the conditions for breakdown caused by sphere bias, magnetic field, chamber size, and secondary emissive properties of the chamber wall. In flight the failure to expose the payload's plasma contactor to the ionosphere caused a bipolar operation with interacting sheaths from both the payload body and the biased conducting spheres. During sphere biasing no breakdowns occurred above 100 km, and the plasma sheath controlling the flow of currents from the ionosphere behaved as a linear resistance with a value in the 100s of kilo-ohms.

Arends, H., Photoelectric uniformity of DAG 213 used as a coating material for a satellite electric field probe, European Space Research and Technology Centre Report ESLAB 81/36, 1981. (7 refs)

Abstract: The uniformity of photo-electric emission has been investigated for DAG 213. DAG 213 is a resin-base graphite dispersion from Acheson. Scans have been made on a hemisphere using a helium discharge lamp. DAG 213 shows photo-electric yield with max. deviation of +-4.5 %.

Beard, D. B., Correction to paper by David B. Beard and Francis S. Johnson, 'Ionospheric limitations on attainable satellite potential', J. Geophys. Res., 71, 4707, 1966. (1 refs)

Abstract: -

Beard, D. B. and F. S. Johnson, Ionospheric limitations on attainable satellite potential, J. Geophys. Res., 66, 4113-4122, 1961. (correction J. Geophys. Res., 71, 4707, 1966.) (2 refs)

Abstract: Artificial satellites orbiting in the ionosphere normally tend to acquire a slight negative charge due to the high velocity of the electrons relative to the ion and satellite velocities. (At very high altitudes where the ionospheric electron concentrations are small, and in the presence of solar radiation, satellites tend to have a small positive charge due to the photoelectric effect, which in this case predominates over the ionospheric effect.) If the attempt is made to change the potential of a satellite by ejecting energetic beams of electrons or positive ions, the ionosphere provides some rather stringent limitations on the potential that can be achieved, for a substantial potential on a satellite causes appreciable currents to flow between the satellite and the ionosphere, which is a highly conducting medium containing a large reservoir of charge. The physics of the ionospheric limitation on satellite charge is investigated, and it is shown how the satellite charge and potential depend on injection currents and ambient charge density for both intermittent and continuous injection currents.

Bergmann, A., Two-dimensional particle simulation of Langmuir probe sheaths with oblique magnetic field, Phys. Plasmas, 1, 3598-3606, 1994. (19 refs)

Abstract: Two-dimensional particle simulations of the sheath in front of a flat Langmuir probe mounted into a particle absorbing plate are performed to study the influence of a strong magnetic field (relation between Larmor radii and Debye length: rho /sub e/> lambda /sub D/) which is oriented obliquely to the probe surface. Ion-attracting probes are considered and the sheath is assumed to be collisionless. The full particle orbits in the homogeneous magnetic field and the self-consistent electric potential are calculated with a particle-in-cell (PIC) code with two spatial coordinates and three velocity components (2d,3 upsilon). The main results are: in the sheath the ion trajectories are bent towards the normal to the probe surface so that the ion flow is focused to the edges of the probe. This leads to an enhancement of the ion current as compared to the current flowing in the flux tube subtended by the probe. As a consequence the current does not saturate at large (negative) probe voltage, because the thickness of the Debye sheath, and thus the effective probe size, grows with increasing probe voltage. This effect is particularly strong if the sheath thickness is about as large, or larger than, the projection of the probe size along the magnetic field lines. These results can help to explain the ion-current nonsaturation found in recent measurements with Langmuir probes in the boundary layer of magnetically confined plasmas.

Bering, E. A., M. C. Kelley, F. S. Mozer, and U. V. Fahleson, Theory and operation of the split Langmuir probe, Planet. Space Sci., 21, 1983-2001, 1973. (43 refs)

Abstract: An instrument, the split Langmuir probe, has been developed to make in situ measurements of current density and plasma bulk flow. The split Langmuir probe consists of two conducting elements that are separated by a thin insulator that shield each other over a 2 pi solid angle, and that are simultaneously swept from negative to positive potentials with respect to the plasma. By measuring the current to each plate and the difference current between plates, information can be obtained on the plasma's current density, bulk flow, electron temperature, and density. The instrument was successfully test flown from Fort Churchill on 2 August 1968, with results in reasonable agreement with those from another experiment on the same rocket. Sources of error indicated by these results include plate area differences, plate work function differences, input resistor differences, and probe wake effects. The error signal in the difference current data due to plate work function differences rose to a sharp maximum at plasma potential, which served he useful purpose of precisely marking plasma potential. Possible changes in probe geometry, sweep rate, and telemetry designed to reduce these errors are discussed.

Bernstein, I. B. and I. N. Rabinowitz, Theory of electrostatic probes in a low-density plasma, Phys. Fluids, 2, 112-121, 1959. (5 refs)

Abstract: The theory of sperical and cylindrical probes immersed in plasmas of such low density that collisions can be neglected is formulated. The appropriate Boltzmann equation is solved, yielding the particle density and flux as functionals of the electrostatic potential, the situation in the body of the plasma, and the properties of the probe. This information when inserted in Poisson's equation serves to determine the potential, and hence the probe characteristic. No a priori separation into sheath and plasma regions is required. Though amenable to a determination of the full probe characteristic, the method is applied in detail and numerical results are presented only for the collection of monoenergetic ions, for the case of negligible electron current. These results indicate that the potential is not so insensitive to ion energy as has been believed, and that if the probe radius is sufficiently small, there enters the possibility of a class of ions which are trapped near the probe in troughs of the effective radial potential energy. The population of these trapped ions is determined by collisions, however infrequent. It is difficult to calculate, and conceivably can have a marked effect on the local potential.

Berthelier, J. J. and M. Pirre, Ion analyzers for measurement of d.c. electric field and ion temperature, Space Sci. Instr., 4, 213-226, 1978. (18 refs)

Abstract: Describes the ion analyzers for measurement of electric fields and ion temperature which were part of the nose-cone experiment package in the ARAKS experiment. Principle of measurement and theoretical models used in numerical calculations are presented as well as a brief description of the electronics and modes of operation. The accuracy of the instrument was estimated to be 2-3 mV m/sup -1/ for the electric field and about 150K for the ion temperature. Results from preliminary tests in a plasma tank are given.

Berthelier, J. J., R. Godard, and J. M. Illiano, Influence of surface effects on electron temperature measurement in the ionosphere, Abstracts of the Belgo-German symposium on surface- vacuum- space, p. 40, Belgian Vacuum and Vacuum Technique Soc, Liege, Belgium, 1971. (0 refs)

Abstract: Within the framework of ionospheric research electrostatic sondes, for electron temperature measurements of the plasma, have been improved. The results obtained and comparative diagnostic studies by different researchers show the surface effects. These will modify the response of the sonde (contact potentials, desorption etc.). The results obtained in a laboratory and in space by sondes coated with different materials are presented.

Bertotti, B., Theory of an electrostatic probe in a strong magnetic field, Phys. Fluids, 4, 1047-1052, 1961. (7 refs)

Abstract: If the magnetic field is so strong as to impair collective transverse drifts, all the charges supplied to the probe come mainly from a long tube of force, whose section is about one Larmor radius larger than the probe; while a diffusion process, more efficient than ordinary drifts, continuously exchanges particles between the inside and the rest of the plasma. A one-dimensional model of this process is proposed, leading to an integro-differential Poisson's equation, which has been studied for the case in which the collected particles are very fast. The solution consists of a chargeless, slowly decaying potential which describes the geometrical screening effect of the probe; while in the

sheath an approximate boundary-layer solution matches with the probe's potential.

Biasca, R. and J. Wang, Ion current collection in spacecraft wakes, Phys. Plasmas, 2, 280-288, 1995. (22 refs)
Abstract: This work investigates the current-voltage characteristics of a highly negatively biased, isolated probe in the plasma wake of a much larger, unbiased object. The system is investigated for both flowing and stationary plasmas. For the stationary plasma case, spherical probe theory and a simple correction factor to account for the shadow of the plate adequately explain the results. The introduction of a flowing plasma leads to a much more complicated situation. In this case, since the potential field is asymmetric and the ions no longer conserve angular momentum, spherical probe theory is inadequate. The current-voltage characteristics take a much more complicated form than in the stationary case. Indeed, the simulations show that the current-voltage characteristics of varying axial probe positions cross each. other. The high complexity of the ion orbits makes theoretical predictions of where the crossover occurs difficult. However, the potential structure around the probe and the ion trajectories are analyzed to explain the current-voltage characteristics. These results show regions can exist in energy-angular momentum space which cause incoming ions to orbit the probe and then escape the simulation. The fraction of particles which do orbit and the escape as a function of probe potential and position determines the shape of the current-voltage characteristic.

Bienkowski, G. K. and Chang, K.-W., Asymptotic theory of a spherical electrostatic probe in a low-density plasma, Phys. Fluids, 11, 784-799, 1968. (16 refs)
Abstract: The electrical effects of a charged sphere immersed in weakly ionized gas of infinite extent are analyzed on the basis of kinetic theory. Approximate solutions are obtained by the application of a moment method in which velocity space is split on the basis of exact orbit calculation, self-consistent with the field. The theory is shown to reproduce both the collisionless and the continuum results in the appropriate limits. Detailed analysis is performed asymptotically for small Debye length but arbitrary mean free path. For quantitative calculations a completely absorbing probe surface at a highly negative potential is chosen. Criteria for the collisionless or collision-dominated sheath are obtained explicitly. The voltage-current characteristics are presented in a form useful for direct prediction of electron number density in terms of measured voltage and current, provided the mean free path is known. If the mean free path is only estimated, the results can be used directly to check self-consistency of that estimate.

Bochev, A. Z. and L. G. Bankov, On an algorithm for processing Langmuir's probe data, Comptes Rendus de l'Academie Bulgare des Sciences, 27, 1647-1649, 1974. (5 refs)
Abstract: Direct investigations of the ionospheric plasma make wide use of measurements by means of the Langmuir cylindrical electrostatic probe mounted to artificial satellites and rockets. The proposed algorithm is for the determination of the electronic temperature T/sub e/, the electron concentration N/sub e/ and the satellite potential U/sub 0/.

Bohm, D., Minimum ionic kinetic energy for a stable sheath, in The Characteristics of Electrical Discharges in Magnetic Fields, Ed. by A. Guthrie and R. K. Wakerling, Ch. 3, pp. 77-86, McGraw-Hill Book Co., New York, 1949. (1 refs)
Abstract: -

Bohm, D., E. H. S. Burhop, and H. S. W. Massey, The use of probes for plasma exploration in strong magnetic fields, in The Characteristics of Electrical Discharges in Magnetic Fields, Ed. by A. Guthrie and R. K. Wakerling, Ch. 2, pp. 13-76, McGraw-Hill Book Co., New York, 1949. (5 refs)
Abstract: -

Bond, R. A., P. M. Latham, A. R. Martin, and M. R. Daymond, Computational analysis of spacecraft--plasma wake phenomena using the simulation code SAPPHIRE, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 9-35, Editrice Compositori, Bologna, 1992. (2 refs)
Abstract: Computational simulations of plasma wakes produced by charged bodies in plasma conditions equivalent to low Earth orbit have been carried out using the plasma simulation code SAPPHIRE. The numerical methods used by the code are described and typical results given. The results are compared to experimental data and very good agreement is shown. Finally, recent improvements to the code are described and some examples given.

Borovsky, J. E., The theory of Langmuir probes in strong electrostatic potential structures, Phys. Fluids, 29, 718-730, 1986. (34 refs)
Abstract: The inside of a time-stationary strong electrostatic potential structure is characterized by the absence of thermal plasma and by the presence of electron and ion beams, the beam energies being related to the local plasma potential. The operation of collecting and emitting Langmuir probes and double probes within such potential structures is analyzed. It is demonstrated that a floating double probe will not measure the electric field of a strong potential structure without substantial reinterpretation of the potential between the probes. A prescription for the interpretation is given. Some probe-surface effects are considered, as are the effects of strong magnetic fields.

Boyd, R. L. F., Langmuir probes on spacecraft, in Plasma Diagnostics, Ed. by W. Lochte-Holtgreven, pp. 732-776, North-Holland Publ. Co., Amsterdam, 1968. (37 refs)
Abstract: -

Brace, L. H., W. R. Hoegy, and R. F. Theis, Solar EUV measurements at Venus based on photoelectron emission from the Pioneer Venus Langmuir probe, J. Geophys. Res. 93, 7282-7296, 1988. (21 refs)
Abstract: The photoelectron current from the Pioneer Venus Langmuir probe has provided measurements of the solar extreme ultraviolet flux at Venus since 1979. This current is the product of the photoelectric yield of the collector and

the solar spectrum at the wavelengths short enough to cause emission. The Langmuir probe provides a direct measure of the total solar EUV flux, including most of the wavelengths that produce the Venus ionosphere and heat and excite neutrals in the thermosphere. The measurement technique is described, and the daily average measurements of photocurrent obtained between 1979 and 1987 are presented. The authors present three indices of EUV based on the measurements: (1) the photoemission current itself, (2) the total EUV flux, and (3) an F/sub 10.7/-like solar index. These are compared with related measurements made simultaneously at Earth. These data may also help solar physicists track the intensity of EUV emission regions on the Sun while they are not visible from the Earth. The EUV flux profile of a solar flare event is also illustrated.

Bradley, D. and K. J. Matthews, Double spherical electrostatic probe continuum theory and electron temperature measurement, Phys. Fluids, 10, 1336-1341, 1967. (5 refs)

Abstract: The continuum theory and numerical solutions of Su and Lam for single spherical probes are used to derive double probe current-voltage characteristics. These characteristics are used to test the validity of the expression for electron temperature derived by Johnson and Malter on the basis of collisionless theory. It is found that the expression gives reabonably accurate values of electron temperature for continuum conditions. Values of correction factor are presented to enable more accurate temperatures to be derived from the expression of Johnson and Malter.

Calder, A. C. and J. G. Laframboise, Terminal properties of a spherical RF electrode in an isotropic Vlasov plasma: results of a computer simulation, Radio Sci., 20, 989-999, 1985. (22 refs)

Abstract: Present results generated using a computer code which has been developed to model an isotropic inhomogeneous Vlasov plasma surrounding a DC-biased spherical or infinite cylindrical electrode to which a radio frequency potential is applied. The initial Maxwellian velocity distribution of the plasma is approximated by a multiple water bag distribution. The instantaneous response of the plasma to a sinusoidal potential applied to the electrode is calculated in the electrostatic approximation. Transient and nonlinear effects can be modeled in this way. Ions are usually treated as fixed, but some runs include ion dynamics with an ion/electrode mass ratio of 16. Data produced by the code include the instantaneous particle flux and electric field at the electrode surface, from which the RF admittance is calculated. Results are presented for spheres in the frequency range 0.1 omega /sub pe/ to 3 omega /sub pe/, RF amplitudes of 1-9 kT/el, and antenna radii of 1 and 10 lambda /sub D/. Here omega /sub pe/ is the electron plasma frequency and lambda /sub D/ is the Debye shielding distance. The code reproduces the admittance behavior expected near the antenna-plasma series resonance. Some evidence for RF modification of the sheath is seen well above the frequency of the series resonance.

Calder, A. C. and J. G. Laframboise, Multiple-water-bag simulation of inhomogeneous plasma motion near an electrode, J. Computational Phys., 65, 18-45, 1986. (22 refs)

Abstract: A computer code has been developed to model an isotropic inhomogeneous collisionless plasma surrounding a DC-based spherical or infinite cylindrical electrode to which a radio-frequency potential is applied. The initial Maxwellian velocity distribution of the plasma is approximated by a multiple-water-bag distribution. The instantaneous response of the plasma to a time-dependent potential applied to the electrode is calculated in the electrostatic approximation. Transient and nonlinear effects can be modeled in this way. Use of the multiple-water-bag method results in less numerical noise than would be seen using, for example, a particle-in-cell method with a comparable number of simulation particles.

Calder, A. C. and J. G. Laframboise, Time-dependent sheath response to abrupt electrode voltage changes, Phys. Fluids B (Plasma Phys.), 2, 655-666, 1990. (27 refs)

Abstract: Results are presented of particle simulations of the time-dependent behavior of the plasma sheath surrounding an electrode whose potential is abruptly changed from 0 to some positive value. The surrounding plasma is assumed isotropic and collisionless and the electrode is a sphere or infinite cylinder. Potential steps in the range 10 to 10/sup 4/ kT/e are considered. The transient behavior of the collected electron current is examined. For small (< 25 kT/e) potential steps, the current quickly settles to a value consistent with existing static probe theory. Persistent Langmuir oscillations launched by larger potential steps significantly perturb the collected current, with the result that static probe theory predictions are in some cases exceeded for times as long as 1000 omega/sub p//sup -1/ after the step. Trapping of electrons during all or part of an oscillation cycle is evident in many cases. The oscillations themselves are examined and evidence for beating between modes at omega/sub p/ and at sheath-plasma parallel resonance frequency is discussed. In some cases, the oscillations are driven to large amplitudes by the ion-electron two-stream instability.

Calder, A. C., G. W. Hulbert, and J. G. Laframboise, Sheath dynamics of electrodes stepped to large negative potentials, Phys. Fluids B (Plasma Phys.), 5, 674-690, 1993. (36 refs)

Abstract: Results are presented of particle simulations of the time-dependent behavior of the plasma sheath surrounding an electrode whose potential is abruptly changed from 0 to some negative value. The surrounding plasma is assumed isotropic and collisionless, and the electrode is a sphere or infinite cylinder. Potential steps up to -1000 kT/sub e//e are considered. The time histories and frequency spectra of sheath and plasma oscillations initiated by the potential step are studied. The energy in sheath oscillations (frequency < omega /sub p/) escapes by nonlinear coupling to plasma oscillations (frequency omega /sub p/). The time scale for sheath-plasma coupling depends strongly on the ion-to-electron mass ratio. Both linear and nonlinear ion acoustic waves are excited by the response to the potential step. Near a small (radius 1 lambda /sub D/) cylindrical electrode, trapped ions are shown to make an important contribution to the sheath ion density. The height of the transient peak in collected current for a cylindrical electrode

displays strong dependence on the ion-to-electron temperature ratio, for sufficiently large potential steps.

Cauffman, D. P., The effects of photoelectron emission on a multiple-probe spacecraft near the plasmapause, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 153-161, D. Reidel Publ. Co., Dordrecht, 1973. (7 refs)

Abstract: Measurements of the potential difference between a pair of symmetrical, spherical, floating probes (14-cm diam, 5-m baseline) flown on the S^3-A (Explorer 45) spacecraft (launched November 1971, with apogee 5.2 R_E, perigee 222 km, and inclination 3.6 deg) have shown a complicated response to sheath potentials in the region near and inside the plasmapause. Typically, with increasing altitude (1) the sunward sphere will first assume a more negative potential than the antisunward sphere, (2) the polarities of the probes in the region L = 2 to 4 may switch, sometimes several times, until finally, (3) the detector output may saturate (Delta phi > 125 mV) with the sunward sphere more negative again. In order to explain the above observations a model has been created in which the system is divided into four floating plasma probes: two identical spheres and the sunlit and dark sides of the spacecraft body. The interaction between the probes is computed considering space charge effects, currents due to the emission and exchange of photoelectrons, and currents from the ambient plasma. The four floating potentials are calculated by requiring current balance for each probe and solving by iteration the resulting four coupled, non-linear equations. The results of the model indicate that the S^3-A measurements described above may be explained in terms of the fact that one sphere is exposed to a greater portion of the sunlit side of the body. This study has found that both space charge and current balance effects must be considered in order to explain the observed sheath potentials. Sheath potentials asymmetries are sensitive to ambient temperature and density, relative photoemission intensities of the spheres and spacecraft body, and the conductivity of the probe surfaces. The effects become significant for Debye lengths greater than about one-tenth of the sphere-body separation. At the present time there is no reliable way to compensate for the effects of sheath potentials. The only way to make good floating probe measurements is to separate the probes by distances very large compared to the shielding length.

Chaganti, V. S. R. K. R. and N. Singh, Current collection by a conducting spherical satellite in the ionosphere, in Proc. IEEE Southeastcon '92 (Cat. No.92CH3094-0), vol. 1, pp. 75-78, IEEE, New York, NY, USA, 1992. (4 refs)

Abstract: The current collection by a conducting satellite moving in a magnetized plasma across the Earth's magnetic field was studied by means of numerical simulations. The satellite was assumed to be a conducting sphere. In the low Earth orbit, the satellite moves with an orbital velocity V/sub 0/ approximately=8 km/s perpendicular to the Earth's magnetic field lines. This orbital velocity V/sub 0/ is relative to the plasma. Numerical simulations have been carried out to determine the current collected by the satellite moving in a magnetized plasma. For comparison, simulations for the plasma having no relative motion between the plasma and the satellite have been also carried out. The results from the later simulations agree with the canonical upper bound on the collected current calculated by Parker and Murphy (1967). Results from simulations with the relative drift of 8 km/s show that the collected current by the satellite is appreciably greater than that collected without the relative drift.

Chang, J.-S. and J. G. Laframboise, Double-probe theory for a continuum low-density plasma, J. Phys. D (Appl. Phys.), 9, 1699-1703, 1976. (19 refs)

Abstract: A theoretical treatment is given for the double-probe method in an electron/positive-ion plasma having mean free path << probe size << Debye length, i.e. under conditions in which neither the collisionless theory of Johnson and Malter or Kojima and Takayama, nor the continuum theories based on either the work of Su and Lam or that of Cohen are applicable. The theory applies to any probe geometry, including spheres and finite cylinders. Applications to plasma diagnostics are developed in detail.

Chang, J.-S. and J. G. Laframboise, Probe theory for arbitrary shape in a large Debye length, stationary plasma, Phys. Fluids, 19, 25-31, 1976. (18 refs)

Abstract: The theory of electrostatic probes of arbitrary shape in a large Debye length, stationary plasma is investigated. It is shown that the analytic solutions for the current-voltage characteristics derived by Fendell (1970) and Chapkis and Baum (1971) for spherical probes are valid for any probe shape. The theory is extended in an approximate way to finite-mean-free-path conditions. It is shown that a cylindrical probe will be more sensitive to departures from collisionless conditions than a spherical probe of equal radius. Computations for spherical probes with a nonzero Debye ratio are also presented. The implication for plasma diagnostics using single and triple probes are discussed. The use of the Druyvesteyn method is examined.

Chang, J.-S. and J. G. Laframboise, Theory of electrostatic probes in a flowing continuum plasma: numerical solutions for cylindrical probes in cross flow, J. Phys. D (Appl. Phys.), 26, 42-48, 1993. (20 refs)

Abstract: An exact numerical treatment has been performed for a cylindrical probe in continuum flowing plasma. This solution for the low-density continuum case, i.e. when the probe radius R/sub p/ << Debye length lambda/sub D/, is calculated for scaled probe potentials phi/sub P//epsilon ln 2L from 0 to +-20, Reynolds numbers (based on probe diameters) Re from 10/sup -1/ to 10/sup 2/ and charged particle Schmidt numbers Sc/sub c/ from 10/sup -1/ to 10/sup 5/, where phi/sub p/ = eV/sub p//kT/sub e/, epsilon = T/T/sub e/ and L is the ratio of probe length to diameter. Numerical solutions provided by other authors are used for the neutral flow. The electric potential profiles used are logarithmic, obtained by using the Laplace potential at the equator of a prolate spheroid, approximated for radii much smaller than the major axis. The numerical results show the following. (1) For retarding potentials, the usual 'retarding potential' method for temperature determination leads to large errors. (2) For a near space potential, the effects of flow are to

smooth the 'knee' of the probe characteristics and to render the determination of the space potential more imprecise. (3) For large attracting potentials, the attracting current characteristics are always linear. Application to the plasma diagnostic method and magnetoplasma are also discussed.

Chang, J.-S., R. Godard, and R. C. Scott, Total heat transfer to cylindrical collectors in collisionless plasma flow, Phys. Fluids, 27, 679-681, 1984. (7 refs)

Abstract: The theory of total heat transfer to cylindrical collectors in collisionless plasma flow with no magnetic field is presented. The treatment has been done for orbit-motion-limited conditions (Debye length sufficiently large), for the speed ratio S=U/(2kT/m)/sup 1/2/ of the flow velocity to the most probable ion thermal velocity from 0 to 10, and for various values of the nondimensional collector potential chi /sub p/=qV/sub p//kT.

Chang, K. W. and G. K. Bienkowski, Effects of electron emission on electrostatic probes at arbitrary pressures, Phys. Fluids, 13, 902-920, 1970. (11 refs)

Abstract: A systematic investigation of the effects on probe response caused by electrons emitted with a half-Maxwellian distribution is performed. In the absence of collisions a theory for spherical and cylindrical probes in a plasma of infinite extent is formulated, based on orbit calculations. A simple method is developed which greatly facilitates the solution of the space-charge-limited emission problem. The structure of the resulting double sheath is analyzed in detail. Solutions of current-voltage characteristics and space- charge-limited current are presented for the case of thermionic emission. Theories in the transitional and continuum regimes for a spherical probe in a weakly ionized gas are formulated by the moment method.

Chen, F. F., Electric Probes, in Plasma Diagnostic Techniques, Ed. by R. H. Huddlestone and S. L. Leonard, pp. 113-200, Academic Press, New York, 1965. (53 refs)

Abstract: -

Chou, Y. S., L. Talbot, and D. R. Willis, Kinetic theory of a spherical electrostatic probe in a stationary plasma, Phys. Fluids, 9, 2150-2167, 1966. (17 refs)

Abstract: The spherical electrostatic probe in a weakly-ionized gas is analyzed on the basis of kinetic theory using a Krook-type model for the collision integral. The charged particle motions are analyzed in energy-angular momentum phase space, and formal integral solutions for the distribution functions are obtained. Moment equations based on the Lees method are derived, taking into account curvature of the particle paths, and sample solutions of these equations are presented. The results obtained agree well with the numerical analysis of Laframboise, in the collisionless limit. It is found that the effect of charged-neutral particle collisions is to extend the probe sheath farther into the plasma and reduce the electron and ion currents, as well as the departures from neutrality within the sheath. Fair agreement is obtained with a moment method based on straight-line trajectories for the charged particles. The continuum limit is not included in the computations.

Chung, K.-S. and I. H. Hutchinson, Kinetic theory of ion collection by probing objects in flowing strongly magnetized plasmas, Phys. Rev. A (Gen. Phys.), 38, 4721-4731, 1988. (29 refs)

Abstract: A new one-dimensional collisionless kinetic model is developed for the flow of ions to probing structures in drifting plasmas. The cross-field flow into the presheath is modeled by accounting consistently for particle exchange between the collection flux tube and the outer plasma. Numerical solutions of the self-consistent plasma-sheath equations are obtained with arbitrary external ion temperature and parallel plasma flow velocity. Results are presented of the spatial dependence of the ion distribution function as well as its moments (density, particle flux, temperature, and power flux). The ion current to the probe is obtained and the ratio of the upstream to downstream currents is found to be well represented by the form R=exp(Ku/sub d/), where K approximately 1.7 and u/sub d/ is the drift velocity in units of square root (T/sub e//m/sub i/). The results are in good agreement with comparable recent fluid calculations but show substantial deviations from other models which ignore particle exchange out of the presheath. No evidence is found of the formation of shocks in the downstream wake, contrary to the implications of some fluid theories.

Chung, P. M., L. Talbot, and K. J. Touryan, Electric probes in stationary and flowing plasmas. I. Collisionless and transitional probes, AIAA J. 12(2), 133-144, 1974. (52 refs)

Abstract: The authors attempt to review the state-of-the-art of electric probes including the response of cylindrical and spherical electrostatic probes in the collisionless and transitional regimes.

Chung, P. M., L. Talbot, and K. J. Touryan, Electric probes in stationary and flowing plasmas. II. Continuum probes, AIAA J., 12(2), 144-157, 1974. (57 refs)

Abstract: For pt. I see ibid., vol.12, 133-144 (1974). The authors review the status of continuum electrostatic probes which represent the limit of many collisions within the sheath.

Chung, P. M., L. Talbot, and K. J. Touryan, Electric probes in stationary and flowing plasmas - Theory and application, Springer-Verlag, New York, 1975. (193 refs)

Abstract: (Preface:) The electric probe has long been used as a fundamental diagnostic tool for measuring the local properties of a plasnna. Since Langmuir first developed the electric-probe technique in 1924, probes have been used to measure electron densities and temperatures in a wide variety of gaseous ionized media, such as electric discharges, afterglows, ionizing shock waves, flames, MHD, and plasmajet flows, reentry vehicle flow fields, and atmospheric and space plasmas. The first systematic account of modern theories of electric-probe behavior was given by Chen (1965), who also provided practical information on experimental techniques. A subsequent survey by Swift and Schwar (1970), which was representative of results contained in the literature through 1969, included additional information on some

of the modern theories and on practical details of probe utilization. The purpose of this volume is to supplement the previously mentioned two works by providing an account of a large body of the up-to-date information available on electric probes, particularly in the areas of transitional and continuum-flow phenomena, and by offering, for all domains of probe application, a critical appraisal of the more significant probe theories and experimental investigations in the literature. The volume is intended for the engineer or scientist who desires a comprehensive survey of probe theories and who wishes to apply the available information to specific diagnostic situations encountered in the laboratory or in flight environments. The emphasis therefore is on practicality and usefulness rather than mathematical rigor. Due consideration is given to the particular limitations inherent in each analysis or experimental correlation described. Chapter I begins with a discussion of the fundamental definitions and key paramclcrs that govcrn the electric-probe measurement technique. Chapter II is devoted to collisionless and transitional probe behavior in quiescent as well as flowing plasmas. Chaptcr III covers the continuum regime of probe operation. Chapter IV deals with special topics such as probe surface integrity, and probe measurements in turbulent flows and magnetic fields. The equations that provide the theoretical basis for the analysis of probe behavior are developed in some detail in the Appendix.

Clements, R. M., Plasma diagnostics with electric probes, J. Vacuum Sci. Tech., 15, 193-198, 1978. (24 refs)
Abstract: The application of Langmuir probes, including double probes, to the determination of electron densities and temperatures in discharges typical of those employed in sputter coating is reviewed. The discharges are classed as collisionless plasmas in view of the low pressure normally employed and the analysis and discussion presented are restricted to these conditions. It is concluded that Langmuir probes are likely to yield useful data for sputter coating systems.

Cohen, I. M., Asymptotic theory of spherical electrostatic probes in a slightly ionized, collision-dominated gas, Phys. Fluids, 6, 1492-1499, 1963. (10 refs)
Abstract: The problem of an electrostatic probe in a dense, slightly ionized gas is treated by techniques of asymptotic analysis. In particular, the asymptotic limits rho_p = r_p/lambda_d --> infinity and epsilon = T_+/T_- --> 0 are treated in considerable detail. (r_p, lambda_d, T_-, T_+ are probe radius, Debye length, electron, and ion temperature, respectively.) Sample integral curves for electrostatic potential, ion and electron density are given. Probe characteristic curves for three values of epsilon (finite and small) and rho_p (large) are also given. It is noted that these characteristics do not saturate for large probe potential because the influence of the probe is felt to very great distances from the probe; the shielding due to the space-charge sheath is incomplete.

Cooke, D. L. and I. Katz, Ionization-induced instability in an electron-collecting sheath, J. Spacecr. Rockets, 25, 132-138, 1988. (25 refs)
Abstract: An analytical and numerical investigation of the electron-collecting sheath about a positively biased probe in the ionosphere has been conducted to determine the effect of ionization within the sheath. This study is directed at the space charge-limited regime appropriate to objects having a characteristic size much larger than the ambient Debye length and a uniform surface potential much greater than the ambient plasma temperature. Computer calculations employing a fluid approximation were performed to study the dynamics of the secondary plasma produced within the sheath. Both the analytical and computational results indicate that, when the secondary ion production exceeds a critical level, the sheath edge propagates outward in an explosive fashion similar to the propagation of double layers. This critical level is shown to occur for sheath dimensions D_s greater than about 1% of an ionization mean free path lambda. This sheath size corresponds to the mean free path for ionization electrons reduced by the square root of the electron mass ratio D_s > alpha lambda sqrt(m_e/m_i), where alpha is a geometrical factor of order unity. At Shuttle altitudes and above, ionosphere probe interactions appear to be subcritical, but experiments with very high voltages, enhanced neutral background, or increased ionization by beam plasma discharge will trigger the instability.

Cooke, D. L., M. F. Tautz, and A. G. Rubin, The sheath of a positive probe in a flowing plasma, Proceedings of the 12th Conference on the Numerical Simulation of Plasmas.Lawrence Livermore Nat. Lab, Livermore, CA, USA, 1987. (11 refs)
Abstract: A plasma will shield a charged object with a charge sheath that, in general, may only be studied by numerical solution of the coupled Poisson and Vlasov equations. If a plasma has a significant drift past a probe, sheath formation becomes difficult to model, and because of the difference in ion and electron thermal speeds, the sheaths of positive and negative probes become dissimilar. The authors have refined the Poisson-Vlasov code, TDWAKE (Parker, 1976) into a new program, MACH, which has the precision to perform self-consistent space charge calculations for finite charged objects in flowing plasma. The physical theory of MACH is identical to TDWAKE, however algorithms have been added to achieve the degree of resolution and stability required for this study. Dimensionally, MACH computes in cylindrical coordinates with two space dimensions, r and z and three velocity dimensions, v/sub r/, v/sub phi /, and v/sub 0/.

Daly, E. J., Simulations of the electrostatic charging of ESA communications satellites, European Space Agency Report ESA STM-239, December 1987. (21 refs)
Abstract: The NASA Charging Analyzer Program (NASCAP) has been used to investigate the electrostatic charging of ECS-type three-axis stabilised communications satellites in geostationary plasma environments. NASCAP's capabilities and limitations are discussed and the use of NASCAP in support of anomaly investigations is described. Three-dimensional charging simulations were performed for eclipse and sunlight conditions, with various plasma environments. The importance of three-dimensional effects is demonstrated. Differential charging on the solar arrays

during eclipse and immediately following eclipse are predicted to be significant. Charging is strongly dependent on exposed material parameter values, since these determine the magnitudes of the seven basic components of the current to a surface. The effects of changes in the properties of some of these exposed materials on the charging of the satellite were investigated in order to try and identify ways of reducing the differential charging. The associated program MATCHG has also been used to investigate material response to charging environments (including laboratory electron beams) at a more basic level, without consideration of the geometrical or photoemission effects. MATCHG has proved a useful tool in aiding the interpretion and reconciliation of material testing results before a full NASCAP run is undertaken.

Diebold, D. A., N. Hershkowitz, J. R. DeKock, T. P. Intrator, and S.-G. Lee, Space charged enhanced, plasma gradient induced error in satellite electric field measurements, J. Geophys. Res., 99, 449-458, 1994. (17 refs)

Abstract: In magnetospheric plasmas it is possible for plasma gradients to cause error in electric field measurements made by satellite double probes. The space charge enhanced plasma gradient induced error is discussed in general terms, the results of a laboratory experiment designed to illustrate this error are presented, and a simple expression that quantifies this error in a form that is readily applicable to satellite data is derived. The simple expression indicates that for a given probe bias current there is less error for cylindrical probes than for spherical probes. The expression also suggests that for Viking data the error is negligible.

Dobrowolny, M., U. Guidoni, E. Melchioni, C. Vannaroni, and J. P. Lebreton, Current-voltage characteristics of the TSS 1 satellite, J. Geophys. Res., 100, 23953-23958, 1995. (14 refs)

Abstract: One of the primary science objectives of the TSS 1 project was that of studying the current-voltage (I-V) characteristics of a highly charged body in the ionosphere. In this paper, using data from experiments of the TSS 1 mission for an event of high satellite charging, the authors derive points of the I-V characteristics of the TSS satellite. These are then compared with different theoretical models. The comparison shows that the measured currents, for given potentials, are well above the predictions of the Parker and Murphy model. The results obtained appear, in fact, to support a model which computes an enhancement of the collected current due to the rapid motion of the collector across the geomagnetic field.

Dote, T., H. Ameniya, and T. Ichimiya, Effect of the magnetic field upon the saturation electron current of an electrostatic probe, Japanese J. Appl. Phys., 3, 789-796, 1964. (9 refs)

Abstract: Some experiments are performed in several kinds of discharged plasma to investigate the influence of a magnetic field upon the saturation electron current of an electrostatic probe. The apparent electron density N_e' and the positive ion density N_+ are calculated from the saturation electron current and the positive ion saturation current respectively and the ratio N_e'/N_+ is plotted against omega tau (omega: the angular cyclotron frequency of electrons, tau: the mean collision time between electrons and molecules). The curve is found to show a clear dependence on the gas pressure, and when the ratio is log-log plotted against the magnetic field, it gives a nearly straight line, showing N_e'/N_+ to be 0.1 at about 1000 gauss. Since the positive ion saturation current is hardly influenced by the magnetic field of the order of several kilogauss, it is apparent that these properties are based upon the behavior of the electrons around the probe. A theory developed by Bohm and others is applied here to explain the experimental results.

Dote, T., H. Ameniya, and T. Ichimiya, Effect of the geomagnetic field upon the saturation electron current of an electrostatic probe, J. Geophys. Res., 70, 2258-2261, 1965. (22 refs)

Abstract: -

Earle, G. D., Electron-temperature determination in space plasmas with sounding-rocket electric field booms, Rev. Sci. Instrum., 60, 3076-3077, 1989. (6 refs)

Abstract: The electron temperature can be determined from an interference signal that occurs in electric field data when the probe passes through the shadow of the vehicle. The technique does not perturb the surrounding plasma and is applicable to any sunlit sounding rocket or satellite experiment that measures the ambient electric field using high-impedence probes.

Enloe, C. L., D. L. Cooke, S. Meassick, C. Chan, and M. F. Tautz, Ion collection in a spacecraft wake: laboratory simulations, J. Geophys. Res., 98, 13635-13644, 1993. (33 refs)

Abstract: Data from a laboratory simulation of current collection by a highly biased body in a spacecraft wake are presented. Ion collection from the plasma stream is investigated for both axisymmetric and nonaxisymmetric geometries. Spatially resolved current collection data, obtained by using a segmented collector, are presented for the first time. A threshold potential for the onset of current collection exists; this threshold depends strongly on the geometry of the problem and is related directly to the details of the trajectories of the collected ions. Results from an approximate numerical model of the laboratory experiment as well as the predictions of a simplified analysis are given. The agreement of the numerical solution with the experimental data is good.

Fahleson, U., Theory of electric field measurements conducted in the magnetosphere with electric probes, Space Sci. Rev., 7, 238-262, 1967. (40 refs)

Abstract: -

Fahleson, U., C.-G. Fälthammar, and A. Pedersen, Ionospheric temperature and density measurements by means of spherical double probes, Planet. Space Sci., 22, 41-66, 1974. (18 refs)

Abstract: Rocket-borne double probes for electric field measurements can be intermittently operated in special, diagnostic modes involving current bias and low-impedance shunts to obtain information on the properties of the

ambient ionospheric plasma along the flight path. Several such modes, and the information that they can provide, are analyzed. The data from four rocket flights from ESRANGE are discussed. Electron emperature was measured in three of these flights. In all cases the temperature profile is in good agreement with theoretically predicted profiles based on the CIRA 1965 reference atmosphere and the solar illumination prevailing during the respective flights (twilight).

Fahleson, U. V., M. C. Kelley, and F. S. Mozer, Investigation of the operation of a d.c. electric field detector, Planet. Space Sci., 18, 1551-1561, 1970. (10 refs)

Abstract: Double Langmuir probe detector systems have been flown successfully on three rockets in the past two years to measure quasi-static and a.c. electric fields in the auroral zone ionosphere. Diagnostic tests and redundancy features associated with these experiments are investigated to show that the instruments performed as expected theoretically and that d.c. electric fields with typical magnitudes of 20-40 mV/m have been measured with 10-20 per cent accuracy. Uncertainties of the measurement that are considered include sheath potential and sheath resistance effects, photoemission, energetic particle currents, wake effects, asymmetries from the presence of the booms and rocket body, contact potentials, and the subtraction of upsilon *B from the rocket data to obtain electric fields in an Earth-fixed frame of reference.

Feuerbacher, B. and B. Fitton, Experimental investigation of photoemission from satellite surface materials, J. Appl. Phys., 43, 1563-1572, 1972. (43 refs)

Abstract: Photoemission properties were measured for the materials gold, aluminium, stainless steel, vitreous carbon, Aquadag, graphite, silica, and semiconducting indium oxide. The surfaces were treated so as to simulate the conditions actually encountered in space missions. Results are presented on the reflectance, photoyield, and photoelectron energy distributions up to 25 eV, and on the work function. From the yield curves the photoelectron flux under solar irradiation is calculated. Methods are discussed for controlling photoemission both with respect to the number of emitted electrons and thir energy distribution.

Friedland, L. and Yu. M. Kagan, The theory of electron current to a spherical probe at intermediate pressures, J. Phys. D (Appl. Phys.), 12, 739-748, 1979. (10 refs)

Abstract: The problem of the calculation of the electron current to a retarding spherical probe at intermediate gas pressures is discussed. The current to the probe is expressed in terms of the probability with which an electron, from the undisturbed plasma, will reach the probe surface. This probability is obtained using the Monte Carlo method. It is shown that the computer simulation has to be performed only for the laser few free paths of the electron near the probe surface. An integral equation for the electron energy distribution function in the undisturbed plasma is received from the expression for the electron current.

Garrett, H. B., The charging of spacecraft surfaces, NASA Report 900-982, Jet Propulsion Laboratory, Pasadena, 1981. (217 refs)

Abstract: The buildup of static charge on satellite surfaces is an important issue in the utilization of satellite systems. The analysis of this phenomena has required important advances in basic charging theory and the development of complex codes to evaluate the plasma sheaths that surround satellites. The results of these theories and calculations have wide application in space physics in the design of systems and in the interpretation of low energy plasma measurements. In this review, those aspects of charge buildup on satellite surfaces relevant to the space physics community are summarized. The types of charging processes, models of charge buildup, satellite sheath theories, and charging observations are described with emphasis on basic concepts.

Gerver, M. J., D. E. Hastings, and M. R. Oberhardt, Theory of plasma contactors in ground-based experiments and low-earth orbit, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 150-189, 1990. (26 refs)

Abstract: Previous theoretical work on plasma contactors as current collectors has fallen into two categories: collisionless double layer theory (describing space charge limited contactor clouds) and collisional quasineutral theory. Ground based experiments at low current are well explained by double layer theory, but this theory does not scale well to power generation by electrodynamic tethers in space, since very high anode potentias are needed to draw a substantial ambient electron current across the magnetic field in the absence of collisions (or effective collisions due to turbulence). Isotropic quasineutral models of contactor clouds, extending over a region where the effective collision frequency nu_e exceeds the electron cyclotron frequency omega_ce, have low anode potentiab, but would collect very little ambient electron current, much less than the emitted ion current. A new model is presented, for an anisotropic contactor cloud oriented along the magnetic field, with nu_e < omega_ce. The electron motion along the magnetic field is nearly collisionless, forming double layers in that direction, while across the magnetic field the electrons diffuse collisionally and the potential profile is determined by quasineutrality. Using a simplified expression for nu_e due to ion acoustic turbulence, an analytic solution has been found for this model, which should be applicable to current collection in space. The anode potential is low and the collected ambient electron current can be several times the emitted ion current.

Godard, R. and J. G. Laframboise, Total current to cylindrical collectors in collisionless plasma flow, Planet. Space Sci., 31, 275-283, 1983. (31 refs)

Abstract: A theory is presented for charged-particle collection by a cylindrical conducting object, such as a spacecraft or an electrostatic probe, which is moving transversely through a collisionless plasma, such as those in the upper atmosphere and space. The calculation is approximate, using symmetric potential profiles which are exact for the infinite-cylinder stationary case. Theoretical current predictions are presented for ratios of collector potential to

electron thermal energy e phi /sub c//kT/sub c/ from 0 to -25, for ion-to-electron temperature ratios T/sub i//T/sub c/=1 and 0.5, ratio of collector radius to electron Debye length r/sub c// lambda /sub D/ from 0 to 100, and ratio of flow speed to ion thermal speed S/sub i/=U/(2kT/sub i//m/sub i/)/sup 1/2/ from 0 to 10. Comparisons with existing exact calculations by other authors show that none of these fulfil all of the requirements for nontrivial comparison. Appropriate parameter ranges for future exact calculations are thereby suggested.

Godard, R. and J. G. Laframboise, Low-frequency sheath admittance of a sphere in a collisionless plasma, Radio Sci., 21, 421-428, 1986. (30 refs)

Abstract: A systematic theoretical study of the low-frequency sheath admittance of a spherical electrode immersed in a collisionless, nonflowing, Maxwellian plasma with no magnetic field is done for the following ranges of plasma parameters: nondimensional electrode DC bias potential -25 ≤ e phi_p/kT_e ≤ 25, ion to electron temperature ratio 0 ≤ T_i/T_e ≤ 1, electrode radius to Debye length ratio 0 ≤ r_p/lambda_D ≤ 200. A surprising feature of the results is the existence of double and triple peaks in the sheath capacitance as a function of dc bias potential at certain values of r_p/lambda_D. The same capacitance results also show discontinuous behavior at space potential. Various plasma-diagnostics applications are suggested.

Godard, R. and J. J. Berthelier, Influence of nonuniform surface potential on the measurement of plasma parameters by electrostatic probes, J. Geophys. Res., 94, 17161-17168, 1989. (30 refs)

Abstract: In the theory of electrostatic probes, it is usually assumed that the probe surface potential is uniform. However, nonuniformity of the surface work function produces a nonuniformity in surface potential which may modify the current collection. This effect is examined for an infinite cylinder immersed in a collisionless plasma in the limit when Debye length is much greater than probe radius, by numerically following charged-particle orbits in a Laplace potential. It is shown that only large-scale, macroscopic potential nonuniformities are important for the current collection, and that microscopic nonuniformities are smoothed over relatively short distance by the Laplace equation character of the potential.

Godard, R. and J. S. Chang, Local and total heat transfer on a sphere in a free molecular ionised gas flow, J. Phys. D (Appl. Phys.), 13, 2005-2012, 1980. (11 refs)

Abstract: The calculation has been done for orbit-motion-limited conditions, for charged-particle speed ratios S=u/(2kT/m)/sup 1/2/ from 0 to 7, and for chi /sub p/=e phi /sub p//kT/sub i/ from 0 to -50, where phi /sub p/ is sphere surface potential. Result show that: (i) for the front axial point, the local heat flux may be a non-monotonic function of speed atio and increases with increasing attracting potential - chi /sub p/; (ii) the non-monotonic dependence of the local heat flux to the speed ratio can also be found up to position theta =90 degrees referred to the front axial point (theta =0 degrees); (iii) for the rear axial point, the local heat flux is a continuously decreasing function of speed ratio; (iv) for ion retarding potentials, chi /sub p/>0, the total heat flux decreases with increasing chi /sub p/.

Godard, R., H. G. James, J. G. Laframboise, B. Macintosh, A. G. McNamara, S. Watanabe, and B. A. Whalen, The OEDIPUS Experiment: analysis of the current/voltage data, J. Geophys. Res., 96, 17879-17890, 1991. (31 refs)

Abstract: The OEDIPUS A rocket was successfully launched from the Andoya Rocket Range on January 30, 1989, toward a moderate auroral arc. In this experiment, the forward and the aft payloads constituted a tether. The forward and the aft payloads were connected through a thin isolated wire and the two subpayloads were used as a plasma diagnostic tool. The authors examine the results obtained in the current/voltage mode, when the two subpayloads act as a large double probe. The measured current/voltage characteristic has been fitted with a theoretical model of the current collected to extract the plasma parameters.

Grard, R. J. L., Properties of the satellite photoelectron sheath derived from photoemission laboratory measurements, J. Geophys. Res., 78, 2885-2906, 1973. (15 refs)

Abstract: The electrical properties of the medium around a probe in the outer magnetosphere and in interplanetary space are modified by photoemission. Information on the energy distribution of the photoemitted electrons is important for the evaluation of the physical parameters of the surrounding photosheath and for the interpretation of the probe measurements. This energy distribution has been determined for various materials exposed to sunlight, at the Earth's orbit, by combining laboratory measurements on photoemission with solar spectrum data collected in space.

Grard, R. J. L. and J. K. E. Tunaley, Photoelectron sheath near a planar probe in interplanetary space, J. Geophys. Res., 76, 2498-2505, 1971. (8 refs)

Abstract: Estimating the extent of the photoelectron sheath and associated electric field around a spacecraft in interplanetary space is important for the interpretation of probe measurements. In addition, the theory is relevant to predicting the sheath parameters around bodies such as the moon. In the present study a general treatment of the photoelectron sheath in a vacuum is undertaken, typical sheath profiles are given, and the effects of a tenuous ambient plasma are also discussed. Finally an attempt is made to ascribe orders of magnitude to the physical parameters of the photoelectron sheath.

Grard, R. J. L., K. Knott, and A. Pedersen, The influence of photoelectron and secondary electron emission on electric field measurements in the magnetosphere and solar wind, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 163-189, D. Reidel Publ. Co., Dordrecht, 1973. (25 refs)

Abstract: The feasibility of electric field measurements by the double sphere technique both inside and outside the magnetosphere is critically reviewed. In particular, influences resulting from photoemission, secondary electron emission and the plasma environment are analyzed. It is concluded that a double sphere aerial, 50-100 m long, can

measure dc electric fields as low as 10^-6 A m^-2 in the magnetosphere, provided certain precautions are taken regarding probe symmetry. In the magnetosphere, for current densities larger than 10^-6 A m^-2, the space charge of photoelectrons escaping from the satellite will give rise to potential asymmetries near the probes, and an electron gun is required for control of the satellite potential and reduction of space charge asymmetries.

Hastings, D. E., Enhanced current flow through a plasma cloud by induction of plasma turbulence, J. Geophys. Res., 92, 7716-7722, 1987. (12 refs)

Abstract: Electrodynamic tethers have been proposed as a means of generating power in low Earth orbit. One of the limitations on the power generated is the relatively low electron current that can be collected. It is proposed that the electron current can be significantly enhanced by means of current-induced plasma turbulence in a plasma cloud around the collecting anode. This is examined for the specific case of ion acoustic turbulence. An important conclusion is that the use of plasma clouds in the ionosphere will entail a high-impedance (no stability) and a low-impedance (ion acoustic instability) mode of operation. The low-impedance mode of operation will have two submodes, one steady and one pulsed.

Hastings, D. E., Theory of plasma contactors used in the ionosphere, J. Spacecr. Rockets, 24, 250-256, 1987. (18 refs)

Abstract: The use of plasma contactors has been proposed as a means of enhancing the current flow through an electrodynamic tether. A simple isothermal spherical model of the plasma cloud around a contactor is outlined for a plasma contactor which is biased positively with respect to the ambient plasma and hence collects electrons. It is shown that for significant current amplification to occur, the plasma cloud must be turbulent. The amount of current amplification is obtained as a function of the ion current through the contactor. For ion currents of several amperes amplification factors of 2-6 can be obtained for potential drops in the range 100-500 V. For smaller ion currents, much larger amplification factors can be obtained.

Hastings, D. E., A review of plasma interactions with spacecraft in low Earth orbit, J. Geophys. Res., 100, 14457-14483, 1995. (52 refs)

Abstract: All spacecraft interact in some manner with the plasma environment in space, either the natural environment or a self-induced environment. Early work on plasma spacecraft interactions focused on geosynchronous altitudes where the primary effect is spacecraft charging from the non-Maxwellian high-energy plasma environment. This work has been extensively reviewed. In the last several years, there have been a number of measurements in low Earth orbit (LEO) which, when combined with models, have revealed a rich variety of plasma interaction phenomena at these low altitudes. These are reviewed in this work. These include charging on polar orbits, ram and wake flows, use of high-voltage power systems in space, arcing on high-voltage solar arrays, noise generation in self induced plasma clouds around large, active spacecraft such as the shuttle, anomalous ionization of emitted neutral gases, use of electrodynamic tethers and plasma contactors and phenomena associated with the use of electrically propelled rockets.

Hastings, D. E. and J. Blandino, Bounds on current collection from the far field by plasma clouds in the ionosphere, J. Geophys. Res., 94, 2737-2744, 1989. (11 refs)

Abstract: Plasma clouds can enhance the collection of current to a charged body in the ionosphere in two ways. The first is by providing a large collection are for ionospheric electrons to be collected. The second is by ionization of neutral gas in the vicinity of the plasma cloud. The collection of electrons across a magnetic field is examined, and it is shown that the effective collection area is limited to the region of the cloud where the magnetic field effects are overwhelmed by the electric field effects from the charged body. Upper and lower bounds are obtained for the radius of the core region where the electric field influence dominates. From these, upper and lower bounds are obtained for the current collection. This work is relevant for electrodynamic tether electric power generation by artificial satellites and ion rockets.

Hershkowitz, N., How Langmuir probes work, in Plasma Diagnostics, Vol. 1, Discharge parameters and chemistry, Ed. by O. Auciello cnd D. L. Flamm, pp. 113-183, Academic, San Diego, CA, 1989. (61 refs)

Abstract: -

Hester, S. D. and A. A. Sonin, Ion temperature sensitive end effect in cylindrical Langmuir probe response at ionosphere satellite conditions, Phys. Fluids, 13, 1265-1274, 1970. (14 refs)

Abstract: Computations and experiments are presented for an end effect which can dominate the ion current response of long, cylindrical Langmuir probes in high-speed collision-free flows. The end effect, which takes the form of an increase in ion current when the probe is aligned with the flow direction, depends strongly on the plasma properties and flow speed, and can be large even for very long probes when the speed ratio is large and the probe radius small compared with the Debye length. Experimental data are compared with theory, and some apparently contradictory results reported in the literature are reconciled in terms of the end effect.

Hilgers, A., Interaction between biased sunlit electron collectors in an infinite-Debye-length magnetoplasma: Electron temperature threshold effect, J. Geophys. Res., 100, 5705-5713, 1995. (10 refs)

Abstract: The interaction between two electrically coupled collectors in an infinite-Debye-length magnetoplasma is studied by the use of a simple semianalytical model of particle collection by spherical probes. Because the ambient magnetic field confines the orbits of the particles, the perturbations due to photoelectron emission or screening of the ambient electron flux are guided along the magnetic field direction. When one collector is used as a density probe, depending on the size of the other collector with respect to the mean Larmor radius of the photoelectrons, the model predicts either an electron temperature threshold or a satellite potential threshold in order that the photoelectron current

between the collectors overcomes the ambient electron current to the probe. This explains the variation of the modulation patterns of biased Langmuir probe measurements observed in the vicinity of auroral arcs.

Hilgers, A. and B. Holback, Some aspects of satellite spin effects on spherical probe measurements in a magnetized plasma, Geophys. Res. Lett., 20, 347-350, 1993. (16 refs)

Abstract: Analyzes the spin modulation of the current of a boom-mounted positive biased spherical Langmuir probe observed with the Viking experiment. The satellite perturbs the probe current by two distinct effects: (1) photoelectrons emitted by the low potential part of the satellite, (2) capture of ambient electrons. These two effects are shown to be strongly dependent on the magnetic field orientation with respect to the probe-satellite axis. The authors quantify experimentally and theoretically the range of such effects. They then deduce some basic rules for electron energy and density measurements and for the design of satellite-borne Langmuir probe experiments.

Hilgers, A., B. Holback, G. Holmgren, and R. Bostrom, Probe measurements of low plasma densities with applications to the auroral acceleration region and auroral kilometric radiation sources, J. Geophys. Res., 97, 8631-8641, 1992. (40 refs)

Abstract: When the Viking satellite crossed the auroral acceleration region, a strong spin modulation of the signal of a positively biased Langmuir probe was observed. A simple model based on the photoelectron orbits in a spherically symmetric electric field, in the presence of a uniform magnetic field, gives the right phase and amplitude of the extrema. As a consequence, the authors deduce important information for the design of experiments devoted to very low density measurements. The study is to facilitate the probe density measurements inside the acceleration structures. In these regions the cold plasma density is depleted, with a density of thermal electrons below 5 cm/sup -3/ at altitudes between 6500 km and 8500 km. When auroral kilometric radiation (AKR) is generated inside such structures, the density of thermal electrons can be below 1 cm/sup -3/. The edges of the structures are sharply defined, and the gradient scale can be as low as 1.4 km.

Hoegy, W. R. and L. E. Wharton, Current to moving spherical and cylindrical electrostatic probes, 9 pp., NASA, Greenbelt, MD, USA, June 1971. (? refs)

Abstract: The current collection characteristics of moving cylindrical and spherical electrostatic probes are evaluated for velocities encountered in earth satellites and planetary probes. Simple asymptotic current formulas are developed for large values of the Mach number and potential. A data reduction technique for determining ion densities from the accelerated ion current is discussed. The current expressions are given in terms of standard functions and recurrence relations, therefore allowing straightforward computer programming of all the probe current formulas.

Hoegy, W. R. and L. E. Wharton, Current to a moving cylindrical electrostatic probe, J. Appl. Phys., 44, 5365-5371, 1973. (9 refs)

Abstract: It is stated that cylindrical electrostatic probes (Langmuir probes) are widely used on rockets and satellites for charged particle temperature and density measurements, and may be used on future planetary missions. An important factor is the ratio of probe speed to particle thermal velocity; this is known as the 'speed ratio'. Current expressions are derived that are valid for a wide range of speed ratios and potentials.

Holmes, J. C. and E. P. Szuszczewicz, Circuit details and application note for a versatile plasma probe, 22 pp., Naval Res.Lab., Washington, DC, USA, 31 Dec. 1974. (? refs)

Abstract: A plasma-probe circuit has been developed which couples standard Langmuir-probe diagnostic procedures with pulse-mode operation. Circuit versatility provides the means for studying time-dependent phenomena and establishes an improved capability for measuring plasma density and temperature. As an application of the circuit, it is shown that pulse-mode operation reduces the time variance of electrode surface conditions which often cause hysteresis in the Langmuir-probe current-voltage characteristics.

Holmes, J. C. and E. P. Szuszczewicz, Versatile plasma probe, Rev. Sci. Instrum., 46, 592-598, 1975. (13 refs)

Abstract: A new Langmuir probe circuit is reported which allows high frequency sweeps of pulse mode operation for studying time-dependent plasma phenomena and gives improved measurements of plasma density and temperature. Applications of the circuits are discussed for measurements in contaminating plasma environments. Pulse operation allows the effects of surface contamination of the probe and the resulting distortions and hysteresis of the Langmuir probe current-voltage characteristics to be sorted out.

Holmes J. C. and E. P. Szuszczewicz, Plasma probe system with automatic sweep adjustment, Rev. Sci. Instrum., 52, 377-381, 1981. (9 refs)

Abstract: The capabilities of the pulsed Langmuir probe technique have been expanded to operate in charged particle environments having spatial and temporal variations in plasma or reference-electrode potentials. This practice is implemented by logic circuitry which tracks local floating potentials and adjusts automatically the sweep center voltage to guarantee a complete current-voltage characteristic with minimum sweep voltage excursion.

Hutchinson, I. H., A fluid theory of ion collection by probes in strong magnetic fields with plasma flow, Phys. Fluids, 30, 3777-3781, 1987. (15 refs)

Abstract: A one-dimensional fluid theory of Langmuir probe operation in strong magnetic fields is presented. Cross-field diffusion of ions both into and out of the collection region is consistently accounted for, in effect taking momentum and particle diffusivity to be equal. The results differ by significant factors from previous analyses, which did not account for outward diffusion but in effect set momentum diffusivity to zero. The differences are especially large when parallel flow of the external plasma is present. It is thus clear that the value assumed for the momentum diffusivity

strongly affects the interpretation of recent probe measurements. It is argued that the present results offer a more reliable basis for this interpretation.

Hutchinson, I. H., Principles of Plasma Diagnostics, Chapter 3: Plasma particle flux, Cambridge University Press, New York, 1987. (6 refs)
Abstract: -

Hutchinson, I. H., Ion collection by probes in strong magnetic fields with plasma flow, Phys. Rev. A (Gen. Phys.), 37, 4358-4366, 1988. (23 refs)
Abstract: Fluid-theory calculations are presented of ion collection by electric probes in strongly magnetized plasmas with parallel flow. In the first calculations the problem is treated in a one-dimensional approximation but the cross-field transport of momentum is included in such a way as to model different ratios of viscosity to diffusivity. The results show that the flow deduced from probe measurements is not particularly sensitive to the assumed viscosity, provided it is finite. However, results with zero viscosity are qualitatively different from those with nonzero viscous momentum transport. The second set of calculations is two dimensional but only for fixed (unity) ratio of viscosity to diffusivity. The results are in remarkably good agreement with the corresponding one-dimensional model.

Illiano, J. M. and L. R. O. Storey, Apparent enhancement of electron temperature in the wake of a spherical probe in a flowing plasma, Planet. Space Sci., 22, 873-878, 1974. (10 refs)
Abstract: An apparent enhancement of electron temperature has been observed in the wake of a conducting sphere in a flowing laboratory plasma, using as a probe a segment of the sphere itself. It is considered to be due to the negative potential well that is present in the wake, and which causes the velocity distribution of the electrons collected on the rear hemisphere to be non-Maxwellian. This interpretation of the phenomenon has been checked by numerical simulation.

Ivanova, T. N., S. K. Chapkunov, and L. G. Bankov, On the measurement of the plasma's electron temperature in Earth's ionosphere, Comptes Rendus de l'Academie Bulgare des Sciences, 31, 421-424, 1978. (8 refs)
Abstract: A method is described that renders more exact the determination of the electron temperature as measured by means of a cylindrical Langmuir probe. The method involves telemetering more points of the probes I-V characteristics for greater accuracy. Suitable circuit diagrams are described.

Johanning, D., Effects of nonMaxwellian electron energy distributions on the orbital-motion limited current of cylindrical Langmuir probes, Beitraege aus der Plasma-Physik, 24, 49-56, 1984. (14 refs)
Abstract: One of the main assumptions of the conventional Langmuir probe theory is the demand for a Maxwellian distribution function. The author calculates probe currents for non-Maxwellian electron energy distributions. The classical Langmuir formula for the orbital-motion-limited current of cylindrical probes is simplified. The new equation permits a very easy detailed examination of the effects of non-Maxwellian electron energy distributions on the current-voltage characteristics in the accelerating region for large probe voltages. It is shown that the electron current is independent of the form of the electron energy distribution, if the mean electron energy E is constant. Variations of the mean energy change the probe current slightly. As examples currents are computed for two non-Maxwellian distribution functions.

Johanning, D., H.-R. Lehmann, and W. Seifert, The calculation of plasma parameters from Langmuir-Probe characteristics by means of digital computers, Beitraege aus der Plasma-Physik, 24, 389-405, 1984. (27 refs)
Abstract: Reviews the physical problems resulting from computer analysis of Langmuir probe-characteristics including the estimation of the measurement range for the temperature and density calculation of the cold magnetospheric plasma. It follows the derivation of the equations for the calculation of the real physical parameters. The authors also consider the computer data-handling.

Johanning, D., W. Seifert, and A. Best, Analytical density correction for cylindrical Langmuir probes showing end effects, Plasma Phys. Contr. Fusion, 27, 159-179, 1985. (38 refs)
Abstract: In this paper, an attempt is made to develop an analytical density correction method for cylindrical Langmuir probes displaying end effects caused by finite probe length and sheath expansion as well as potential inhomogeneities within the plasma. The correction depends on probe as well as plasma parameters such as the Debye length, the probe radius, the probe length, and the applied potential, and is based on the deviation of the measured characteristics from the ideal form according to orbit-limited conditions. A comparison with real data was carried out and showed the applicability of the method. Based on this correction, accurate Langmuir-probe electron-density measurements are possible in the very low density ionospheric plasma (n/sub e/>or approximately=10/sup 3/ cm/sup -3/) as well as in the disturbed regions around rockets or satellites, especially in the wake. The range of application of the conventional Langmuir-probe method has been extended.

Johnson, E. O. and L. Malter, A floating double probe method for measurements in gas discharges, Phys. Rev., 80, 58-68, 1950. (5 refs)
Abstract: Measurements with a Langmuir probe in varying or decaying plasmas are defeated by the fact that the plasma potential in these cases follows that of the most positive electrode it can contact. The difficulties can be obviated by the use of a pair of probes joined by a variable potential source. The double probe system "floats" with respect to the discharge system. From the measurements of probe current vs. differential probe voltage, electron temperature and plasma densities can be determined. The method is also applicable to "going" discharges where it has the advantage over the single probe of exerting a negligible influence on the discharge.

Josephson, D., Interpretation of current-voltage characteristics from spherical Langmuir probes in the ionosphere, Royal Institute of Technology, Department of Plasma Physics Internal Report 77-104, 44 pp., 1977. (16 refs)
Abstract: Langmuir probes are often used tools for measurements of electron temperature, plasma density and other properties of the ionospheric plasma. This work contains studies of the interpretation of the current-voltage characteristic from a spherical Langmuir probe experiment performed from a rocket in the ionosphere of the auroral zone. For the positive ion current to a negatively biased probe an explicit expression is formed which gives a good approximation to numerical results from self-consistent solutions of the problem. Influence of some sources of inaccuracy are studied, like the effect of the magnetic field, the wake of the probe and surface problems. Tables have been constructed for evaluation of the current -voltage characteristics for different parameter combinations. The tables have been studied to investigate the influence of different plasma properties, for example the plasma translation velocity relative to the probe and differing electron and ion temperature. Using these tables characteristics from two flights of sounding rockets have been evaluated. Practical experience from the evaluation is reported in the last chapter.

Katz, I., A. Wilson, L. W. Parker, P. L. Rothwell, and A. G. Rubin, Static and dynamic behaviour of spherical probe and satellite plasma sheaths, IEEE Trans. Nucl. Sci., NS-23, 1814-1819, 1976. (6 refs)
Abstract: A number of electrical anomalies in satellites are ascribed to environmentally-caused charging, particularly where the satellites are in geosynchronous orbits. The time-dependence of the interactions between spacecraft plasmas is considered. Using specially designed particle simulation codes, the authors have examined how an idealized satellite dynamically interacts with several different plasmas. These particle codes follow the time-dependent trajectories of representative charges in their self-consistent electric field. The results are compared with previously reported calculations. By defining regions where an equilibrium model applies, that is, regions where dynamical aspects of sheath behavior do not persist as strong space charge fluctuations, multi-dimensional steady state calculations are performed. Four numerical examples are discussed.

Katz, I. and D. E. Parks, Space Shuttle orbiter charging, J. Spacecr. Rockets, 20, 22-25, 1983. (25 refs)
Abstract: Considers the charging of the Space Shuttle orbiter by energetic particles of environmental origin and from emission by accelerators. The theoretical results indicate that precipitating electrons quickly induce large voltages. High voltages may also occur when onboard accelerators inject energetic beams into the high-altitude plasma. A significant conclusion from electron beam experiments is that the rocket is charged to positive potentials much less than anticipated from the theory of probes in a quiescent plasma. Such theories, however, predict the large negative potentials observed by firing energetic ions.

Katz, I. and V. A. Davis, Physical processes associated with current collection by plasma contactors, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 190-200, 1990. (9 refs)
Abstract: Recent flight data confirms laboratory observations that the release of neutral gas increases plasma sheath currents. Plasma contactors are devices which release a partially ionized gas in order to enhance the current flow between a spacecraft and the space plasma. Ionization of the expellant gas and the formation of a double layer between the anode plasma and the space plasma are the dominant physical processes. A theory is presented of the interaction between the contactor plasma and the background plasma. The conditions for formation of a double layer between the two plasmas are derived. Double layer formation is shown to be a consequence of the nonlinear response of the plasmas to changes in potential. Numerical calculations based upon this model are compared with laboratory measurements of current collection by hollow cathode-based plasma contactors.

Katz, I., D. E. Parks, and K. H. Wright, Jr., A model of the plasma wake generated by a large object, IEEE Trans. Nucl. Sci., NS-32, 4092-4096, 1985. (8 refs)
Abstract: A theoretical model is developed to determine the structure of the wake behind a large object, such as a spacecraft, moving with mesosonic speeds through a rarefied plasma. The model takes account of the electric field that develops as a result of separation between electrons and ions. Calculated and experimental results are in good agreement.

Katz, I., D. E. Parks, M. J. Mandell, J. M. Harvey, S. S. Wang, and J. C. Roche, NASCAP, a three-dimensional charging analyzer program for complex spacecraft, IEEE Trans. Nucl. Sci., NS-24, 2276-2280, 1977. (4 refs)
Abstract: A computer code, NASCAP (NASA Charging Analyzer Program), has been developed to simulate the charging of a complex spacecraft in geosynchronous orbit. The capabilities of the NASCAP code include a fully three-dimensional solution of Poisson's equation about an object having considerable geometrical and material complexity, particle tracking, shadowing in sunlight, calculation of secondary emission, backscatter and photoemission, and graphical output. A model calculation shows how the NASCAP code may be used to improve understanding of the spacecraft-plasma interaction.

Katz, I., D. L. Cooke, D. E. Parks, M. J. Mandell, and A. G. Rubin, Three-dimensional wake model for low Earth orbit, J. Spacecr. Rockets, 21, 125-127, 1984. (5 refs)
Abstract: A calculation of the electrostatic potential surrounding a model satellite which ignores ion focusing effects on space charge density is presented. The calculation is a three-dimensional simulation of a finite length right octagonal cylinder moving at Mach 8 through neutral plasma of ambient density n_0 = 10^5 cm^-3 and a temperature theta = 0.1 eV. The object and the plasma parameters represent well the low-altitude AE-C satellite and its measured environment. Surface potentials on the object were obtained using a theory by Parks and Katz (1983). This yields a

mean spacecraft potential in general agreement with that reported by Samir et al. (1979). Wake ion densities were calculated assuming straight line trajectories, while electron densities were taken to be n_o exp(e phi/theta). The electrostatic potential phi was found by solving self-consistently Poisson's equation in the space surrounding the object. Wake potentials and ion currents predicted by this model are compared with those obtained from AE-C data by Samir. Good agreement between observation and the predictions of this relatively simple theory is found.

Katz, I., G. A. Jongeward, V. A. Davis, M. J. Mandell, R. A. Kuharski, J. R. Lilley, Jr., W. J. Raitt, D. L. Cooke, R. B. Torbert, G. Larson, and D. Rau, Structure of the bipolar plasma sheath generated by SPEAR I, J. Geophys. Res., 94, 1450-1458, 1989. (20 refs)

Abstract: The Space Power Experiment Aboard Rockets I (SPEAR I) (Raitt et al., 1988) was specifically designed to measure whether or not the Earth's magnetic field impedes electron collection. SPEAR I had two 10-cm radius spherical electron collectors each at the end of a 3-m boom. The planned experiment was to apply a positive voltage on the spheres, up to 46000 V with respect to the rocket body, and measure the current of collected electrons from the ionosphere. During the SPEAR I flight, the cover failed to come off the plasma contactor. Because of this, the plasma contactor was unable to expel the electrons collected by the spheres. The rocket body was driven thousands of volts negative to achieve current balance. Instead of a single, nearly spherical electron-collecting sheath around the spheres, a second, larger ion-collecting sheath was formed around the rocket body, partially enveloping the electron sheath. The structures and interactions between the two sheaths and the closure of the circuit through the ionosphere are the subjects of the paper.

Katz, I., M. J. Mandell, G. W. Schnuelle, D. E. Parks, and P. G. Steen, Plasma collection by high-voltage spacecraft at low Earth orbit, J. Spacecr. Rockets, 18, 79-82, 1981. (11 refs)

Abstract: A computer model, NASCAP/LEO, of the three-dimensional sheath formation and plasma current collection by high-voltage spacecraft has been developed. By using new space charge density and plasma collection algorithms, it is practical to perform calculations for large, complex spacecraft. The model uses objects and geometries compatible with the NASA Charging Analyzer Program (NASCAP). Results indicate that ion focusing observed in the laboratory during high-voltage collection experiments is probably due to voltage gradients on the collecting surfaces.

Katz, I., M. Mandell, G. Jongeward, and M. S. Gussenhoven, The importance of accurate secondary electron yields in modeling spacecraft charging, J. Geophys. Res., 91, 13739-13744, 1986. (27 refs)

Abstract: Spacecraft charging has commonly been attributed to electrons with several kilovolts of energy impinging upon spacecraft surfaces. Recent experimental evidence from the SCATHA satellite has shown that charging correlates well with electrons of energies greater than 30 keV. In this paper it is shown that the SCATHA observations are consistent with the model of charging in which a satellite is immersed in a Maxwellian plasma, particle collection is orbit limited, and dominant surface effects are the emission of secondary and backscattered electrons. The energy dependence of the secondary yield for multikilovolt incident electrons determines the charging threshold. In the past, inadequate representations of the secondary yield have led experimenters to question the validity of the charging model. The accuracy of the secondary electron yield formulation based on electron stopping power gives good agreement with the SCATHA results.

Kojima, S. and K. Takayama, Probe measurements for high frequency discharge II, J. Phys. Soc. Japan, 5, 357-358, 1950. (2 refs)

Abstract: -

Kwan, J. and B. Ahlborn, Electrical potentials at probes in supersonic plasma flow, Phys. Fluids, 27, 499-505, 1984. (15 refs)

Abstract: Pressure and temperature gradients across a bow shock in a (partially) ionized gas induce an electrical potential that can be used to drive a current. The voltage difference between electrodes (probes) mounted upstream and downstream of the shock depends on the shock strength (flow Mach number and oblique shock angle) and on the difference of the electrode work functions. Such a bow shock generator has an internal resistance that is governed by the electron temperature and the ion current density at the colder (upstream) electrode. The bow shock generator voltage (typically one volt) was measured as a function of the extracted current (up to 0.8 A/cm/sup 2/) for different shock angles and flow Mach numbers. A theoretical model that agrees well with these experiments has been developed.

Laakso, H., T. L. Aggson, and R. F. Pfaff, Jr., Plasma gradient effects on double-probe measurements in the magnetosphere, Ann. Geophys., 13, 130-146, 1995. (20 refs)

Abstract: The effects on double-probe electric field measurements induced by electron density and temperature gradients are investigated. We show that on some occasions such gradients may lead to marked spurious electric fields if the probes are assumed to lie at the same probe potential with respect to the plasma. The use of a proper bias current will decrease the magnitude of such an error. When the probes are near the plasma potential, the magnitude of these error signals, delta E, can vary as delta E approximately=T/sub e/(Delta n/sub e//n/sub e/)+0.5 Delta T/sub e/, where T/sub e/ is the electron temperature, Delta n/sub e//n/sub e/ the relative electron density variation between the two sensors, and Delta T/sub e/ the electron temperature difference between the two sensors. This not only implies that the error signals will increase linearly with the density variations but also that such signatures grow with T/sub e/, i.e., such effects are 10 times larger in a 10-eV plasma than in a 1-eV plasma. This type of error is independent of the probe separation distance provided the gradient scale length is much larger than this distance. The largest errors occur when the probes are near to the plasma potential. At larger positive probe potentials with respect to the plasma potential, the error becomes smaller if the probes are biased, as is usually the case with spherical double-probe experiments in the

tenuous magneto-spheric plasmas. The crossing of a plasma boundary (like the plasmapause or magnetopause) yields an error signal of a single peak. During the crossing of a small structure (e.g., a double layer) the error signal appears as a bipolar signature. Our analysis shows that errors in double-probe measurements caused by plasma gradients are not significant at large scale (>>1 km) plasma boundaries, and may only be important in cases where small-scale (<1 km), internal gradient structures exist. Bias currents tailored for each plasma parameter regime (i.e., variable bias current) would improve the double-probe response to gradient effects considerably.

Laframboise, J. G., Theory of spherical and cylindrical Langmuir probes in a collisionless, maxwellian plasma at rest, University of Toronto, Institute for Aerospace Studies Report No. 100, June 1966. (30 refs)

Abstract: A method has been developed and used to obtain theoretical predictions of the current collected from a collisionless, fully Maxwellian plasma at rest by an electrically conducting Langmuir probe having spherical or cylindrical symmetry. The probe characteristic, or functional relation between current ard probe potential, has been determined for both geometries for probe radii up to 100 times the Debye shielding distance of the hotter species of charged particle, for a complete range of ion-to-electron temperature ratios and for probe potentlals from -25 to +25 times the thermal energy of the hotter species. Each current collection result is computed to a relative accuracy of 0.002 or better in an average time of approximately two minutes on the IBM 7094. Maxwellian velocity distributions and finite current collection are assumed for both ions and electrons. The infinite plasma is replaced by an outer boundary at a finite radius, beyond which a power-law potential is specified. The resulting nonlinear system of integral equations is solved by an iterative numerical scheme which incorporates an extension of the Bernstein and Rabinowitz method to provide charge densities for ions and electrons. No a priori separation into sheath and quasi-neutral regions is assumed. Explicit comparison is made between the results for a completely Maxwellian plasma and those for a plasma mono-energetic in attracted particles, as treated by Bernstein and Rabinowitz, Lam, and Chen. It is shown that in certain cases, the mono-energetic plasma does not adequately simulate the Maxwellian plasma. It is also shown that difficulties encountered by Bernstein and Rabinowitz in computing the ion current for the cylinder in the zero-ion-temperature limit are illusory, and that the computations of Chen for this case do not take into account the fact that the ion temperature acts as a singular perturbation. Computed charge density and potential functions are presented graphically. Computed probe characteristics are presented in graphical and tabular form. A listing is included of the Fortran programs used to obtain these results.

Laframboise, J. G., Incident velocity distributions on sampling electrodes of spacecraft plasma instruments, in Proc. of 17th ESLAB Symp. on 'Spacecraft/Plasma Interactions and their Influence on Field and Particle Measurements', Noordwijk, The Netherlands, 13-16 Sept. 1983, ESA SP-198, 101-108, 1983. (21 refs)

Abstract: We present information which is applicable to the sampling conditions of low-energy charged-particle detectors (including electrostatic energy analyzers and flush disc probes) located on the surfaces of spacecraft or sounding rockets. The information is in the form of numerically-conputed velocity-space cutoff boundaries for incident particles. It is obtained from the results of self-consistent sheath potential calculations for spherical slectrodes in collisionless, Maxwellian plasmas. We also apply it to the calculation of current-voltage characteristics of retarding-potential analyzers. Two features of the results may cause surprise. The planar-sheath approximation does not provide the correct result in the limit of small Debye length; this is because of the presence of a quasineutral presheath outside the electrode. Also, the sheath potential can influence the velocity-space cutoff boundaries at particle energies up to many times the thermal energy of the surrounding plasma.

Laframboise, J. G., Calculation of secondary-electron escape currents from inclined spacecraft surfaces in a magnetic field, in Proc. USAF/NASA Spacecraft Environmental Interactions Technology Conference, 1984. (6 refs)

Abstract: In low Earth orbit, the geomagnetic field B is strong enough that secondary electrons emitted from spacecraft surfaces have an average gyroradius much smaller than typical dimensions of large spacecraft. This implies that escape of secondaries will be strongly inhibited on surfaces which are nearly parallel to B, even if a repelling electric field exists outside them. This effect is likely to make an important contribution to the current balance and hence the equilibrium potential of such surfaces, making high-voltage charging of them more likely. We present numerically-calculated escaping secondary electron fluxes for these conditions. For use in numerical spacecraft-charging simulations, we also present an analytic curve-fit to these results, accurate to within 3% of the emitted current.

Laframboise, J. G., Calculation of escape currents of electrons emitted from negatively charged spacecraft surfaces in a magnetic field, J. Geophys. Res., 93, 1933-1943, 1988. (45 refs)

Abstract: In low Earth orbit, the geomagnetic field B is strong enough that secondary electrons or photoelectrons emitted from spacecraft surfaces have an average gyroradius much smaller than typical dimensions of large spacecraft. This implies that escape of these electrons will be strongly inhibited on surfaces which are nearly parallel to B, even if a repelling electric field exists outside them. This effect is likely to make an important contribution to the current balance and hence the equilibrium potential of such surfaces, making high-voltage charging of them more likely. The author presents numerically-calculated escaping electron fluxes for these conditions, based on the approximations of uniform fields and Maxwellian emission-velocity distributions. The author also presents an analytic curve-fit to the results for the important case of normal electric field (uniformly-charged surfaces) and proposes a simple approximate method for calculating the space-charge-density distribution of escaping electrons.

Laframboise, J. G., Current collection by a positively charged spacecraft: Effects of its magnetic presheath, J. Geophys. Res., 102, 2417-2432, 1997. (31 refs)

Abstract: We analyze the processes governing current collection by a positively charged spacecraft moving through the ionospheric magnetoplasma. We consider separately the situations of a suborbital rocket payload and a tethered subsatellite, such as TSS-1. For the latter case, our analysis leads to an upper-bound electron-current prediction of 6.4 times the Parker and Murphy [1967] current, independently of the subsatellite's voltage with respect to its surroundings. This result is consistent with the experimental results of Dobrowolny et al. [1995], who measured currents which were generally between 4 and 6 times the Parker and Murphy [1967] values. Iterative refinement of our model decreases our prediction to about 6.0 and 5.9 times their values, for ambient electron-to-ion temperature ratios Te/Ti = 1 and 2, respectively. Incorporation of estimated effects of thermalization due to instabilities in the magnetic presheath decreases our prediction further to about 5.6 and 5.3 times their values, respectively. Features of our model include a quasineutral "magnetic presheath" region, which extends very far from the spacecraft in both directions along the geomagnetic field; an ion-enhancement region, also quasineutral, which is located forward of the magnetic presheath and concentrates the electron flux passing into it; and an elongation, along the drift direction, of the electron collection area perpendicular to the geomagnetic field. This elongation further increases the electron collection.

Laframboise, J. G. and J. Luo, High voltage polar-orbit and beam-induced charging of a dielectric spacecraft: a wake-induced barrier effect mechanism, J. Geophys. Res., 94, 9033-9048, 1989. (71 refs)

Abstract: Proposes a mechanism which permits high-voltage auroral zone charging to occur on a large enough mostly dielectric spacecraft in situations where one would otherwise expect such charging to be suppressed by secondary-electron or photoelectron escape from exposed surfaces. This extends, to lower equivalent temperatures, the range of auroral electron plasma conditions in which one can expect such a spacecraft to undergo high-voltage charging. It should also permit such charging to occur in sunlight. The results presented yield a tentative prediction that the Shuttle Orbiter is large enough spacecraft for such charging, if one adopts the usual definition of 'high-voltage' charging as involving surface potentials at least 100 V different from space potential. The mechanism involves the suppression of electron emission by a potential barrier in the spacecraft wake.

Laframboise, J. G. and J. Rubinstein, Theory of a cylindrical probe in a collisionless magnetoplasma, Phys. Fluids, 19, 1900-1908, 1976. (9 refs)

Abstract: A theory of a cylindrical electrostatic probe in a collisionless plasma is developed. The probe is inclined at an angle to a uniform applied magnetic field. The theory yields an upper bound and an adiabatic limit for current collection. Sheath potential profiles are presented. The case of a probe aligned with the magnetic field is treated separately.

Laframboise, J. G. and J. S. Chang, Theory of charge deposition on charged aerosol particles or arbitrary shape, J. Aerosol Sci., 8, 331-338, 1977. (16 refs)

Abstract: The theory of charge collection by a charged equipotential aerosol particle in a stationary medium having no external electric fields, is re-examined for arbitrary particle shape. In the continuum limit, when the particle size is much greater than the ion mean free path, it is shown that the expression for charging current dependence on particle surface potential, previously derived for spheres has the same form for all particle shapes. In the general (finite mean-free-path) case, an approximate charging current expression is obtained and evaluated for oblate and prolate spheroids. This result is shown to have the correct form in the collisionless (large mean-free-path) limit if the particle surface potential is repelling for the ion species considered.

Laframboise, J. G. and L. J. Sonmor, Current collection in a magnetoplasma, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 13-49, 1990. (56 refs)

Abstract: We present a survey of a very incomplete subject. Our presentation is intended in part as an introduction to topics to be covered in greater detail by others later in this Workshop. The best-developed and simples theories for current collection are steady-state collisionless theories, and these must be understood before departures from them can be analyzed usefully, so we begin with a review of them. We include some recent numerical results by one of us (L.J.S.) which indicate that steady-state collisionless Laplace-limit currents remain substantially below the Parker-Murphy (1967) canonical upper bound out to very large electrode potentials, and approach it as a limit only very slowly if at all. Attempts to correct this theory for space-charge effects lead to potential disturbances which extend to infinite distance along the electrode's magnetic shadow, unless collisional effects are also taken into account. However, even a small amount of relative plasma drift motion, such as that involved in a typical rocket experiment, can change this conclusion fundamentally. It is widely believed that time-averaged current collection may be increased by effects of plasma turbulence, and we review the available evidence for and against this contention. Steady-state collisionless particle dynamics predicts the existence of a toroidal region of trapped orbits which surrounds the electrode. Light emissions from this region have been photographed, indicating that collisional ionization may also occur there, and this, and/or scattering by collisions or possibly turbulent fluctuations in this region, may also increase current collection by the electrode. We also discuss effects on particle motions near the electrode, associated with "breakdown of magnetic insulatoin" in the region of large electric fields near it.

Laframboise, J. G. and L. J. Sonmor, Physics of current collection by electrodes in space magnetoplasmas: A review, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 77-92, Editrice Compositori, Bologna, 1992. (37 refs)

Abstract: We present a survey of a very incomplete subject. The best-developed and simplest theories for current collection in magnetic fields are steady-state collisionless theories, and these must be understood before departures

from them can be analyzed usefully, so we begin with a review of them. We include some recent numerical results by one of us (L.J.S.) which indicate that steady-state collisionless Laplace-limit currents remain substantially below the Parker-Murphy (1967) canonical upper bound out to very large electrode potentials, and approach it as a limit only very slowly if at all. Attempts to correct this theory for space-charge effects lead to potential disturbances which extend to infinite distance along the electrode's magnetic shadow, unless collisional effects are also taken into account. However, even a small amount of relative plasma drift motion, such as that involved in a typical rocket experiment, can change this conclusion fundamentally. It is widely believed that time-averaged current collection may be increased by effects of plasma turbulence, and we review the available evidence for and against this contention. Steady-state collisionless particle dynamics predicts the existence of a toroidal region of trapped orbits which surrounds the electrode. Light emissions from this region have been photographed, indicating that collisional ionization may also occur there, and this, and/or scattering by collisions or possibly turbulent fluctuations in this region, may also increase current collection by the electrode. We also discuss effects on particle motions near the electrode, associated with "breakdown of magnetic insulation" in the region of large electric fields near it.

Laframboise, J. G. and L. J. Sonmor, Current collection by probes and electrodes in space magnetoplasmas: a review, J. Geophys. Res., 98, 337-357, 1993. (124 refs)

Abstract: We present a survey of a very incomplete subject. The best-developed and simplest theories for current collection in magnetic fields are steady-state collisionless theories, and these must be understood before departures from them can be analyzed usefully, so we begin with a review of them. Recent numerical results indicate that steady-state collisionless Laplace-limit currents remain substantially below the Parker-Murphy (1967) canonical upper bound out lo very large electrode potentials and approach it as a limit only very slowly if at all. Attempts to correct this theory for space-charge effects lead to potential disturbances which extend to infinite distance along the electrode's magnetic shadow, unless collisional effects are also taken into account. However, even a small amount of relative plasma drift motion, such as that involved in a typical rocket experiment, can change this conclusion fundamentally. It is widely believed that time-averaged current collection may be increased by effects of plasma turbulence, and we review the available evidence for and against this contention. Steady-state collisionless particle dynamics predicts the existence of a toroidal region of trapped orbits which surrounds the electrode. Light emissions from this region have been photographed, indicating that collisional ionization may also occur there, and this, and/or scattering by collisions or possibly turbulent fluctuations in this region, may also increase current collection by the electrode. We also discuss effects on particle motions near the electrode, associated with "breakdown of magnetic insulation" in the region of large electric fields near it.

Laframboise, J. G. and L. W. Parker, Probe design for orbit-limited current collection, Phys. Fluids, 16, 629-636, 1973. (14 refs)

Abstract: A new type of electrostatic probe design is proposed, which should permit temperature and density measurements for both ions and electrons from the accelerated current regions of the probe characteristics. The method is based on the use of one of a family of nonspherical probe geometries which can be operated as if they were spheres collecting orbit-limited current. The proposed probe operation is theoretically justified by rederiving the usual orbit-limited current expressions for spheres and circular cylinders, without making any assumption regarding particle angular momentum conservation. Some aspects of probe use in flowing plasmas and in magnetic fields are also discussed.

Laframboise, J. G. and R. Godard, Perturbation of an electrostatic probe by a spacecraft at small speed ratios, Planet. Space Sci., 22, 1145-1155, 1974. (22 refs)

Abstract: The disturbance of an electrostatic probe, by the spacecraft carrying it, in the intermediate altitude regime from about 3000 km to the plasmapause, is discussed. Three kinds of disturbance are studied: loss of symmetry of the probe sheath due to the presence of the spacecraft; creation of potential barriers around the probe; and particle removal, or shadowing, by the spacecraft. The first of these effects is shown to be nonexistent for a spherical probe if it is operated under orbit-limited conditions. Methods are given for estimating the magnitudes of the other two effects in any given situation. The electrostatic probe experiment on the Explorer 31 spacecraft is discussed as an example.

Laframboise, J. G., J. Rubinstein, and F. H. Palmer, Theory of topside sounder transmission effects on antenna quasistatic sheath impedance, Radio Sci., 10, 773-784, 1975. (13 refs)

Abstract: A quasistatic theory has been constructed for predicting low frequency sheath impedance changes produced on spacecraft sounding antennas as a result of intense RF emissions by them. The theory involves calculation of time-independent sheath profiles around an infinite cylindrical antenna in a collisionless plasma, using a numerical scheme which includes the time-averaged repelling force on electrons due to a nonlinear effect of the RF near field of the antenna. These sheath profiles then yield values of static sheath capacitance and conductance per unit antenna length. The theory includes a more accurate electron current calculation than in the earlier treatments by the authors. Results are presented for a range of parameters applicable to spacecraft sounding conditions. In most cases, the quasistatic sheath capacitance and conductance both decrease as RF amplitude increases. At large enough RF amplitudes, the capacitance becomes almost independent of antenna potential and Debye ratio.

Laframboise, J. G., L. J. Sonmor, and B. A. Whalen, The magnetic presheath of a positively charged spacecraft: Estimating its potential, in Proc. of the Fourth International Conference on Tethers in Space, pp. 1025-1034, Smithson. Inst., Washington, D. C., 1995. (? refs)

Abstract: ...

Lai, S. T., An improved Langmuir probe formula for modeling satellite interactions with near-geostationary environment, J. Geophys. Res., 99, 459-467, 1994. (24 refs)
Abstract: The Langmuir plasma probe model is an important tool in spacecraft current collection and charging calculations. In ideal geometries, such as a sphere or an infinitely long cylinder, the model is well understood. However, the realistic geometries of current collectors, or spacecraft, are nonideal. An empirical formula for a Langmuir probe with a given nonideal geometry would be useful. We derive such a formula for the SCATHA satellite by using the SC10 potential data obtained during electron beam emissions. The satellite rotated perpendicular to sunlight with the SC10 booms in the equatorial plane. We choose one special mode of operation during a quiet space environment. In this mode the beam current increased continuously, while the energy remained constant. We analyzed the variations of the vehicle potential responding to the unique driving factor, the beam current. To provide physical explanation to the behavior of the SC10 potential data, we model the interactions between the beam, photoelectron, and ambient currents. We present an algorithm which successfully yields an empirical Langmuir probe formula for SCATHA, from which we obtain improved estimate of ambient electron temperatures and densities. The results predicted by the improved Langmuir probe model compare favorably with the very few published measurements from the region.

Lai, S. T., H. A. Cohen, T. L. Aggson, and W. J. McNeil, Boom potential of a rotating satellite in sunlight, J. Geophys. Res., 91, 12137-12141, 1986. (23 refs)
Abstract: An interpretation is provided for the behavior of long boom potential measurements taken on the spinning P78-2 (SCATHA) satellite at near geosynchronous altitudes. This study uses data taken during a quiet day, with the satellite in sunlight. The data show periodic variations with a maximum amplitude of 6 V. The theory explains why the variations correlate well with Sun direction but not with the geomagnetic field. A current balance model, assuming a Maxwellian distribution of photoelectrons, is studied. The photoelectron temperature, the degrees of positive charging of the boom and of the satellite, and the ambient electron flux are calculated. Deviations from the model are discussed.

Lai, S. T., H. A. Cohen, T. L. Aggson, and W. J. McNeil, The effect of photoelectrons on boom-satellite potential differences during electron beam ejection, J. Geophys. Res., 92, 12319-12325, 1987. (27 refs)
Abstract: Data taken on the SCATHA satellite at geosynchronous altitudes during periods of electron beam ejection in sunlight showed that the potential difference between an electrically isolated boom and the satellite main body was a function of beam current, energy, and boom-Sun angle. The potential difference decreased as the boom area illuminated by the Sun increased; the maximum and minimum potential differences were measured when minimum and maximum boom areas, respectively, were exposed to the Sun. It is shown that photoelectrons, created on the boom, could be engulfed in the electrostatic field of the highly charged satellite main body. Theoretical calculations made using a simple current balance model showed that these electrons could provide a substantial discharging current to the main body and cause the observed variations in the potential difference between the main body and the booms.

Lam, S. H., Unified theory for the Langmuir probe in a collisionless plasma, Phys. Fluids, 8, 73-87, 1965. (9 refs)
Abstract: An asymptotic analysis is presented of the Langmuir-probe problem in a quiescent, collisionless plasma in the limit of large body dimension to Debye length ratio. The structures of the electric potential distribution about spheres and cylinders are analyzed and discussed in detail. It is shown that when the probe potential is smaller than a certain well defined value, there exists no sheath adjacent to the solid surface. At large body potentials, for which a sheath is present, the electric potential distribution is given in terms of several universal functions. Master current-voltage characteristic diagrams are given which exhibit clearly the effects of all the pertinent parameters in the problem. An explicit trapped-ion criterion is presented. The general problem with an arbitrary boy dimension to Debye length ratio is qualitatively discussed.

Langmuir, I. and K. B. Blodgett, Currents limited by space charge between coaxial cylinders, Phys. Rev., 22, 347-356, 1923 (also in The Collected Works of Irving Langmuir, Ed. by C. Guy Suits, Vol. 3: Thermionic Phenomena, Pergamon Press, Oxford, 1961, pp. 115-124). (6 refs)
Abstract: Limiting current between coaxial cylindcrs; calculation of the function beta = f(r/r/sub 0/) in the, space charge equation i = (2sqrt(2)/9)sqrt(e/m)V^(3/2)(r,beta^2). Two different infinite senes were obtained for beta and the coefficients for fourteen terms of each were determined. These two sets were checked against each other. Thus values of beta^2 were obtained accurateto 1 in 10,000. These were checked by an integration method which was also used to calculate values in the region where the series failed. For a cathode of radius r/sub 0/ inside an anode of radius r, beta^2 increases from 1 at r/r/sub 0/ = 11.2 to a maximum value 1.0946 at r/r/sub 0/ = infinity, decreases to a minimum value 0.9990 at r/r/sub 0/ = 30,000, and becomes 1 at r/r/sub 0/= infinity. The customary assumption beta^2 = 1 has therefore led to errors up to 9.5 per cent in previous calculations, but this error is usually about equal and opposite to that introduced by neglecting the effect of initial velocities. For the cathode outside the anode, beta^2 is given very closely by the equation beta^2 = 4.6712 (r/sub 0/r) [log/sub 10/(r/sub 0/r)—log/sub 10/(sqrt(2))]^(3/2), for r/sub 0//r > 10. The empirical constant sqrt(2) is interpreted to mean that the potential distribution near the anode is unaltered if the hot cathode is replaced by a cold cylinder having one half the cross-section of the original cathode. The correction for initial velocities is less for a cylindrical cathode inside an anode than for parallel planes. In the inverted case it is much greater than in the case for parallel planes, and the effect of the tangential component of the initial velocity may greatly decrease the current that flows.

Langmuir, I. and K. B. Blodgett, Currents limited by space charge between concentric spheres, Phys. Rev., 24, 49-59, 1924 (also in The Collected Works of Irving Langmuir, Ed. by C. Guy Suits, Vol. 3: Thermionic Phenomena, Pergamon Press, Oxford, 1961, pp. 125-135). (3 refs)

Abstract: Limiting current betwcen concentric spheres; calculation of the function alpha = f(r/r/sub 0/) in the space charge equation i = (4 sqrt(2/9) sqrt(e/m) V^(3/2)/alpha^2. - The coefficients of the first six terms of a series for alpha were determined, and alpha^2 calculated from this series. The results were checked by an integration method which was also used to calculate values in the region where the series faileds For an emitter of radius r/sub 0/ inside a collector of radius r, values of alpha^2 when log (r/r/sub 0/) > 6.4 are given by the equation 1/2 alpha^2 = 0.112 log (log r/r/sub 0/) + 1/3 log (r/r0) + 0.152. Where the collector is the inside sphere, values of alpha^2 for r/sub 0//r > 9 are given by the equation (1/2 alpha^2)^(2/3) = 1.11 (r/sub 0//r) - 1.64. It is shown that when the collector is the inside sphere the potential distribution near the collector is unaltered if the emitter is replaced by a non-emitting sphere with a diameter .677 times the original diameter. Limiting current between coaxial cylinders and between concentric spheres. - Equations are derived for the current in terms of the radius of curvature of the emitter. It is shown that at a surface in space four-fifths of the distance from the emitter to the collector the current density is independent of the radius of curvature when r/r/sub 0/ or r/sub 0//r < 2; and in the case of coaxial cylinders with the emitter inside this holds true even when r/r/sub 0/ = 20.

Latham, P. M., R. A. Bond, A. J. Pearce, and A. R. Martin, Laboratory simulation of spacecraft-plasma wake phenomena and comparison with computational predictions, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 37-62, Editrice Compositori, Bologna, 1992. (4 refs)
Abstract: Experimental simulation of plasma wakes produced by simple bodies, under plasma conditions equivalent to low Earth orbit, have been carried out, with the aim of cross-validating computer codes, in particular the code SAPPHIRE. The simulation methods used are described, concentrating on critical areas necessary for valid simulation. Typical results are given for illustration, and a comparison is made with the predictions of the code SAPPHIRE. Generally, both qualitative and quantitative agreement are good. Examples are also given showing how SAPPHIRE may be applied to other experimental problems.

Li, W. and J. G. Laframboise, The space-charge effect on current collected by a spherical probe in a magnetized plasma, in Proc. of the Fourth International Conference on Tethers in Space, pp. 993-1007, Smithson. Inst., Washington, D. C., 1995. (? refs)
Abstract: ...

Li, W.-W., Current collection in an anisotropic plasma, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 74-87, 1990. (4 refs)
Abstract: A general method is given to derive the current-potential relations in anisotropic plasmas. Orbit limit current is assumed. The collector is a conductive sphere or an infinite cylinder. Any distribution which is an arbitrary function of the velocity vector can be considered as a superposition of many mono-energetic beams whose current-potential relations are known. The results for two typical pitch angle distributions are derived and discussed in detail. The general properties of the current potential relations are very similar to that of a Maxwellian plasma except for an effective temperature which varies with the angle between the magnetic field and the charging surface. The conclusions are meaningful to generalized geometries.

Lilley, J. R., Jr., I. Katz, and D. L. Cooke, v x B and density gradient electric fields measured from spacecraft, J. Spacecr. Rockets, 23, 656-658, 1986. (7 refs)
Abstract: -

Lindqvist, P.-A., The potential of ISEE-1 in different plasma environments, in Proc. of the 17th ESLAB Symp. on Spacecraft/Plasma Interactions and their Influence on Field and Particle Measurements, Noordwijk, The Netherlands, 1983, ESA SP-198, 25-33, 1983. (5 refs)
Abstract: The quasistatic electric field experiment on the ISEE-1 satellite uses spherical double probes to measure the electric field. The potential of each probe relative to the plasma potential can be controlled with a bias current, which compensates for the loss of photoelectrons from the probe. Thus, keeping a probe at plasma potential, the satellite potential is measured with reference to the probe. The dependence of the satellite potential on the ambient plasma electron flux is discussed. It is shown theoretically and found experimentally that the satellite potential is proportional to log(n_e·T_e^1/2). Examples are given of how the satellite potential varies in different regions of space penetrated by ISEE: the solar wind, the magnetosheath, the dayside magnetosphere, the magnetotail lobes, the plasmasheet and the plasmasphere. Typical values of the satellite potential are from one or a few Volts in the dense plasmasphere or magnetosheath to nearly 50 Volts in the tenuous tail lobes.

Lindqvist, P.-A., G. T. Marklund, and L. G. Blomberg, Plasma characteristics determined by the Freja electric field instrument, Space Sci. Rev., 70, 593-602, 1994. (11 refs)
Abstract: A new approach to the study of ionospheric plasma characteristics is presented using data from the Freja double probe electric field instrument. Plasma characteristics are derived from continuous measurements of the satellite potential and from intermittent Langmuir sweeps. These provide information on both relative variations in the plasma density and absolute density and temperature, useful for comparisons with other plasma measurements on Freja, and essential for the interpretation of the electric field measurements. The on-board memory makes it possible to obtain full-orbit coverage of this type of information, which is a new feature of the Freja measurements. The memory is also used for high time resolution Langmuir sweeps which allow for the first time detailed studies of the time behavior of the probe response and computation of the probe-plasma capacitance. Comparisons are also made with similar measurements on earlier missions.

Linson, L. M., Current-voltage characteristics of an electron-emitting satellite in the ionosphere, J. Geophys. Res.,

74, 2368-2375, 1969. (14 refs)

Abstract: Thc limitation on the current that a sphere at a large positive potential phi can draw from a surrounding collisionless plasma in a magnetic field is investigated. The model assumes that for e phi >> kT/sub e/ the ions are excluded from the region of high potential surrounding the sphere. This region increases in size as phi increascs, thus presenting an increased cross-sectional area through which electrons can be collected from the plasma along magnetic field lines. We suggest that there is a critical density at which the electron cloud surrounding the positive body is sufficiently turbulent that the electrons are free to diffuse radially across magnetic field lines and thereby reach the sphere. This model yields a saturation current I prop phi/ln phi, which scales directly with the density, thermal velocity, and square of the body size. At high background electron concentrations a space-charge-limited flow model that predicts I prop phi/sup 6/7/ is more applicable, whereas the more pessimistic limit of I prop phi/sup 1/2/, which follows from thc conservation of the single particle constants of motion in time invariant electric and magnetic fields, is appropriate when the background electron concentration is low.

Linson, L. M. and J. D. Daugherty, On the solution of Langmuir's spherical space-charge equation, Proc. IEEE, vol. 57, no.1, 87-88, 1969. (7 refs)

Abstract: An analytic (parametric) integration of Langmuir's equation for spherical space-charge flow has been obtained. The analytic expression for the voltage distribution is expanded for both large and small ratios of cathode to anode radius. The usefulness and form of the resulting precise expressions are compared with Langmuir's empirical results.

Loeb, L. B., Basic Processes of Gaseous Electronics, University of California, Berkeley, 1961. (? refs)

Abstract: ...

Maassberg, H. and U. Isensee, Symmetric theory of probe-plasma interactions, Planet. Space Sci., 29, 555-577, 1981. (16 refs)

Abstract: The theory of interactions between a probe and the surrounding plasma at rest is developed in a spherically and in a cylindrically symmetric model (probe theory). The theory is based on the Vlasov-Poisson system an iterative procedure. Various ambient plasma and charged particle emission properties are described by the complete set of boundary conditions for the distribution functions in the phase space. Potential and space charge density in the whole surroundings of the probe as well as the current densities of all plasma constituents are calculated self-consistently. The regions of the phase space with particle trajectories of the same kind can be approximated depending on the plasma properties. The current densities can be estimated analytically. The approach yields self-consistent approximations and is the only stringent derivation of the thick sheath and of the thin sheath approximation of the classical Langmuir theory. These approximations are generalized with respect to the charged particle emission from the surface. The symmetric probe theory is applied problems of spacecraft environment and spacecraft charging.

Makita, H. and K. Kuriki, Comparative study of spherical and cylindrical drift probes, in Proc. of 10th Int. Symp. on Rarefied Gas Dynamics, Ed. by J. Leith Potter, pp. 1007-1014, American Inst. Aeronautics & Astronautics, New York, 1977. (6 refs)

Abstract: Current-voltage characteristics of drifting spherical and cylindrical Langmuir probes has been studied experimentally in collisionless plasmas. In order to single out the effect of probe speed on the ion current collection, the experiment was carried out by a rotating arm facility. With this device the probe speed as well as properties of the undisturbed plasma was accurately determined. The ion current of the spherical probe showed good agreements with the theory of Godard at very large or small R/sub p// lambda /sub D/, the ratio of probe radius to Debye length. Discrepancies were found at the intermediate R/sub p// lambda /sub D/, where the spherically asymmetric potential distribution was important. At some R/sub p// lambda /sub D/, the ion current collection by the cylindrical probe was dominated by the end effect, which was explained in terms of the spherical probe

Makita, H. and K. Kuriki, Current collection by spherical Langmuir probes drifting in a collisionless plasma, Phys. Fluids, 21, 1279-1286, 1978. (10 refs)

Abstract: Current-voltage characteristics of drifting spherical Langmuir probes have been studied experimentally in collisionless plasmas. In order to single out the effect of probe speed on the ion current collection, the experiment was carried out by a rotating arm facility. With this device the probe speed as well as the properties of the undisturbed plasma were accurately determined. As predicted by the theories of Kanal, Godard, and Laframboise, the ion current at small speed ratios was found to be either increased or decreased as the probe speed increased depending on the ratio of probe radius to Debye length R/ lambda /sub D/. As for the absolute magnitude of ion current change by the probe speed, however, discrepancies were found between these theories and the present experiment. The difference is considered to originate from asymmetry in the potential distribution, which is not included in these theories.

Makita, H. and K. Kuriki, Determination of ion temperature by two-probe method, Rev. Sci. Instrum., 49, 371-373, 1978. (7 refs)

Abstract: The ion temperature is determined from graphs derived from the theory of Laframboise and the results of measurements of the ion currents collected from a spherical and a cylindrical Langmuir probe. A procedure for determining the electron temperature, the ion number density, the nondimensional ion currents, and the floating potentials of the two probes is described. A graph showing the normalized ion current as a function of the ratio of the probe radius to the Debye length for various values of the square root of the ratio of the ion temperature to the electron temperature is presented. The results of an experiment on a low density xenon plasma, corrected for the end effect of the cylindrical probe, are compared with results derived from Laframboise's theory. An additional verification of the

method is provided by a drift probe experiment.

Mandell, M. J., I. Katz, and D. L. Cooke, Potentials on large spacecraft in LEO, IEEE Trans. Nucl. Sci., NS-29, 1584-1588, 1982. (9 refs)

Abstract: A system-oriented computer code is used to predict surface charging due to voltages generated within a satellite operating in the typical dense plasma environment of LEO. The use of this code is demonstrated by predicting the expansion of electric fields onto a Kapton surface from a pinhole over a biased conductor in a LEO environment. The results are compared to a more-exact solution and experimental data.

Mandell, M. J., I. Katz, G. W. Schnuelle, and J. C. Roche, Photoelectron charge density and transport near differentially charged spacecraft, IEEE Trans. Nucl. Sci., NS-26, 5107-5111, 1979. (5 refs)

Abstract: The effects of photoelectron space charge and current density on differentially charged spacecraft are studied. The steady-state potentials of a sunlit cylinder are calculated using a two-dimensional computer code with a fully self-consistent treatment of space charge and an effective surface conductivity treatment of photoelectron currents. It is found that under conditions of strong differential charging the results do not differ greatly from NASCAP results, which neglect photosheath space charge and currents.

Mandell, M. J., I. Katz, V. A. Davis, and R. A. Kuharski, NASCAP/LEO calculations of current collection, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 334-351, 1990. (9 refs)

Abstract: NASCAP/LEO is a 3-dimensional computer code for calculating the interaction of a high-voltage spacecraft with the cold dense plasma found in Low Earth Orbit. Although based on a cubic grid structure, NASCAP/LEO accepts object definition input from standard CAD programs so that a model may be correctly proportioned and important features resolved. The potential around the model is calculated by solving the finite element formulation of Poisson's equation with an analytic space charge function. Five previously published NASCAP/LEO calculations for three ground test experiments and two space flight experiments are presented. The three ground test experiments are a large simulated panel, a simulated pinhole, and a 2-slit experiment with overlapping sheaths. The two space flight experiments are a solar panel biased up to 1000 volts, and a rocket-mounted sphere biased up to 46 kilovolts. In all cases, we find good agreement between calculation and measurement.

Medicus, G., Theory of electron collection of spherical probes, J. Appl. Phys., 32, 2512-2520, 1961. (4 refs)

Abstract: The theory of spherical probes based on the impact parameter of the carriers is given. Fields outside the sheath are neglected, which is justified for electron collection. The results are obtained more directly than by Mott-Smith and Langmuir. General results such as the transition from impact parameter limitation to sheath limitation for the spherical case, not previously treated in the literature, are derived. The case of a uniformly accelerated half-Maxwellian (as opposed to a Maxwellian with drift) is treated in detail; the case of a pure Maxwellian superimposed on the accelerated half-Maxwellian and the superposition of two different accelerated half-Maxwellians is discussed briefly.

Morita, M., T. Ogawa, K. Tsuruda, and T. Obayashi, Electric-field measurements with double probes in a space-simulation plasma-chamber, J. Geomag. Geoelectr., 30, 55-64, 1978. (16 refs)

Abstract: -

Mott-Smith, H. M. and I. Langmuir, The theory of collectors in gaseous discharges, Phys. Rev., 28, 727-763, 1926 (also in The Collected Works of Irving Langmuir, Ed. by C. Guy Suits, Vol. 4: Electrical Discharge, pp. 99-132, Pergamon Press, Oxford, 1961). (14 refs)

Abstract: When a cylindrical or spherical electrode (collector) immersed in an ionized gas is brought to a suitable potential, it becomes surrounded by a symmetrical space-charge region or sheath of positive or of negative ions (or electrons). Assuming that the gas pressure is so low that the proportion of ions which collide with gas molecules in the sheath is negligibly small, the current taken by the collector can be calculated in terms of the radii of the collector or sheath, the distribution of velocities among the ions arriving at the sheath boundary and the total drop of potential in the sheath. The current is independent of the actual distribution of potential in the sheath provided this distribution satisfies certain conditions. Orbital Motion equations for spherical and cylindrical collectors.— General formulas for the current are derived and the calculations are then carried out for collectors in a group of ions having velocities which are (A) equal and parallel; (B) equal in magnitude but of random direction; (C) Maxwellian; (D) Maxwellian with a drift velocity superimposed. In all cases the collector current becomes practically independent of the sheath radius when this radius is large compared with that of the collector. Thus the volt-ampere characteristics of a collector of sufficiently small radius can be used to distinguish between thc different types of velocity distribution. General equations are also given by means of which the velocity distribution can be calculated directly from the volt-ampere characteristics of a sphere or cylinder. Special properties of the Maxwellian distribution.—For a collector of any shape having a convex surface, the logarithm of the current taken from a Maxwellian distribution is a linear function of the voltage difference between the collector and the gas when the collector potential is such as to retard arriving ions, but not when this potential is accelerating. This is a consequence of the following general theorem: Supposing for simplicity of statement that the surface of an electrode of any shape immersed in a Maxwellian distribution is perfectly reflecting, then the ions in the surrounding sheath will have a tistribution (called D/sub M/) of velocities and densities given by Maxwell's and Boltzmann's equations, even in the absence of collisions between the ions, provided that there are in the sheath no possible orbits in which an ion can circulate without reaching the boundary; but if such orbits exist, thc distribution will be D/sub M/ except for the absence of such ions as would describe the circulating orbits. As another

corollary of this theorem there is deduced an equation relating the solution of problems having inverse geometry. Finally it is indicated how the theorem can be applied to calculate the volt-ampere characteristic of A. F. Dittmer's "pierced collector" when placed in a Maxwellian distribution. The effect of reflection of ions at the collector surface in modifying currents calculated by the preceding equations is discussed.

Neubert, T., M. J. Mandell, S. Sasaki, B. E. Gilchrist, P. M. Banks, P. R. Williamson, W. J. Raitt, N. B. Meyers, K. I. Oyama, and I. Katz, The sheath structure around a negatively charged rocket payload, J. Geophys. Res., 95, 6155-6165, 1990. (29 refs)

Abstract: Investigates the sheath structure around a rocket payload charged up to 460 V negative relative to the ambient ionospheric plasma. The results come from the CHARGE 2 sounding rocket experiment in which the payload was split into two separate sections (mother and daughter) connected by a conducting, insulated tether. In one of the experimental modes, the voltage between the payloads was increased linearly from 0 to 460 V in 2.5 s. The tethered mother/daughter functioned as a double probe, the negative probe (mother) reaching large negative potentials, while the positive probe (daughter) stayed close to the ambient plasma potential. The measurements contain signatures, which are interpreted resulting from the outward expansion of the ion sheath with increasing negative mother potential. At high potentials, the observed ion current flowing to the mother increased strongly with bias potential. The enhancement of the current may be generated by secondary electrons emitted by the ions bombarding the payload skin.

Nilsson, K., Calculation of magnetospheric plasma density and temperature from probe current sweeps on the Swedish satellite Viking, Royal Institute of Technology, Alfvén Laboratory, Division of Plasma Physics Internal Report ALP-1994-104, 27 pp. + app., 1977. (16 refs)

Abstract: A simple model has been used for adjusting theoretical sweeps to measured current-voltage-sweeps obtained by electrostatic probes on the Viking satellite. From these sweeps, plasma electron density and temperature, n_e and T_e, photoelectron temperature, T_ph, and photoemissivity, i_ph0 are calculated. An asymmetry in the electronic circuits of the experiment causes the determination of the probe floating potential, V_fl, to be uncertain, which also affects the determination of the plasma parameters. An iteration procedure has been used in an attempt to handle this problem. A larger value of T_ph than used in earlier works, was found to give best results. An attempt to include ion currents in the model has not been successful. This is partly due to a lack of information in the literature, treating this problem for the conditions prevailing at VIKING altitudes, and partly due to the asymmetry problem, making it difficult to find a unique combination of as many as six plasma parameters (n_e, T_e, T_ph, i_ph0, n_i, T_i). For most of the measured sweeps, the ion current is not important except in dense plasma regions, where the ion current component is responsible for the linear slope in the current-voltage characteristic observed for negative probe potential.

Norman, K. and R. M. Freeman, Energy distribution of photoelectrons emitted from a surface on the OGO-5 satellite and measurements of satellite potential, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 231-244, D. Reidel Publ. Co., Dordrecht, 1973. (10 refs)

Abstract: A Langmuir probe experiment, provided by the Mullard Space Science Laboratory and flown on the OGO-5 spacecraft, is used to measure the energy distribution of photo-electrons emitted from the gold surface of the spherical probe in sunlight. Comparison with laboratory measurements and theoretical models yields a most probable energy of 1.1 eV for the distribution. The effects of photo and secondary emission on the potential of the spacecraft are discussed and some examples are shown.

Olsen, R. C. and E. C. Whipple, An unusual charging event on ISEE 1, J. Geophys. Res., 93, 5568-5578, 1988. (36 refs)

Abstract: Electrostatic cleanliness requirements on ISEE 1 were expected to prevent negative charging in sunlight. This has largely been true, but on three occasions, ISEE 1 has been observed to charge to significant negative potentials in sunlight. Data from the two electric field experiments and from the plasma composition experiment on ISEE 1 show that the spacecraft charged to close to -70 V in sunlight at about 0700 UT on March 17, 1978. Data from the electron spectrometer experiment show that there was a potential barrier of some -10 to -20 V about the spacecraft during this event. The potential barrier was effective in turning back emitted photoelectrons to the spacecraft. Potential barriers can be formed by differential charging on the spacecraft or by the presence of excess space charge in the plasma. The shape of the barrier suggests that it is due to the former, even though electrostatic cleanliness specifications imposed on ISEE were intended to eliminate differential charging. Modeling of this event showed that the barrier could not be produced by the presence of space charge but that it was most likely produced by differential charging of the solar arrays.

Oyama, K. and K. Hirao, Discussion for the evaluation of the charged particle temperatures on satellite, Bull. of the Institute of Space and Aeronautical Sci., Univ. of Tokyo A, 9, no.2(A),305-316, 1973. (12 refs)

Abstract: Contamination effect of a retarding potential trap is discussed on the evaluation of electron and ion temperatures. Discussion shows that the contaminated electrodes of the retarding potential trap give the ion and electron temperatures higher than the true value as in the case of a Langmuir probe. It is strongly noted that the discrepancy which has been reported in satellite observations may be solved from the consideration above.

Oyama, K. and K. Hirao, Underestimation of electron density by a contaminated Langmuir probe, Bull. of the Institute of Space and Aeronautical Sci., Univ. of Tokyo B, 10, no.3B, 561-568, 1974. (6 refs)

Abstract: Contamination layer model predicts that a contaminated electrode gives a lower electron density than the true value and that the effect is increased with the increase of the ambient plasma density. The above prediction was

confirmed in laboratory plasma as well as in space plasma. Consequently the serious influence of the probe contamination upon the density evaluation is emphasised.

Palop, J. I. F., J. Ballesteros, V. Colomer, and M. A. Hernandez, Theoretical ion current to cylindrical Langmuir probes for finite ion temperature values, J. Phys. D (Appl. Phys.), 29, 2832-2840, 1996. (19 refs)
Abstract: A new theoretical model for the potential distribution in the surroundings of a cylindrical conductor placed inside a neutral plasma permitted us to analyse the effect of the positive ion thermal motion on the ion current collected by a cylindrical Langmuir probe. The new theoretical model includes the ABR (Allen-Boyd-Reynolds) theory as a limiting case, which is that of a negligible ion temperature compared to the electron temperature.

Parker, L. W., Theory of the circular planar (guardring) Langmuir probe, 85 pp., Mount Auburn Research Associates, Inc., Cambridge, MA, USA, 12 April 1968. (0? refs)
Abstract: A computational study is presented of the properties of a guardring-type circular planar Langmuir probe, commonly flush-mounted in the skin of a satellite. This geometry results in a three-dimensional potential distribution which cannot be treated analytically, even in axially symmetric problems. Given arbitrary particle velocity distributions at infinity, the current-voltage characteristics of the external aperture grid and of an internal repelling collector may be determined by particle trajectory calculations. The electric field and charge density distributions in the vicinity of the probe are defined at the nodes of a grid. The charge density and collected currents in the Poisson equation are evaluated by summing trajectory contributions. The Poisson field is computed self-consistently by an iterative technique.

Parker, L. W., Computer solutions in electrostatic probe theory, I, Spherical symmetry with collisions, Report AFAL-TR-72-222, part 1, Air Force Avionics Lab., Wright-Patterson Air Force Base, Ohio, 1973. (? refs)
Abstract: ...

Parker, L. W., Computer method for satellite plasma sheath in steady-state spherical symmetry, Report AFCRL-TR-75-0410, Air Force Geophys. Lab., Hanscom Air Force Base, Mass., 1975. (? refs)
Abstract: ...

Parker, L. W., Theory of electron emission effects in symmetric probe and spacecraft sheaths, Report AFGL-TR-76-0294, Air Force Geophys. Lab., Hanscom Air Force Base, Mass, 1976. (17 refs)
Abstract: A computer model developed earlier (Parker, 1975), for the rigorous theory of the steady-state sheath structure about and the particle fluxes to a conducting spherical body in a collisionless plasma, has been extended to include the contributions of emitted electrons in addition to those of the ambient plasma. The method is also extended to apply to long cylinders as well as to spheres. Potential barriers and general non-monotonic potential distributions are taken into account. Solutions are possible for wide ranges of the values of the relevant parameters, and are applicable to the sheaths and fluxes of emitting spacecraft, and to the current-voltage characteristics of emitting Langmuir probes. The formulation for the orbit analysis is based on the author's (1973, 1975) Turning-Point function method, as opposed to the method of the Effective-Potential function exemplified by the work of Laframboise and Bernstein and Rabinowitz. The two formulations have been shown by the author to be analytically equivalent and to produce identical numerical results. However, in the author's opinion the Turning-Point formulation is simpler and more efficient than the Effective-Potential formulation. Solutions are presented for a number of sample problems, using a code PARKSS based on the analysis of this report. The discussion of these solutions includes comparisons with previous theories and experimental data. In these problems the effects are considered of variations of (a) surface potential, (b) monoenergetic versus Maxwellian emission velocity distributions, (c) emission flux relative to ambient fluxes ("weak" versus "strong" emission), and (d) emission temperature relative to ambient temperatures ("cold" versus "warm" emission), on potential barriers and net surface fluxes.

Parker, L. W., Differential charging and sheath asymmetry of nonconducting spacecraft due to plasma flows, J. Geophys. Res., 83, 4873-4876, 1978. (25 refs)
Abstract: The differential charging of a nonconducting spacecraft is modeled numerically by following charged-particle trajectories in a self-consistent space-charge-less sheath. In the presence of a plasma flow but independent of any photoelectric or secondary emission a potential difference between the front and wake surfaces of the spacecraft is generated, resulting in an asymmetric sheath and in the creation of a potential barrier for electrons. The potential difference can amount to volts in the ionosphere and kilovolts in the solar wind. As in the more familiar case of photoelectric charging, the asymmetric sheath and potential barrier produced by the plasma flow can lead to erroneous interpretations of experiments measuring space electric fields and low-energy particle spectra.

Parker, L. W., Plasmasheath-photosheath theory for large high-voltage space structures, in Space Systems and Their Interaction With Earth's Space Environment, Prog. Astronaut. Aeronaut., vol. 71, Ed. by H. B. Garrett and C. P. Pike, pp. 477-522, Amer. Institute of Aeronautics and Astronautics, New York, 1980. (39 refs)
Abstract: This work presents a new method for rigorously computing sheath structures of large spherical bodies with high-voltage surfaces and with photoelectric/secondary emission. This method, using the author's Turning-Point Formulation, is transparently simple and results in a compact computer program. Self-consistency of the Poisson and Vlasov solutions is achieved through iteration. The power and flexibility of the method is illustrated through four sample sheath solutions, including a) the sheath of a large body (radius 100 Debye lengths) with a voltage 400,000 kT/e, the most extreme combination of size and voltage solved rigorously to date, and b) the "presheath" of an extremely large body, a nontrivial and heretofore unsolved problem in a warm plasma. In addition, two approximate models are considered, a) a linearized space charge model (leading to the Debye potential for spheres), and b) the

Langmuir-Blodgett spherical diode. Both approximate models tend to underestimate current collection.

Parker, L. W. and B. L. Murphy, Potential buildup on an electron-emitting ionospheric satellite, J. Geophys. Res., 72, 1631-1636, 1967. (10 refs)

Abstract: An experiment in which a continuous beam of high-energy electrons is emitted from a satellite (e.g. to produce an artificial aurora or to map the geomagnetic field) will be jeopardized if the satellite potential becomes so high that the beam electrons cannot escape. The potential acquired by the satellite is determined by the condition that it collect from the ambient ionosphere a current equal to the emitted current. The geomagnetic field severely limits the current that may be emitted. For example, a spherical satellite of radius 1.5 meters in the F layer cannot emit more than about 40 ma of kilovolt electrons.

Parker, L. W. and J. G. Laframboise, Multi-electrode plasma probe for orbit-limited-current measurements. II. Numerical verification, Phys. Fluids, 21, 588-591, 1978. (11 refs)

Abstract: For pt.I see ibid., vol.16, p.629 (1973). A multi-electrode electrostatic plasma probe design intended to simulate the operation of an orbit-limited spherical probe is discussed. A flat-ended variation of an earlier design by the same authors is proposed and a numerical calculation of its operating characteristics is made. It is found that the desired operating conditions are possible.

Parker, L. W., E. G. Holeman, and J. E. McCoy, Sheath shapes: A 3-D generalization of the Child-Langmuir sheath model for large high-voltage space structures in dense plasmas, in Proc. Air Force Geophys. Lab. Workshop on Natural Charging of Large Space Structures in Near-Earth Polar Orbit, Ed. by R. C. Sagalyn, D. E. Donatelli, and I. Michael, Rep. AFGL-TR-83-0046/Envifon. Res. Pap. 825, pp. 337-389, Air Force Geophys. Lab., Hanscom Air Force Base, Mass., 1983. (? refs)

Abstract: ...

Parrot, M. J. M., L. R. O. Storey, L. W. Parker, and J. G. Laframboise, Theory of cylindrical and spherical Langmuir probes in the limit of vanishing Debye number, Phys. Fluids, 25, 2388-2400, 1982. (55 refs)

Abstract: A theory has been developed for cylindrical and spherical probes and other collectors in collisionless plasmas, in the limit where the ratio of Debye length to probe radius (the Debye number lambda /sub D/) vanishes. Results are presented for the case of equal electron and ion temperatures. On the scale of the probe radius, the distributions of potential and density in the presheath appear to have infinite slope at the probe surface. The dimensionless current-voltage characteristic is the same for the cylinder as for the sphere, within the limits of error of the numerical results, although no physical reason for this is evident. As the magnitude of probe potential (relative to space) increases, the current does not saturate abruptly but only asymptotically; its limiting value is about 45% larger than at space potential. Probe currents for small nonzero lambda /sub D/ approach those for zero lambda /sub D/ only very slowly, showing power-law behavior as function of lambda /sub D/ in the limit as lambda /sub D/ to 0, with power-law exponents less than unity, resulting in infinite limiting derivatives with respect to lambda /sub D/.

Pedersen, A., Solar wind and magnetosphere plasma diagnostics by spacecraft electrostatic potential measurements, Ann. Geophys., 13, 118-129, 1995. (18 refs)

Abstract: Several satellites (GEOS-1, GEOS-2, ISEE-1, Viking Sweden and CRRES) carried electric field experiments on which probes were driven by a current from the satellite to be close to the plasma potential. The potential difference between an electric field probe and its spacecraft (with conductive surfaces) can be used to determine the ambient electron density and/or electron flux with limited accuracy but with high time resolution, of the order of 10-100 ms. It is necessary for the development of this diagnostic method to understand the photoemission characteristics of probes and satellites. According to the electric field experiments on the above-mentioned satellites, all materials develop very similar photoemission properties when they are beyond the influence of atmospheric oxygen. The photoelectron yield steadily increases over the first few months in space and reaches values well above those measured on clean surfaces in the laboratory. The method can be used for solar radiation levels corresponding to distances from 0.4 to 5 AU from the Sun.

Prokopenko, S. M. L. and J. G. Laframboise, High-voltage differential charging of geostationary spacecraft, J. Geophys. Res., 85, 4125-4131, 1980. (31 refs)

Abstract: A numerical prediction is presented for upper bounds on negative floating potentials of shaded spacecraft surfaces, using a local-current-balance formulation.

Prokopenko, S. M. L., J. G. Laframboise, and J. M. Goodings, Evaluation of an orifice probe for plasma diagnostics, J. Phys. D (Appl. Phys.), 5, 2152-2160, 1972. (17 refs)

Abstract: An orifice probe is described which permits determination of bulk plasma parameters, including temperatures, number densities and space potential, and which also permits simultaneous use of a mass spectrometer sampling through the same orifice.

Prokopenko, S. M. L., J. G. Laframboise, and J. M. Goodings, Orifice probe for plasma diagnostics. II. Multi-parameter analysis, J. Phys. D (Appl. Phys.), 7, 355-362, 1974. (14 refs)

Abstract: For part I see abstr. A1520 of 1973. A multi-grid orifice probe and analysis technique are presented which permit the measurement of the following plasma parameters: the electron number density n/sub e/, the electron temperature T/sub e/, the space potential V/sub s/, and an estimate of the metastable number density n/sub m/. Simultaneously, the analysis allows the quantitative determination of instrumental parameters such as the reflection

coefficient for slow (<30 eV) electrons from the grids R/sub e/(G) and roughened (gold-black) collector R/sub e/(C), the secondary electron emission coefficient for fast (>30 eV) electrons gamma /sub e/(G) from the grids, and the effective transparencies of the grids for electrons tau /sub e/(G), and for metastables tau /sub m/(G). Experimental ion, electron and total current characteristics are presented for a low-pressure (1.8*10/sup -2/ Torr) RF discharge in argon. The major part of the analysis method involves the use of ten 'plateau values' obtained from the characteristics to determine values for most of these parameters. T/sub e/ and V/sub s/ are determined separately.

Prokopenko, S. M. L., J. G. Laframboise, and J. M. Goodings, Orifice probe for plasma diagnostics. III. Experimental considerations, J. Phys. D (Appl. Phys.), 7, 563-568, 1974. (10 refs)

Abstract: For part II see ibid., vol.7, 355 (1974). Experimental methods are presented for the detection and elimination of instrumental effects arising from field penetration and space charge for an orifice probe in low-pressure (<2*10/sup -2/ Torr) RF discharge plasmas in argon. A channel orifice, compared with a sharp-edged orifice, provides number density values which are too low. The creation of a dual-peak ion distribution by the insertion of a grid into the plasma is illustrated.

Raitt, W. J., N. B. Myers, J. A. Roberts, and D. C. Thompson, SPEAR-1, an experiment to measure current collection in the ionosphere by high voltage biased conductors, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 214-231, 1990. (8 refs)

Abstract: An experiment will be described in which a high electrical potential difference, up to 45 kV, was applied between deployed conducting spheres and a sounding rocket in the ionosphere. Measurements were made of the applied voltage and the resulting currents for each of 24 applications of different high potentials. In addition, diagnostic measurements of optical emissions in the vicinity of the spheres, energetic particle flow to the sounding rocket, DC electric field and wave data were made. The ambient plasma and neutral environments were measured by a Langmuir probe and a cold cathode neutral ionization gauge, respectively. In this presentation, the payload will be described and examples of the measured current and voltage characteristics will be presented. The characteristics of the measured currents will be discussed in terms of the diagnostic measurements and the in-situ measurements of the vehicle environment. In general, it was found that the currents observed were at a level typical of magnetically limited currents from the ionospheric plasma for potentials less than 12 kV, and slightly higher for larger potentials. However, due to the failure to expose the plasma contactor, the vehicle sheath modified the sphere sheaths and made comparisons with the analytic models of Langmuir-Blodgett and Parker-Murphy less meaningful. Examples of localized enhancements of ambient gas density resulting from the operation of the attitude control system thrusters (cold nitrogen) were obtained. Current measurements and optical data indicated localized discharges due to enhanced gas density reduced the vehicle-ionosphere impedance.

Riemann, K.-U., The Bohm criterion and sheath formation, J. Phys. D (Appl. Phys.), 24, 493-518, 1991. (121 refs)

Abstract: In the limit of a small Debye length (lambda_D -> 0) the analysis of the plasma boundary layer leads to a two-scale problem of a collision free sheath and of a quasi-neutral presheath. Bohm's criterion expresses a necessary condition for the formation of a stationary sheath in front of a negative absorbing wall. The basic features of the plasma-sheath transition and their relation to the Bohm criterion are discussed and illustrated from a simple cold-ion fluid model. A rigorous kinetic analysis of the vicinity of the sheath edge allows one to generalize Bohm's criterion accounting not only for arbitrary ion and electron distributions, but also for general boundary conditions at the wall. It is shown that the generalized sheath condition is (apart from special exceptions) marginally fulfilled and related to a sheath edge field singularity. Due to this singularity smooth matching of the presheath and sheath solutions requires an additional transition layer. Previous investigations concerning particular problems of the plasma-sheath transition are reviewed in the light of the general relations.

Rodgers, D. J., Computer simulations of plasma wakes in LEO and laboratory environments, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 63-75, Editrice Compositori, Bologna, 1992. (0 refs)

Abstract: Simulations of the equilibrium plasma flow onto and around a satellite in Low Earth Orbit require the solution of the Poisson and Vlasov equations to determine the potential and density distributions respectively. Four codes that solve the Vlasov step in different ways are described. Some simple test cases are examined to compare their output. Agreement is generally good but artifacts can be created if the simulations do not comply with assumptions implicit in the code. Different codes become most useful under different circumstances.

Rubinstein, J. and J. G. Laframboise, Plasma sheath around a floating cylindrical antenna at high power, Can. J. Phys., 48, 1882-1893, 1970. (9 refs)

Abstract: At high frequencies the time-averaged effect of the radiation field near an antenna is to add the term a/sup 2/E/sup 2/(r) to the Coulomb potential. Its inclusion in a self-consistent Poisson-Vlasov analysis shows that a high power antenna may acquire a strong positive floating potential. Electron, ion, and potential profiles are obtained from numerical calculations and the effect on antenna impedance is discussed. Time taken to acquire positive potentials is estimated.

Rubinstein, J. and J. G. Laframboise, Ion temperature diagnostic using a high power alternating current probe, Phys. Fluids, 15, 1259-1267, 1972. (8 refs)

Abstract: A method is presented for using a probe to measure ion temperature in a plasma. The method involves

applying a strong rf alternating potential to the probe, of sufficient amplitude to drive electrons away from it via a nonlinear effect which acts preferentially on them. The ion current, essentially unaffected, can then be seen separately and the usual retarding potential techniques can be used to measure ion temperature. The method requires electron motion to be sufficiently collisionless. Self-consistent sheath profiles are calculated in order to show that there exists a useful range of driving frequencies and amplitudes over which the dc potential is ion repelling and monotonic and hence the ion current is exponential in probe dc potential. Since the ion and electron fluxes to the probe can simultaneously be made very small, the technique should permit measurements in some environments which would otherwise overheat and destroy probes. An ion heating mechanism based on the electron repulsion effect is also proposed.

Rubinstein, J. and J. G. Laframboise, Upper-bound current to a cylindrical probe in a collisionless magnetoplasma, Phys. Fluids, 21, 1655-1656, 1978. (4 refs)

Abstract: Upper bounds are presented for ion or electron collection by a cylindrical electrostatic probe inclined at an angle to a uniform magnetic field. Over a wide range of conditions, these bounds are better than those derived earlier by Laframboise and Rubinstein (1976).

Rubinstein, J. and J. G. Laframboise, Theory of a spherical probe in a collisionless magnetoplasma, Phys. Fluids, 25, 1174-1182, 1982. (22 refs)

Abstract: A theory is presented for a spherical electrostatic probe in a collisionless, Maxwellian plasma containing a uniform magnetic field. The theory yields two upper bounds and an adiabatic limit for collection of the attracted particle species (either electrons or ions). For the repelled species, it yields a lower and an upper bound. The theory is similar in concept to existing theories for cylindrical probes by Laframboise and Rubinstein (1976). It is applicable when the ratio of probe radius to Debye length is small enough, and/or the probe potential is large enough, that no potential barriers exist near the probe. Otherwise a theory of Sanmartin (1970) applies. The attracted-particle current in the adiabatic limit, i.e. when mean gyroradius << Debye length, shows negative-resistance behavior. One of the upper bounds is based on the use of particle canonical angular momentum conservation to define allowed and forbidden regions for particle orbits, and generalizes an existing theory by Parker and Murphy to include particle thermal motion.

Rubinstein, J. and J. G. Laframboise, Theory of axially symmetric probes in a collisionless magnetoplasma: aligned spheroids, finite cylinders, and disks, Phys. Fluids, 26, 3624-3627, 1983. (6 refs)

Abstract: A theory is presented for current collection by electrostatic probes in a collisionless, Maxwellian plasma containing a uniform magnetic field B, where the probes are spheroids or finite cylinders whose axis of symmetry is aligned with B, or disks perpendicular to B. The theory yields upperbound and adiabatic-limit currents for the attracted particle species. For the repelled species, it yields upper and lower bounds. This work is an extension of existing theory for spherical probes by Rubinstein and Laframboise (1982).

Samir, U., R. H. Comfort, K. H. Wright, Jr., and N. H. Stone, Intercomparison among plasma wake models for plasmaspheric and ionospheric conditions, Planet. Space Sci., 35, 1477-1487, 1987. (32 refs)

Abstract: Intercomparisons are made among the angular distributions of ions in the wake of a body moving through a space plasma as computed from three different expressions (models). Both subsonic and supersonic relative flows are considered in order to examine the wake current depletion ratios under conditions realistic for the topside ionosphere and plasmasphere. Results of these comparisons demonstrate the importance of ion detectors in making theory-experiment comparisons. Gradients in the angular variations of the fluxes are found to be steeper near the wake-ambient interface than closer to the maximum rarefaction region of all models, although quantitatively there is considerable variation among the models.

Sanmartin, J. R., Theory of a probe in a strong magnetic field, Phys. Fluids, 13, 103-116, 1970. (14 refs)

Abstract: An asymptotic analysis of the Langmuir-probe problem is performed for electron cyclotron radius and Debye length much smaller than probe radius, and this not larger than either ion cyclotron radius or mean free path. The electric potential, which is not confined to a sheath, controls the diffusion far from the probe; inside the magnetic tube bounded by the probe cross section the potential overshoots to a large value before decaying to its value in the body of the plasma. The electron current is independent of the shape of the body along the field and increases with ion temperature; owing to the overshoot in the potential, the current at negative voltages does not vary exponentially, its magnitude is strongly reduced by the field, and the usual sharp knee at space potential disappears. In the regions of the C-V diagram studied the ion current is negligible or unaffected by the field.

Sanmartin, J. R., End effect in Langmuir probe response under ionospheric satellite conditions, Phys. Fluids, 15, 1134-1143, 1972. (6 refs)

Abstract: A theory is presented for an end effect in the current response of a highly negative cylindrical Langmuir probe in a collisionless plasma flow. Under conditions where the ratio of probe radius to Debye length is small and the ion-acoustic Mach number is large, the current exhibits a strong peak when the probe axis is brought into alignment with the flow direction. Closed formulas are given for the maximum and angular half-width of the peak, and universal graphical results are presented for the entire peak structure. The theory shows very good agreement with experimental data. The use of the end effect for diagnostic purposes, in particular, for the determination of the ion temperature, is discussed.

Sanmartin, J. R., Ion-temperature-sensitive effect in transient Langmuir probe response, Phys. Fluids, 15, 391-401, 1972. (4 refs)

Abstract: A theory is presented for a method, recently proposed by Hester and Sonin, of determining the ion

temperature in a plasma by measuring the transient current to a cylindrical Langmuir probe after applying a potential V/sub p/(-eVp>> kappa T/sub e/) under conditions where collection is collision free and the ratio of probe radius to Debye length is small. The ion component of the current does not approach its final steady-state value monotonically, but exhibits a strong, ion-temperature-dependent overshoot in the first few ion-plasma periods following the biasing of the probe. Analytical formulas are derived for the case of a Maxwellian plasma, and convenient graphical results are presented. The possible masking of the overshoot by a transient displacement current is discussed; it is shown how to avoid such displacement effects. For the overshoot to be sensitive to the ion temperature T/sub i/, the probe must be near plasma (zero) potential before applying V/sub p/(eV/sub p//sup -/<0.1 kappa T/sub e/, V/sub p//sup -/ being that initial potential); this is not a drawback of the method, but, on the contrary, it can be used to accurately determine plasma potential along with T/sub i/.

Sanmartin, J. R. L., The theory of electrostatic probes in strong magnetic fields, Univ.of Colorado, CO, USA, 186 pp., 1968. (? refs)

Abstract: A theory is developed of an electrostatic probe in a fully ionized plasma in the presence of a strong magnetic field. The ratio of electron Larmor radius to probe transverse dimension is assumed to be small. Poisson's equation, together with kinetic equations for ions and electrons are considered. An asymptotic perturbation method of multiple scales is used by considering the characteristic lengths appearing in the problem. The leading behaviour of the solution is found. The results obtained appear to apply to weaker fields also, agreeing with the solutions known in the limit of no magnetic field. The range of potentials for which results are presented is limited.

Schmidt, R. and A. Pedersen, Long-term behaviour of photo-electron emission from the electric field double probe sensors on GEOS-2, Planet. Space Sci., 35, 61-70, 1987. (17 refs)

Abstract: A double probe instrument with a probe separation of 42 m was used aboard the geostationary spacecraft GEOS-2 to measure the ambient electric field. The probes contained built-in pre-amplifiers and were current biased in order to clamp their surface potential near to the local plasma potential while the spacecraft surfaces floated at a more positive potential. Depending on the plasma density, the potential difference ranges between 1 V (dense plasma, N/sub e/>10 cm/sup -3/) and more than 10 V (tenuous plasma, N/sub e/<3 cm/sup -3/). In the case of the more tenuous plasma, photoelectrons emitted from the probes tend to be picked up by the more positive potential of the wire booms. The net current of this electron flow depends on the spin orientation of the wire booms with respect to the Sun. It occurs in the data as a spurious signal. Using data from quiet magnetospheric conditions (K/sub p/ < 2) the long-term variation of the spurious signal was investigated. It turned out to be constant during the initial 3 y of the mission. Afterwards, as from 1981, it became smaller and did so for the following 2 y. In 1983, the end of the 5-y period of available data, it had vanished. The cause of the drop is suggested to be a consequence of the partial removal of the conductive surface layer of the wire booms by sputter effects. Expressions were derived that yield more accurate correction of the data. A valuable by-product of this work is the statistical relation of spacecraft potential and total electron flux.

Schott, L., Electrical probes, in Plasma Diagnostics, Ed. by W. Lochte-Holtgreven, pp. 668-731, North-Holland Publ. Co., Amsterdam, 1968. (98 refs)

Abstract: -

Schröder, H., Spherically symmetric model of the photoelectron sheath for moderately large plasma Debye lengths, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 51-58, D. Reidel Publ. Co., Dordrecht, 1973. (12 refs)

Abstract: The potential distribution is investigated in the neighbourhood of a conducting, slectron emitting sphere surrounded by a collisionless plasma. The energy dependence of the emitted electrons is adopted in accordance with data on photoemission, but their spatial distribution is treated spherically symmetric. A Maxwellian velocity distribution with zero average velocity is assumed for the undisturbed plasma electrons. The ions are taken as uniform background. A self consistent solution of the collisionless Boltzmann equation for the electrons and Poisson's equation has been obtained by an iterative scheme. Results are presented for plasma parameters corresponding to solar wind conditions at 1 AU. The role of the plasma- and photoelectron Debye lengths is discussed.

Seifert, W., D. Johanning, H.-R. Lehmann, and N. Bankov, Methods for the numerical calculation of the plasma potential from measured Langmuir probe characteristics, Beitraege aus der Plasma-Physik, 26, 237-253, 1986. (33 refs)

Abstract: The authors review different methods to determine the plasma potential especially under the circumstances of computer application. The test of the well developed methods shows, that for the evaluation of the applied methods a number of different factors have to be taken into account. The used test data are model values or real measurements obtained by the satellite missions IK-10, IK-18 and BULGARIA-1300. The computer procedures have been realized as FORTRAN-subroutines for the BESM 6 and ES 1040.

Self, S. A. and C. H. Shih, Theory and measurements for ion collection by a spherical probe in a collisional plasma, Phys. Fluids, 11, 1532-1545, 1968. (29 refs)

Abstract: Following a discussion of existing probe theories for a collisional plasma, a new theory is given for ion collection by a spherical probe in a collisional plasma which goes over to the collisionless theory of Allen et al. at low pressures and the continuum theory of Su and Lam at high pressures. Numerical results are given in a form suitable for interpretation of experimental data. Experimental probe studies in the positive column (1 cm diameter) of a He discharge in the pressure range 0.05-0.4 Torr are described and the use of the electron and ion saturation methods for determining plasma concentration compared with one another and with a microwave cavity perturbation method. The

limitations of the theories and experimental methods are discussed.

Sharma, V. and J. S. Verma, Theoretical analysis of spherical probe characteristics for collisional plasma, Indian J. Pure Appl. Phys., 10, 619-623, 1972. (12 refs)

Abstract: The dynamic conditions in a collisional plasma fitted with an electrical probe have been mathematically formulated on the basis of the kinetic theory and solved using the two-sided distribution function of Wasserstrom et al. (1965) and assuming a curved trajectory for the charged particles in their transit from the last collision to the probe surface. Solutions have been obtained for a small potential on the probe and for an arbitrary potential on the probe in the quasi-neutral region. It is not possible to obtain direct solutions to the mathematical formulation in the sheath region for an arbitrary probe potential but exact solutions can be obtained using numerical analysis on a computer. For a small as well as an arbitrary probe potential, the analysis shows an increase in the value of the current and a decrease in the value of the potential compared to those reported by Wasserstrom et al.

Sharma, V. and J. S. Verma, Theory of spherical Langmuir probe in partially ionized plasma with uniform magnetic field, Indian J. Pure Appl. Phys., 10, 721-727, 1972. (14 refs)

Abstract: A theory to explain the characteristics of a spherical probe in a partially ionized plasma subject to a uniform magnetic field has been developed. The problem has been approached from the postulates of the kinetic theory and the theory has been developed for a small as well as for an arbitrary potential on the probe under symmetric condition. Both curved and straight line trajectories for particles between collisions have been taken into consideration. The solution obtained in the case of the sheath region indicates that the density and potential distributions are not modified by the magnetic field. The plasma characteristics obtained assuming a curved trajectory for the particles between collisions agree closer with the experimental data as compared to the case of a straight line trajectory.

Sharma, V. and J. S. Verma, Cylindrical probe in a collision dominated plasma with a uniform magnetic field, J. Inst. of Telecomm. Engineers, 19, 291-295, 1973. (13 refs)

Abstract: Langmuir probe theory for a slightly ionized, collisional plasma is extended to a problem formulation to allow for the presence of uniform magnetic field. Plasma potential and density distributions have been studied to the cylindrical probe whose axis is aligned with the direction of the magnetic field. The saturation current for the negatively biased probe has been calculated. The derivation of the mathematical formulation is based upon the technique of taking the conservation of mass to the number flux equations for diffusion with electric and superposed magnetic field solved with the help of Poisson's equation for sheath and quasi-neutral regions.

Sharma, V. and J. S. Verma, Theory of Langmuir probe used in a re-entry vehicle surrounded by a flowing continuum plasma, J. Inst. of Electr. and Telecomm. Engineers, 19, 378-386, 1973. (11 refs)

Abstract: Reviews the theory of Langmuir probes in free-molecular stream with a classical approach and attention is accorded to the interpretation of data for the probes in saturation region, and to plasmas in which the characteristic ion and electron energies are comparable. Various methods for interpreting the probe measurements are enumerated. This information is employed in the subsequent theory for cylindrical and spherical probe in a flowing continuum plasma for an electron current collection, which concerns the performance of Langmuir probes flown abroad re-entry vehicles. The sheath region is considered as collisionless and rest of the plasma as collision dominated. The one-dimensional fluid equations are solved with the proper Poisson's equation. The theory deals with the probe at arbitrary potential and under the conditions that the drift velocity is greater than the electron thermal velocity and probe radius to mean free-path ratio, greater than unity. The theoretical results are in agreement with some known experimental results.

Sherman, C. and L. W. Parker, Potential due to a circular double disk, J. Appl. Phys., 42, 870-872, 1971. (3 refs)

Abstract: The authors present a solution for the potential due to an inner circular disk of radius a at constant potential V/sub a/ surrounded by an annular ring of inner radius a and outer radius b at another constant potential V/sub b/. This geometry is a useful one in measuring the properties of both laboratory and atmospheric plasmas.

Shiah, A., K. S. Hwang, S. T. Wu, and N. H. Stone, Three-dimensional simulation of current collection in space, Planet. Space Sci., 45, 475-482, 1997. (23 refs)

Abstract: A three-dimensional, time-dependent, two species (i.e. electron and ion) particle simulation model is developed for a better understanding of current collection by a charged body in space. The Low Earth Orbit (LEO) space environment is simulated by a super-particle simulation (SUPS) model to study the plasma transient response and its asymptotic behavior around a three-dimensional, highly biased satellite. The SUPS model is developed because the plasma particles in LEO are so abundant that it is impossible to simulate each single particle in the study. Each simulation particle in this study represents a particle cloud. The potential distribution and the electric field are dependent on the density distribution of the particles. The simulation could apply to the spacecraft mission such as the first Tether Satellite System Re-fly (TSS-1R) mission for current collection analysis. Both positive and negative charge particles are considered in this study. The potential distribution surrounding the satellite is determined self-consistently by Poisson's equation. This study is both time-dependent and space dependent simultaneously. Results indicate that when the satellite is charged to a highly positive potential, a plasma torus is formed in the plane perpendicular to the geomagnetic field. A dumbbell-shaped potential distribution is formed along the geomagnetic field lines and plasma oscillations are observed within the plasma sheath during the early stage of simulation. The effect of the local geomagnetic field on current collection is also presented.

Shih, C. H. and E. Levi, The effect of collisions on cold ion collection by means of Langmuir probes, AIAA

Bull., 7, no. 7, 334, 1970. (? refs)

Abstract: Langmuir probes are used extensively to sample plasma densities. However, the interpretation of probe data is well established only for collisionless plasmas. In particular, collision effects cannot be neglected when the ions are colder than the electrons, as in some fluid-dynamic situations and in all electrical discharges. A simple theory that accounts separately for the effects of collisions between charged particles and collisions with neutrals is presented. When this theory is applied to all published experimental data taken with cylindrical as well as spherical probes, the calculated and actual plasma densities agree within 10%.

Shih, C. H. and E. Levi, Determination of the collision parameters by means of Langmuir probes, AIAA J., 9, 2417-2421, 1971. (13 refs)

Abstract: The current collected by Langmuir probes in the ion saturation regime is affected by collisions among ions and with neutrals. This effect depends not only on the Debye length and the various mean free paths involved, but also on the geometry size and biasing potential of the probes. Two or more independent measurements thus afford means for determining the collision parameters and related qualities, such as the ion temperature, the chemical composition of the plasma, and the density of the background gas. Good agreement is obtained between theory and experimental data taken in quite distinct physical situations, by using a hypersonic shock tunnel and a laser discharge. The method presented here could be applied to the determination of ion-neutral collision cross sections in the transition region between the D and E layers of the ionosphere.

Shih, C. H. and E. Levi, Effect of collisions on cold ion collection by means of Langmuir probes, AIAA J., 9, 1673-1680, 1971. (22 refs)

Abstract: Langmuir probes are extensively used to sample plasma densities. However, the interpretation of probe data is well established only for the limiting cases of collisionless and collision-dominated plasmas. A transitional regime prevails in some fluid-dynamic situations and electrical discharges in which the ions are colder than the electrons. A simple theory which accounts separately for the effects of collisions between charged particles and collisions with neutrals is presented. When the theory is applied to recent experimental data taken with cylindrical as well as spherical probes the electron densities calculated from the probe measurements by using the present theory agree within 10% with those inferred from existing numerical solutions, an empirical formula, and microwave cavity data.

Shih, C. H. and E. Levi, Transient performance of negatively biased Langmuir probes, AIAA J., 10, 104-110, 1972. (22 refs)

Abstract: Analytical expressions are derived for the response of planar and cylindrical sheaths to large amplitude and sudden changes in the biasing potentials. It is found that the speed of response depends on the sign of the charge, this asymmetry being particularly pronounced with planar probes. The analysis also leads to equivalent circuits describing the dynamic response of probes in the collisional, as well as the collisionless regimes. These circuits show that the time constants characterizing the probe response can exceed 1/ omega /sub pi/ by one order of magnitude and that, at frequencies below 10/sup -2/ omega /sub pi/, a frequency range of great interest for the study of turbulence, the dynamic performance of the probes can be approximated by a succession of quasi-steady states. Expressions are derived which relate the mean and the variance of the probe current to the auto and cross correlations of fluctuations in density, temperature, and potential. The theoretical findings are examined in the light of experimental evidence reported in the published literature. Although qualitative agreement is good, the simplicity of the analytical model and the inadequate characterization of the experimental conditions, often limit the quantitative agreement to one order of magnitude.

Shimizu, T., T. Dote, J. S. Chang, and J. G. Laframboise, Comments on 'Probe theory for arbitrary shape in a large Debye length, stationary plasma' (and reply), Phys. Fluids, 22, 2250-2252, 1979. (5 refs)

Abstract: The authors comment on Chang and Laframboise's (see ibid., vol.19, p.25, 1976) method for locating the space potential in a large-Debye-length plasma. The present authors disagree with Chang and Laframboise's method of utilising the fact that i/sub e//sup / has a maximum at the space potential zero (where i/sub e/ is the normalised probe current). Chang and Laframboise reply to the present authors and state that i/sub e/ cannot be used for attractive potentials in the collisionless limit.

Shkarofsky, I. P., Accuracy of Langmuir probe measurements and skin potential on satellites, J. Geophys. Res., 76, 3746-3754, 1971. (10 refs)

Abstract: Using a model ionosphere, the author deduces realistic values for satellite floating potential for various geographic locations. The deviation in this floating potential is then calculated as one varies a Langmuir probe potential between +or-3 volts. Using Theis's reduction program acting on an I-V probe characteristic, the author deduces electron density and temperature values, compares these with input values, and obtains the errors involved.

Singh, N. and B. I. Vashi, Current collection in a flowing magnetoplasma, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 50-73, 1990. (8 refs)

Abstract: Effects of plasma drift on the current collection by a long conducting cylinder in a magnetized plasma is studied by means of a 2 1/2 dimensional PIC code. It is found that for the drift velocity (V_o) perpendicular to the magnetic field B_o, the electron current collected by a positive cylinder is considerably enhanced depending on the drift velocity. The distributions of plasma and the potential structure around the cylinder for several relative orientations between V_0 and B are presented along with the comparisons of current with and without the magnetic field. Simulations with the magnetic field in the simulation plane show that the potential structures around the cylinder are two-dimensional double layers with dimension (L_perp) perpendicular to B much smaller than the dimension

(L_parallel) parallel to B. In fact, L_perp is found to be approximately determined by the current limiting radius given by the Parker-Murphy model. However, it is found that the collected currents in the simulations are generally higher than those given by this model.

Singh, N. and C. R. Baugher, Sheath effects on current collection by particle detectors with narrow acceptance angles, Space Sci. Instrum., 5, 295-305, 1981. (13 refs)

Abstract: Restricting the aperture acceptance angle of an ion or electron trap on an attracting spacecraft substantially alters the voltage-current characteristics of the instrument in a low Mach number plasma. It is shown that if the criterion (e mod phi /sub s/ mod /kT) tan/sup 2/ alpha /sub p/>>1 applies, where phi /sub s/ is the spacecraft potential and alpha /sub p/ is the trap's angular acceptance, then the total current and RPA characteristics of the instrument are nearly independent of the Debye length of the ambient medium. For the case of an ambient isotopic Maxwellian distribution, the total current to the probe is shown to be proportional to (1+q mod phi /sub s/ mod /kT) even when the Debye length is zero. Expressions for the RPA characteristics for both a thin sheath and thick are derived; and it is shown that as the aperture is narrowed, the curves tend toward equivalence.

Singh, N. and K. S. Hwang, Electrostatic charging of spacecrafts in response to electron beam injection, Physica Scripta, 40, 295-301, 1989. (15 refs)

Abstract: Electrostatic charging level of a conducting surface in response to injections of electron beams into space plasma is investigated by means of one dimensional Vlasov code. Injection of Maxwellian beams into a vacuum shows that the surface can charge up to an electric potential phi /sub s/>W/sub b//e, where W/sub b/ is the average electron beam energy. Since Maxwellian beams have extended tails with electrons having energies >W/sub b/, it is difficult to quantify the charging level in terms of the energies of the injected electrons. In order to quantitatively understand the charging in excess of W/sub b/, simulations were carried out for water-bag types of beam with velocity distribution functions described by f(V)=A for V/sub min/ < V < V/sub max/ and f(V) = 0 otherwise, where A is a constant making the normalized beam density unity. It is found that V/sub max/ does not directly determine the charging level. The pressure distribution in the electron sheath determines the electric field distribution near the surface. The electric field in turn determines the electrostatic potential of the vehicle. The pressure distribution is determined by the beam parameters such as the average beam velocity and the velocity spread of the beam. The effect of the ambient plasma on the charging level is investigated showing that when the ambient plasma density n/sub a/ << n/sub b/, the beam density, the surface potential phi/sub s/ remains greatet than W/sub b//e, but it undergoes oscillations at the plasma frequency corresponding to the ambient plasma density n/sub a/. Under such plasma conditions the majority of the injected electrons return to the surface. Only when the ambient plasma density n/sub a/ is sufficiently large so that the time average surface potential phi/sub sa/ < W/sub b//e, the injected beam penetrates into the ambient plasma. However, the beam propagation speed depends on phi/sub sa/; when phi/sub sa/ /approx/ W/sub b//e, the propagation velocity is /approx/ V/sub t/, the electron thermal speed. Under such a case the propagating beam head is accompanied by a triple-charge-layer potential structure. The net potential drop across such a structure is about phi/sub sa/. When phi/sub sa/ << W/sub b//e, the beam propagates with the injection velocity V/sub b/.

Singh, N. and V. S. Chaganti, Electron collection by a highly positive satellite in the ionosphere: Test particle simulation, J. Geophys. Res., 99, 469-478, 1994. (15 refs)

Abstract: Collection of electrons by a satellite at a relatively high potential in low Earth orbit, including the effects of the satellite's orbital motion, remains a poorly understood phenomenon. Using a test particle simulation in which charged particle motion is tracked in prescribed electric fields, the authors calculate the current collection and the nature of the energy distribution function of the electrons collected by the satellite, including the effects of the satellite orbital motion. The calculated features of the distribution functions result from the chaotic motion of electrons and the associated coupling between the electron motions parallel and perpendicular to the magnetic field in strongly nonuniform electric fields near the satellite.

Singh, N., B. I. Vashi, and L. C. Leung, Three-dimensional numerical simulation of current collection by a probe in a magnetized plasma, Geophys. Res. Lett., 21, 833-836, 1994. (15 refs)

Abstract: A three-dimensional numerical model for current collection in a magnetized plasma is reported. The model is based on an electrostatic particle-in-cell code. The model yields a self-consistent sheath structure including the distributions of plasma and the electric potential around the body and the collection of electrons. The analytical theory of current collection by a body in a magnetized plasma yields an upper bound for the collected current determined by the conservation of energy and canonical angular momentum. The theory shows that the collected charged particles come from a cylindrical volume aligned with the magnetic shadow of the body. The simulations reveal when and why the collected current becomes equal to its upper-bound value. The sheath size in the radial direction perpendicular to the axial ambient magnetic field is determined by the current-limiting radius r/sub 0/. The simulation yields a time-average current in good agreement with its upper bound.

Singh, N., B. I. Vashi, and W. C. Leung, Sheath structure of a probe in a flowing magnetoplasma, IEEE Conference Record - Abstracts. p. 201 in 1995 IEEE International Conference on Plasma Science (Cat. No.95CH35796), iv+308 pp. IEEE, New York, NY, USA; 1995. (0 refs)

Abstract: Summary form only given, as follows. Distributions of plasma and fields near an electric probe in a flowing magnetoplasma are studied by means of 3-D numerical simulations. The flowing plasma represents the relative motion of the probe with respect to the plasma, as imposed by the orbital motion in space. The field distribution shows wings of a resonance-cone like structure, analogous to the whistler wings created by a moving electrode in a laboratory

plasma. In addition to the wings, the relative motion also creates a bow shock in front of a positive probe along with the usual wake with negative potentials in its back. The relative motion significantly enhances the current collection by the positive probe over the upper-bound current given by the conservation of the canonical angular momentum in a stationary plasma.

Singh, N., W. C. Leung, and B. I. Vashi, Potential structure near a probe in a flowing magnetoplasma and current collection, J. Geophys. Res., 102, 195-208, 1997. (17 refs)

Abstract: The distributions of plasma and potential near an electric probe in a relative motion with respect to a magnetized plasma are studied by means of three-dimensional (3D) numerical simulations. The relative motion is simulated by a plasma flowing past the probe across the ambient magnetic field. The plasma flow is imposed by a convection electric field E_/sub 0/. A probe with a positive potential bias is considered. The prominent features of the potential distribution include (1) wings of positive potential perturbations extending along the magnetic field and swept forward in the direction of the plasma flow and (2) a fan shaped structure in planes transverse to the magnetic field in the region where the convection electric field E_/sub 0/ is opposed by space charge electric fields. The wing-like structure can be interpreted in terms of electrostatic plasma waves belonging to the oblique resonance cone in a magnetized plasma. The relative flow causes the formation of a bow shock in front of the probe, where plasma density is enhanced due to the combined effect of the retardation of the flowing ions and the modification in the E_*B_ drift of the electrons in the sheath of a positive probe. The electron collection by the probe is significantly enhanced above the theoretical upper bound current obtained from the conservation of energy and the canonical angular momentum for the case without the relative motion. The current in the plasma, contributing to the collection of electrons by the probe, flows in a magnetic field-aligned channel in the vicinity of the probe where electric fields parallel to the magnetic field are relatively strong. Electron flux is fed into the channel all along its length by the E_*B_ drift in the self-consistent electric field, considerably enhancing the current collected by the probe. The field-aligned current channel is localized near the probe where parallel electric fields dominate; it does not extend to infinity along the probe's magnetic shadow, unlike that for the case of a nonflowing plasma.

Sonmor, L. J., Current collection in a collisionless, large-Debye-length magnetoplasma, 214 pp., Ph. D. Thesis, York Univ., North York, Ontario, 1991. (50 refs)

Abstract: Exact calculations of the steady-state current drawn from a collisionless, Maxwellian plasma in a uniform magnetic field by a spherical, perfectly absorbing electrode are presented for a range of dimensionless electrode potentials and magnetic field strengths; these calculations are valid in the limit of large Debye length. Also presented are two techniques which were developed to increase the efficiency of this large calculation: (1) a set of modified Runge-Kutta-Fehlberg and Runge-Kutta-Nyström-Fehlberg algorithms which solve concurrently a large number of initial-value problems, in order to achieve efficient use of vector computers; and (2) an approximate analysis of the orbits, which replaces the numerical initial-value problem solver in the region far from the probe. The results of the calculation are compared with the theory of Rubinstein and Laframboise, which gives upper and lower bounds for both the attracted-species and the repellec-species current. It is found that as the electrode potential increases from space potential with magnetic field strength fixed, the electron (i.e. attracted-species) current decreases, but not as quickly as the adiabatic-limit (effectively lower-bound) current. The ion current also diverges immediately from the adiabatic-limit current. As the electrode potential increases further, the electron current rises and moves monotonically toward the canonical upper bound, which is the warm-plasma generalization of the well known Parker and Murphy upper bound. Although a quantitative large-potential extrapolation has not been found, the results appear to indicate that the current approaches a limiting fraction of the canonical upper bount which varies with magnetic field strength. The dependence of the current on the magnetic field strength is more complicated. As expected for small fixed electrod potentials, the attracted-species current approaches the adiabatic-limit current monotonically as the magnetic field strength increases. However, for large electrode potentials this pattern reverses: the current approaches the canonical upper bound monotonically as the magnetic field strength increases. These patterns are expected to persist when the Debye length is finite. Interpretation of these results leads to an inference that for large electrode potentials, the effect of decreasing the Debye length may be to reduce the current, as in the nonmagnetic case.

Sonmor, L. J. and J. G. Laframboise, Chaotic particle motion in Coulomb electric and uniform magnetic fields, Geophys. Res. Lett., 18, 1619-1622, 1991. (13 refs)

Abstract: In an attractive Coulomb electric field and a uniform magnetic field B, some particle orbits originating at infinite distance undergo many reflections, repeatedly approaching the Coulomb centre. The chaotic nature of these orbits is discussed and illustrated by numerical calculations. The divergence of neighbouring trajectories is shown to occur mostly in discrete jumps when they pass near the Coulomb centre. The appearance of chaotic behaviour in this simplified situation suggests that it will also occur when space-charge effects significantly perturb the potential distribution around a current-collecting electrode in a space magnetoplasma.

Sonmor, L. J. and J. G. Laframboise, Exact current to a spherical electrode in a collisionless, large-Debye-length magnetoplasma, Phys. Fluids B (Plasma Phys.), 3, 2472-2490, 1991. (36 refs)

Abstract: Exact calculations of the steady-state current drawn from a collisionless, Maxwellian plasma in a uniform magnetic field by a spherical, perfectly absorbing electrode are presented for a range of dimensionless electrode potentials and magnetic-field strengths. These calculations are valid in the limit of large Debye length. The results are compared with the theory of Rubinstein and Laframboise, which gives upper and lower bounds for both the attracted-species and the repelled-species current. It is found that as the electrode potential increases from space potential with

magnetic-field strength fixed, the electron (i.e., attracted-species) current decreases, but not as quickly as the adiabatic-limit (effectively lower-bound) current. The ion current also diverges immediately from the adiabatic-limit current. As the electrode potential increases further, the electron current rises and moves monotonically toward the canonical upper bound, which is the warm-plasma generalization of the well-known Parker and Murphy upper bound. It is unclear whether the current approaches the upper bound asymptotically as the electrode potential becomes large, or instead a constant proportion of the upper bound which varies with magnetic-field strength. The dependence on magnetic-field strength is more complicated. As expected for small fixed electrode potentials, the attracted-species current approaches the adiabatic-limit current monotonically as the magnetic-field strength increases. However, for large electrode potentials this pattern reverses: the current approaches the canonical upper bound monotonically as the magnetic-field strength increases. These patterns are expected to persist when the Debye length is finite. Interpretation of these results leads to an inference that for large electrode potentials, the effect of decreasing the Debye length may be to reduce the current, as in the nonmagnetic case.

Soop, M., Report on photo-sheath calculations for the satellite GEOS, Planet. Space Sci., 20, 859-870, 1972. (4 refs)

Abstract: This is a report on the first results of computations made on the iBM 360/65 computer at ESOC (Darmstadt) in order to determine the perturbation of the electric field around the satellte GEOS arising from the photo-emission. Special emphasis is laid upon comparing a satellite with an electrically conducting surface and an insulating surface. The result is partly presented as a cinematographic film showing the electric field and the movements of the electrons.

Soop, M., Numerical calculations of the perturbation of an electric field around a spacecraft, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 127-136, D. Reidel Publ. Co., Dordrecht, 1973. (3 refs)

Abstract: This report deals with some aspects of the electric field around a body having an insulating surface, with or without an inner conductor. Surface charge accumulation due to flow of ambient plasma electrons and emission of photoelectrons is considered. The purpose of this study is to estimate the magnitude of perturbations due to the spacecraft body on the electric field measurements made at the geostationary orbit.

Spitzer, L., Jr., Physics of Fully Ionized Gases, 2nd ed., Interscience Publ., New York, 1962. (128 refs)

Abstract: -

Su, C. H. and S. H. Lam, Continuum theory of spherical electrostatic probes, Phys. Fluids, 6, 1479-1491, 1963. (8 refs)

Abstract: A continuum theory for spherical electrostatic probes in a slightly ionized plasma is developed. The density of the plasma is taken to be sufficiently high such that both the ions and electrons suffer numerous collisions with the neutrals before being collected by an absorbing probe. A general discussion of probes at an arbitrary potential is given. It is found that for very negative probe potentials the sheath thickness can be comparable to the probe radius. Two explicit forms of current-voltage characteristics are given; one for very negative probes, the other for probes at nearly plasma potential. Both of these are based on the assumption that the probe radius is large compared with the Debye length. Numerical computation is also given for negative probes of a wider range of probe sizes.

Sugawara, M. and Y. Hatta, Effects of a magnetic field on electron probe currents, J. Appl. Phys., 36, 314-315, 1965. (6 refs)

Abstract: In the absence of a magnetic field, probe characteristics are so well understood that they can be used to furnish information about the plasma. In the presence of a magnetic field, this simple picture is no longer true. A theory of electron collection in a magnetic field was developed by Bohm under assumptions that so-called drain diffusion and plasma electric fields are the cause for making up depletion of electrons in the neighborhood of the probe. Their formula is not appropriate for interpreting results obtained here because of taking into account the drain diffusion. On the other hand, Spiwak, using the assumption that electrons within one mean free path away from the probe surface obey orbital motion, developed a theory, which is not consistent with our experimentally observed pressure dependence. The object of this paper is to report some measurements on the electron probe current across a magnetic field and to interpret the electron probe current only in terms of diffusion of electrons.

Swift, J. D. and M. J. R. Schwar, Electrical probes for plasma diagnostics, Iliffe Books, London, 1971. (227 refs)

Abstract: (Preface:) The electrical probe method of plasma diagnostics may be used to study steady and periodic glow and arc discharges, after-glows, ionised gas layers in the upper atmosphere, ionised combustion products of flames and M.H.D. systems, ionised exhaust gases and plasma jets. It is the purpose of this book to provide research workers in the field of partially ionised gases with a sufficiently comprehensive background for the successful measurement of such plasma parameters as charge carrier concentration and temperature. A considerable advance has been made in this method of plasma diagnostics since Langmuir's pioneering work during the years 1923 to 1935. It is now no longer necessary to restrict its application to the study of low pressure d.c. gas discharges where the mean free path of the charge carriers is very much greater than the dimensions of the probe, and where the sheath dimensions are known. Modern probe theories cover a great variety of plasma states. However, because of the increasing complexities of these analyses theoretical probe current characteristics can often be obtained only through a process of numerical integration. As a consequence of this many of the results are presented in graphical form and by including these it is hoped that this book will be a useful reference source to the experimentalist. An advantage of the electrical probe method over other methods of plasma diagnostics is that the diagnosis can be performed using very simple apparatus.

Carrier concentrations and temperatures can be deduced by correctly interpreting the current-potential characteristic of a small electrical probe immersed in the plasma. Another advantage of the method is that it is frequently capable of measuring local values of the above mentioned parameters; most other diagnostic methods give, in the first instance, values of the parameters averaged over a relatively large volume. In the following chapters it is hoped to present the theoretical basis of the behaviour of electrical probes under a wide variety of plasma conditions. The probe theories are arranged in asystematic manner so that the theory applicable to a particular type of plasma can easily be selected. The major part of this book is concerned with systems in which the relative velocity between the probe and the partially ionised gas flow is zero or much less than the ion's mean thermal velocity, and in which magnetic fields are sufficiently small for the electron's radius of gyration to be much greater than their mean free path. However, some discussion of these more complex situations is given in Chapter 13. The beginning of the book is concerned with the basic principles governing the motion of ions and electrons through a neutral gas in the presence of an electric field. In general, the motion of the carriers can be described by Maxwell's transfer equations. In many cases, however, the flow of carriers is accurately described by 'free-fall' considerations or by mobility and diffusion flow. Chapters 3 to 8 deal with the collection of ions and electrons by a probe that is very much smaller than their mean free path. It is also assumed that the carriers come under the influence of the probe's electric field within a distance less than a free path from the probe's surface. Plasmas that satisfy these two conditions are referred to in this book as 'low pressure' plasmas and are the kind originally investigated by Langmuir. 'High pressure plasmas' are treated in Chapters 9 and 10. Here the probe perturbs the plasma in which it is immersed as the mean free path of the carriers is now comparable to, or less than, the probe's dimensions. This is the most difficult regime to analyse theoretically, even when one restricts the analysis to an ideally homogeneous plasma. Only the two extreme cases where the probe's electric field penetrates into the plasma to a depth very much less than and very much more than a mean free path are considered in any detail. The remainder of the book is concerned with the more practical aspects of plasma diagnostics and deals with perturbation effects, probe measuring techniques including the use of probes under more complex conditions such as in the presence of magnetic fields and directed carrier flow, and other methods of plasma diagnostics.

Szuszczewicz, E. P., Area influences and floating potentials in Langmuir probe measurements, J. Appl. Phys., 43, 874-879, 1972. (16 refs)

Abstract: An analysis has been conducted on the influence of a relatively small reference electrode in a Langmuir probe measurement of plasma density and temperature. The analysis employs the collisionless theory of Laframboise for spherical and cylindrical probes and in a completely general way includes a broad range or electron densities and temperatures as well as all possible values of ion mass and multiplicity of ionization. It is shown that a ratio alpha of reference-electrode area to probe area of 10/sup 4/ will guarantee no distortion of the measurement as a result of a shifting reference-electrode potential. It is further shown that the constraint on alpha can be relaxed by approximately two orders of magnitude when the charge-normalized ion mass m/sub i//Z/sup 2/ is decreased from 100 to 0. An additional result of the analysis is the dependence of a probe's floating potential on its geometry and radius as well as on the properties of the plasma. A discussion of the latter result reveals in a striking manner the significant differences between the ion-current responses of spherical and cylindrical probes.

Szuszczewicz, E. P., Fixed-bias floating double-probe technique with simple Langmuir-probe characteristics, J. Appl. Phys., 43, 3328-3335, 1972. (11 refs)

Abstract: A new floating double-probe method is presented which has advantages over other floated-probe systems heretofore described in the literature. The method utilizes two electrodes, one of constant area and the other with a variable area, and the two-electrode configuration is separated by a fixed bias voltage. The current- voltage characteristics of the new technique, which are generated by varying the area of one electrode, are identical to those of a simple Langmuir probe, thus coupling all the advantages of a floated probe system with the simple analysis scheme generally applied to the Langmuir probe for the determination of plasma density and temperature.

Szuszczewicz, E. P., Transition region response of the symmetric double probe and its application in the lower ionosphere, Rev. Sci. Instrum., 44, 115-119, 1973. (10 refs)

Abstract: The technique of the symmetric double probe lends itself to the in situ measurement of plasma temperature in the ionospheric D region because it can lead to meaningful results under relatively high collision frequencies where the Langmuir probe has been observed to fail. For a completely thermalized plasma and over parametric ranges 0.1 <= lambda_ia/R_p <= 100 and 0.004 <= R_p/lambda_D <= 100, where lambda_ia is the ion atom mean free path, R_p is the probe radius, and lambda_D is the Debye length, it is shown that the modification of electron temperature is never greater than 12%, with a value of (8 +- 2)% nominally applicable in the case of D-region diagnostics. This technique has been successfully operated on a Nike-Cajun payload flown at midday from White Sands, N. M. to an apogee of 78.5 km. The associated electronics and deployed double-probe configuration are presented, and a current-voltage characteristic collected in the ascent stage at 73.7 km is briefly discussed. The values of electron temperature indicated by the sampled data are ~30% higher than those predieted by theory for the anticipated state of thermal equilibrium with the ambient neutrals. The importance, however, is that these values, of the order of 300 K, are the lowest ever reported by any in situ method operative in the D region and consequently point out the strong potential of the double-probe technique in future applications to regions of planetary ionospheres which cannot be considered as collision-free.

Szuszczewicz, E. P., Measurement realities of current collection in dynamic space plasma environments, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 88-124, 1990. (63 refs)

Abstract: Theories which describe currents collected by conducting and non-conducting bodies immersed in plasmas have many of their concepts based upon the fundamentals of sheath-potential distributions and charged-particle behavior in superimposed electric and magnetic fields. Those current-collecting bodies (or electrodes) may be Langmuir probes, electric field detectors, aperture plates on ion mass spectrometers and retarding potential analyzers, or spacecraft and their rigid and tethered appendages. Often the models are incomplete in representing the conditions under which the current-voltage characteristics of the electrode and its system are to be measured. In such cases, the experimenter must carefully take into account magnetic field effects and particle anisotropies, perturbations caused by the current collection process itself and contamination on electrode surfaces, the complexities of non-Maxwellian plasma distributions, and the temporal variability of the local plasma density, temperature, composition and fields. This set of variables is by no means all-inclusive, but it represents a collection of circumstances guaranteed to accompany experiments involving energetic particle beams, plasma discharges, chemical releases, wave injection and various events of controlled and uncontrolled spacecraft charging. This paper attempts to synopsize these diagnostic challenges and frame them within a perspective that focuses on the physics under investigation and the requirements on the parameters to be measured. Examples will include laboratory and spaceborne applications, with specific interest in dynamic and unstable plasma environments.

Szuszczewicz, E. P. and A. H. Weber, On the behaviour of low density plasma in the presence of cylindrical Langmuir probes, in International conference on gas discharges, xiv+599 pp., pp. 147-151, IEE, London, UK, 1970. (9 refs)

Abstract: An experimental investigation of the behaviour of low-density plasmas in the presence of ion-attracting cylindrical Langmuir probes has been conducted in the region R/sub p// lambda /sub D/<2.7. Under conditions for which (R/sub p// lambda /sub ee/)/sub max/=1/330, (R/sub p// lambda /sub ii/)/sub max/=1/82 and (R/sub p// lambda /sub ia/)/sub max/=1/36 evidence is presented which supports the existence of an orbital-motion-limited current region as predicted by the collisionless theory of Laframboise. Evidence is also presented which indicates that previous discrepancies between the theory and experiment reported by other investigators resulted from collision frequencies which were too high to properly qualify their experimental probe-plasma systems as collision- free.

Szuszczewicz, E. P. and J. C. Holmes, Surface contamination of active electrodes in plasmas: Distortion of conventional Langmuir probe measurements, J. Appl. Phys., 46, 5134-5139, 1975. (14 refs)

Abstract: When the conventional continuous-sweep method is used, contamination of the probe surface produces hysteresis in Langmuir probe current-voltage characteristics. The paper describes the development of a pulse plasma probe technique, the use of which eliminates hysteresis when layers of contaminants build up on the probe surface. The results are superior to those obtained with the continuous mode even when resistive heating is used to clean the probe surface.

Szuszczewicz, E. P. and P. Z. Takacs, Magnetosheath effects on cylindrical Langmuir probes, Phys. Fluids, 22, 2424-2429, 1979. (21 refs)

Abstract: The response of cylindrical Langmuir probes in magnetoplasmas is studied from a perspective which focuses on the relative magnitudes of Larmor radius and sheath size. The approach results in a classification for magnetic field effects which involves not only the magnetic field strength but also the plasma parameters of density, temperature, and the applied probe potential. It is specifically shown that a 0.25 G field can have similar effects on the current collection properties of the probe in an ionospheric plasma (N/sub e/ approximately=10/sup 6/ cm/sup -3/) as a 30 kG field would have in a hot, dense laboratory plasma (N/sub e/ approximately=10/sup 15/ cm/sup -3/). The classifications are found to agree with new experimental results collected in an ionospheric plasma. The data also show: (a) the effects of probe orientation on electron current collection from magnetoplasmas; (b) that these effects can be important even when the electron Larmor radius is larger than the radius of the probe; and (c) that substantial magnetic field effects occur when the probe's sheath is comparable to or greater than the Larmor radius.

Talbot, L. and Y. S. Chou, Langmuir probe response in the transition regime, in Rarefied Gas Dynamics, Ed. by L. Trilling and H. Y. Wachman, Vol 2., pp. 1723-1737, Academic Press, New York, 1969. (13 refs)

Abstract: A simple approximate analysis is presented which permits the calculation of the effect of collisions on the ion saturation current and the electron current in the electron-retarding region for both cylindrical and spherical electrostatic probes. The results are applicable over a wide range of the Debye length-probe radius ratio, and span the regime from the continuum to the collisionless limit. Comparisons with experiment are made.

Thornton, J. A., Comparison of theory and experiment for ion collection by spherical and cylindrical probes in a collisional plasma, AIAA J., 9, 342-344, 1971. (10 refs)

Abstract: -

Touryan, K. J. and P. M. Chung, Flush mounted electrostatic probe in the presence of negative ions, 38 pp., Sandia Corp., Albuquerque, NM, USA, July 1970. (0? refs)

Abstract: To study the change in current voltage characteristics of a continuum flush electrostatic probe caused by the introduction of negative ions in the boundary layer by ablation surfaces during reentry, a simple flat plate flow analysis was made. The analysis shows that the negative ions affect the ambipolar region only. Two equations are obtained; one for positive ions and one for negative ions. The results show that the negative ions suppress the electron saturation current and increase the positive ion saturation current slightly. The effect of increasing the beta ratio for negative ions is to increase the positive saturation current slightly and further suppress the negative saturation current. The effect of varying the beta ratio for negative ions as well as the ion to electron temperature ratio is discussed in detail. It is

concluded that, in general, when negative ions are present in the boundary layer in quantities greater than trace amounts, measurements by electrostatic probes alone cannot determine electron density concentrations.

Trøim, J. and B. N. Maehlum, Laboratory simulatoin studies on charging of spacecraft, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 305-313, Editrice Compositori, Bologna, 1992. (2 refs)

Abstract: The objective is an investigation of how insulated areas on the wake-side of a large spacecraft model embedded in a streaming plasma is charged when being exposed to beams of fast electrons. The investigation is relevant to differential charging on satellites in orbits crossing polar latitudes where they may be subjected to high energy electron precipitation. The measurements are performed in a medium sized vacuum system (1 m dia x 2 m) with a thermal plasma generator and an electron accelerator. The velocity of the plasma stream can be varied to simulate the relative speed of a spacecraft moving through the ionosphere. The electron accelerator can operate at energies corresponding to typical auroral electron energies. Results are presented showing charging of different materials as a function of plasma density and electron energy. It is also demonstrated how a charged area on the wake-side of the spacecraft model to a large extent modifies the plasma density structure downstream on the wake side of the model.

Troy, B. E., Jr. and E. J. Maier, Electron temperatures in the wake of an ionospheric satellite, J. Geophys. Res., 80, 993-997, 1975. (16 refs)

Abstract: Presents measurements of electron temperature T/sub e/ made by a retarding potential analyzer and a Langmuir probe (both flush-mounted on the spin-stabilized satellite Explorer 31) to investigate the variation of T/sub e/ around the satellite. Most of the time there is a T/sub e/ variation, which repeats for given ionospheric conditions. The variation is strongly controlled by the angle between the velocity vector and the probe normal, T/sub e/ usually being enhanced in the near wake of the satellite. Magnetic field control of T/sub e/, if it is present, is hidden by the stronger velocity vector control. Results indicate that the magnitude of the T/sub e/ enhancement in the wake does not depend on the average ion mass M, although the electron density depletion in the wake is strongly correlated with M.

Tunaley, J. K. E. and J. Jones, The photoelectron sheath around a spherical body, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 59-71, D. Reidel Publ. Co., Dordrecht, 1973. (5 refs)

Abstract: An approximate method based on a variational principle is used to calculate the surface electric field on a spherical body which is photo-emitting electrons. The treatment also allows an estimate of the sheath dimensions to be made. It is shown that, unlike the planar surface, the sheath parameters are strongly dependent on the velocity distributions of the emitted electrons. As the sphere radius is reduced the sheath shrinks for a monoenergetic, isotropic, velocity distribution of electrons at the surface. However for monoenergetic electrons directed radially it expands. The technique is developed because of the possibility of employing it for more complex systems where other methods would prove intractable. The same problems will be treated by solving the differential equations using a series expansion; this will indicate the accuracy of the method for small spheres.

Tunaley J. K. E. and R. J. L. Grard, The impedance of a probe in a warm plasma, Ann. Geophys., 25, 55-65, 1969. (19 refs)

Abstract: When electrons pass in the vicinity of a probe, they interact with the electric field during their transit and on average they abstract energy from the system. Also the thermal movement of the electrons causes an electric dipole to radiate electroacoustic waves. It is shown that the transit effect of the particles and the radiation of electroacoustic waves are two different aspects of the same phenomenon. A straight-forward approach to this problem is presented and the theory is applied to simple geometries. It is also shown how this treatment can be extended to the case of a magnetoplasma.

Urrutia, J. M. and R. L. Stenzel, Anomalous currents to an electrode in a magnetoplasma, Phys. Rev. Lett., 57, 715-718, 1986. (10 refs)

Abstract: An electrode pulsed to a large positive potential (V>>kT/sub e//e) in a collisionless magnetoplasma is observed to draw currents far in excess (10*) of the field-aligned electron saturation current prescribed by probe theory. The large currents are inferred to be due to anomalous cross-field transport caused by current-driven instabilities. Electric fields due to space-charge separation and anomalous conductivity lead in time to ion expulsion from the current channel and hence to density depletion and current collapse.

Vashi, B. I., Numerical simulation of the electrodynamic interaction between a satellite and plasma, Ph.D. Thesis, University of Alabama in Huntsville, Huntsville, AL, 1992. (? refs)

Abstract: ...

Vashi, B. I. and N. Singh, Numerical simulations of current collection for the electrodynamic tether in space, in Proc. The Twenty-First Southeastern Symposium on System Theory (Cat. No.89CH2714-4), pp. 376-379, IEEE Comput. Soc. Press, Washington, DC, USA, 1989. (3 refs)

Abstract: Numerical simulations have been carried out to determine the I-V characteristic of a long cylindrical body moving in a magnetized plasma. Results show that at relatively small body voltages, the collected current in the presence of magnetic field is insignificantly small compared with the situation without a magnetic field. However, when the body voltage increases, the current collection increases almost linearly, even in the magnetized plasma. The result is important in the sense that it is found to be true under the most restrictive condition for current collection, when the plasma particles are allowed to move only in a plane perpendicular to the ambient magnetic field. The result is also important for future power plants in space capable of supplying civilian and military systems, as these are likely

to employ electrodynamics tethers for power generation.

Vashi, B. I. and N. Singh, Current collection by a long conducting cylinder in a flowing magnetized plasma, J. Spacecr. Rockets, 28, 592-598, 1991. (10 refs)

Abstract: The current collection by a conducting cylinder in a plasma through the transport of charged particles across a magnetic field is studied by means of numeerical simulations. An electrostatic particle-in-cell code is used. The conducting cylinder is oriented along an ambient magnetic field B_0. The plasma flows with an unperturbed relative velocity v_0 perpendicular to B_0. The potential structure near a positively biased cylinder is investigated. The potential structure is found to be dynamic and azimuthally asymmetric. The electric fields in the structure create a two-cell circulation of the electrons near the cylinder. The cell in the wake region has negative potentials. A fan-shaped circulation cell forms around the cylinder, and in this cell, the potential is generally positive. The geometry and the size of this positive cell affects the current collection. The time-averaged current is found to vary approximately linearly with the flow velocity v_0. For the typical value of v_0 (approximately 8 km/s) in the low Earth orbit, the enhancement in the current through the cross-field plasma transport is found to be an appreciable fraction of the current predicted by the existing models, which do not account for the cross-field plasma transport. The cross-field transport is affected primarily by the asymmetrical sheath structure.

Vashi, B. I., N. Singh, and W. C. Leung, Numerical simulations for the current collection by a tethered-satellite in the ionosphere, in Proc. IEEE Southeastcon '92 (Cat. No.92CH3094-0), vol. 1, pp. 79-82, IEEE, New York, NY, USA, 1992. (7 refs)

Abstract: The current collection by a tethered-satellite in the low earth orbit and the processes involved in this ionospheric phenomenon were investigated by numerical simulations. Initially a 2-D case was considered in the cylindrical geometry. The system solved numerically was comprised of a long cylindrical satellite, an arbitrary orientation of the ambient magnetic field, and a Maxwellian plasma flowing relative to the satellite. The relative orientation between the magnetic field and the plasma flow has been found to effect the current collection, with an enhancement achieved when the magnetic field and the plasma flow are perpendicular. Results presented include the plasma distributions, sheath structures, and the current-voltage characteristics. For a typical orbit in the ionosphere, the relative plasma flow is perpendicular to the magnetic field, and this case has been extensively studied.

Vashi, B. I., N. Singh, and W. C. Leung, A three-dimensional numerical model for satellite-magnetoplasma interactions, 4 pp., in Proc. IEEE Southeastcon '93 (Cat.No.93CH3295-3), 684 pp., IEEE, New York, NY, USA, 1993. (14 refs)

Abstract: Numerical simulations of the electrodynamic interactions between a satellite and magnetoplasma have been carried out using a 3-D particle-in-cell code in cylindrical geometry. The satellite is a finite-sized cylindrical body, which can be biased at a potential. The plasma is composed of electrons and hydrogen ions. Included in the model is a magnetic field, which represents the geomagnetic field and which in the model can be oriented as required. Simulation cases have been performed with and without a relative motion between the satellite and plasma. For the situation without motion, the current collected by the satellite is compared with the available analytical prediction from the theory of L. W. Parker and B. L. Murphy (1967). The current collected in the simulations is found to be in agreement with the theoretical value for different system parameters. For simulations with a relative motion, the current collected by the satellite is found to be enhanced over the nondrifting case.

Virmont, J. and R. Godard, Spherical Langmuir probe: complements to the 'classical' theory, Plasma Phys., 14, 793-803, 1972. (18 refs)

Abstract: Considers the classical model of a perfectly absorbing spherical probe in a maxwellian plasma at rest, without collisions or magnetic field. After a brief review of the corresponding probe theory, the numerical results of Laframboise (1966) are completed to cover all the values of the ratios: ion temperature/electron temperature and probe radius/Debye length. It is shown that these results permit both a precise value of the plasma and an order of magnitude estimate for the ion temperature to be obtained from the saturation branches of the probe characteristic.

Wagner, H. W. and D. Johanning, Consideration of ion current in electron temperature measurements with a cylindrical Langmuir probe, Cosmic Res., 19, 471-478, 1981 (Translated from: Kosmicheske Issledovaniya, 19, 695-703, 1981). (9 refs)

Abstract: Two different methods for approximate determination of I/sub i/ are described and used for processing readings from a cylindrical Langmuir probe aboard the satellite Interkosmos 10. The effect of I/sub i/-correction on resulting values of the electron temperature T/sub e/ is considered. The theory of a Langmuir probe moving within a plasma gives a complex dependence of ion current on voltage I/sub i/(U/sub p/). Therefore, generally speaking, it is impossible to use the least-squares method to determine the coefficients in the equations defining the theoretical dependence, and it is impossible to extrapolate I/sub i/ into the region U/sub p/ ≤ U/sub plasma/.

Wahlström, M. K., E. Johansson, E. Veszelei, P. Bennich, M. Olsson, and S. Hogmark, Improved Langmuir probe surface coatings for the Cassini satellite, Thin Solid Films, 220, 315-320, 1992. (15 refs)

Abstract: A Langmuir probe will be used on the space research satellite Cassini in order to monitor electron density fluctuations in space plasmas surrounding Saturn. In order to obtain well-defined Langmuir probe characteristics a surface with uniform work function Phi must be used. A graphite coating has been used on earlier Earth-bound satellite probes, but for the Cassini satellite to Saturn a hard, wear-resistant coating must be used. In this contribution the authors report on investigations of titanium nitride (TiN) coatings for this purpose. TiN coatings have been prepared

by high temperature nitriding in pure nitrogen gas and by reactive magnetron sputtering on plane titanium and titanium alloy substrates. The resulting coatings have been analysed with laterally resolved, relative Phi measurements, and the tribological properties of the coatings have been investigated by solid particle erosion and scratch testing. In the tests, the graphite coating DAG 213 used on earlier satellite probes was used as a reference material. The results show that the TiN coatings are superior to the graphite coating with regard to both photoelectric properties and solid particle impact resistance. The sputtered coatings exhibit the lowest lateral work function variation, but the nitrided coatings have superior erosion resistance. The results demonstrate that TiN coatings are suitable for space plasma probes.

Walker, E. H., Plasma sheath and screening of charged bodies, in Photon and Particle Interations with Surfaes in Space, Ed. by R. J. L. Grard, pp. 73-89, D. Reidel Publ. Co., Dordrecht, 1973. (18 refs)

Abstract: The potential and charge density distributions are derived quite generally for both a stationary charged sphere and a charged body moving rapidly through a plasma. The work of previous authors on the screening of stationary charged bodies has been limited by a failure to recognize the importance of spiral orbits on the space charge density and a failure to cope with the problem of bound orbits. Most of the previous work on the screening of rapidly moving bodies has been limited to cases where the body, the potential, or the Debye length are assumed small. In the few cases where these assumptions have not been made, iterative solutions have been calculated, but here the uniqueness of the solution is seriously in question. We have calculated the potential and charge density as a function of position about a stationary charged sphere, using both monoenergetic and Maxwellian velocity distributions for the ions and electrons of the ambient plasma. The potential decreases with distance more slowly than in the case of local thermodynamic equilibrium; the density of the ions (if the body is negative, electrons if positive) is generally much smaller than given by the barometric formula and varies in a complicated way. We also calculate the ion and electron voltage-current probe characteristics and the equilibrium potential as a function of the radius of the body. We find the Mott-Smith and Langmuir equations for the ion current (if the body is negative, electrons if positive) are unsatisfactory unless the sheath thickness is expressed as a function of the potential and radius of the body. For a spherical body, the appropriate expression for the sheath thickness a is found to be sigma = 1.17 psi_s^(1/2) rho_s^(1/3) - 0.34 psi_c^(1/2) rho_c^(1/3) inside the pericritical surface and 0.83 psi_s^(1/2) rho_s^(1/3) outside the pericritical surface, where psi_s and rho_s are the nondimensional potential and radius for the body, and psi_c, rho_c are the values of these quantities on the pericritical surface. Eigenvalue solutions are obtained if the charged body neutralizes most of the ions and electrons that strike its surface; i.e., if the reflection coefficients for the surface of the body are small. Under these conditions, the potential is found to vary more slowly than r^-2 for small values of the potential. For a rapidly moving body, we have developed a self-consistent method for solving the screening problem which does not require iterative calculations. Equations for the solution of the screening of axially symmetric bodies are derived for plasmas in which the thermal motion of the ions can be neglected and for plasmas with a Maxwellian velocity distribution. We have calculated the potential and density variation in the wake, the probe characteristics, and the impact and electric drag characteristic curves for various bodies. These calculations show that there is a trough in the ion density surrounding a highly charged body. The drag calculations show that under certain conditions a negative drag is obtained if the potential on the body is large and if the ions are neutralized and elastically reflected at the surface of the body.

Wasserstrom, E., C. H. Su, and R. F. Probstein, Kinetic theory approach to electrostatic probes, Phys. Fluids, 8, 56-72, 1965. (24 refs)

Abstract: A spherical electrostatic (Langmuir) probe in a slightly ionized plasma is studied from a kinetic theory point of view. The two-sided distribution function method of Lees, which embodies the Mott-Smith approach, is used. The velocity space is divided into two regions along the straight cone tangent to the spherical probe, and different distribution functions are defined in the two regions. On satisfying the two relevant moments of the distribution function (continuity and number density flux) three simultaneous ordinary nonlinear differential equations, which are appropriate to all values of the Debye length, collision mean free path and probe potential, are obtained for determining the ion and electron number densities, and the potential. These equations reduce to the usual linear flux equations when the mean free path is much shorter than the probe radius and the Debye length. The equations are first linearized and solved for the case of small probe potential. Explicit solutions are given for the current-voltage characteristics and the distributions of the number densities and the potential. The general case of arbitrary probe potential is also studied. Results for the characteristics and for the potential drop in the sheath are presented for some representative cases. Many of the results which are obtained do not appear in the original simplified Langmuir model. The modifications required to take into account the curvilinear orbits of the charged particles are dlscussed.

Whipple, E. C., Potentials of surfaces in space, Rep. Prog. Phys., 44, 1197-1250, 1981. (372 refs)

Abstract: In the last decade, large electrostatic potentials of the order of tens of kV have been measured on spacecraft in the Earth's magnetosphere. Observations in space have led to the inference of large potentials on natural objects in the solar system. The result for spacecraft can be material damage and operational interference caused by electrostatic discharges. Natural objects such as dust grains can be disrupted, and their motion influenced by electromagnetic forces. The potential of a body in space is determined by a balance between various charging currents. The most important are transfer of charge from plasma particles, photoemission, and secondary electron emission, with other charging mechanisms sometimes contributing. The currents are affected by the body's charge and motion and by local magnetic and electric fields. Dielectric surfaces may have surface potential gradients which can affect the current balance through the creation of potential barriers. These processes are evaluated for bodies in the solar system and in interstellar space. Expected equilibrium potentials range from a few tenths of a volt negative in the ionosphere to a few volts positive in the quiet magnetosphere and in interplanetary space. However, large negative potentials can occur in hot plasmas such

as in the disturbed magnetosphere, especially on shaded surfaces. Potentials in interstellar space can be positive or negative, depending on the properties of the local radiation field and plasma. In regions where there have been measurements of spacecraft potentials the results generally agree with these expectations. Deviations can be attributed to the effects of biased or dielectric surfaces or to the magnetic induction effect in large structures such as antennae. An intensive research effort has been initiated to measure material properties, to study charging and discharge processes, to model the current balance to realistic spacecraft configurations, and to obtain additional data in space. Spacecraft potential control experiments have been carried out using passive methods, such as careful surface material selection, and active methods, such as emission of charged-particle beams. The review closes with a survey of possible astrophysical applications where charging effects may be important.

Whipple, E. C., Current collection from an unmagnetized plasma: A tutorial, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 1-12, 1990. (16 refs)

Abstract: The current collected by a body in an unmagnetized piasma depends in general on: (1) the properties of the plasma; (2) the properties of the body; and (3) the properties of any neutral species that are present. The important plasma properties are the velocity distributions of the plasma particles at a location remote from the body (at infinity), and the Debye length which determines the importance of plasma space charge effects. The important body properties are its surface characteristics, namely the conductivity and secondary yield coefflcients. The neutral species affect the current through collisions which impede the flow of current and possibly through ionization of the neutrals which can enhance the current. The technique for calculating the current collected by a body in a plasma will reviewed with special attention given to the distinction between orbit limited and space charge limited regimes, the asymptotic variation of the potential with distance from a body, and the concept of a sheath.

Whipple, E. C., Jr., The equilibrium electric potential of a body in the upper atmosphere and in interplanetary space, Rep. X-615-65-296, NASA Goddard Space Flight Center, Greenbelt, MD, 1965. (? refs)

Abstract: ...

Whipple, E. C., Jr. and L. W. Parker, Theory of an electron trap on a charged spacecraft, J. Geophys. Res., 74, 2962-2971, 1969. (11 refs)

Abstract: A theory is developed for the behavior of an electron trap on a slowly moving charged spacecraft in the limit of large Debye length and no magnetic field. Analytic expressions are obtained for the current to an aperture electrode and to an internal retarding electrode for all values of aperture and spacecraft potentials. When the spacecraft is repulsive and the aperture grid is less repulsive (or even attractive), a potential barrier exists that reduces the current. The extent of the reduction can be orders of magnitude. As a consequence, the 'knee' in the current-voltage curve can yield incorrect values for the electron concentration and spacecraft potential if conventional Langmuir probe theory is used. When the spacecraft is more attractive than the aperture grid, the current is unaffected by the spacecraft potential. It is shown that in the so-called exponential regime of the retarded current, it is impossible for secondary electrons to reach the collector from the adjacent external surfaces of the spacecraft. In the attractive regime the current characteristic is linear for large aperture potentials.

Wilbur, P. J. and J. D. Williams, Laboratory experiments on plasma contactors, in Current Collection From Space Plasmas, Ed. by N. Singh, K. H. Wright, Jr., and N. H. Stone, NASA Conf. Publ. 3089, pp. 128-149, 1990. (17 refs)

Abstract: Experimental results describing the operation of hollow cathode plasma contactors collecting and emitting electrons from and to an ambient plasma at current levels of order one ampere are presented. The voltage drops induced between a contactor and an ambient plasma are shown to be a few tens of volts at such current levels. The development of a double sheath and the production of substantial numbers of ions by electrons streaming across it are associated with the electron collection process. The development of a complex potential structure including a high potential hill just downstream of the cathode orifice is shown to characterize a typical contactor emitting electrons.

Williamson, P. R., Modeling vehicle capacitance, in Physics of Charged Bodies in Space Plasmas, Ed. by M. Dobrowolny and E. Sindoni, pp. 297-304, Editrice Compositori, Bologna, 1992. (6 refs)

Abstract: Most spacecraft are covered with a material, such as paint or insulation, in order to control radiation absorbed and emitted by the spacecraft. These materials are normally electrical insulators and, consequently, have an associated capacitance. The effective capacitance of a spacecraft varies depending on the surrounding plasma. In low earth orbit, where the debye length can be relatively small compared to the dimensions of the spacecraft, the effective capacitance is determined by the dielectrics covering the vehicle. For the Space Shuttle, the effective capacitance on orbit is about 12 microfarads. This capacitance is large enough that it will affect the rate of rise of vehicle potential during the first Tethered Satellite System mission. The effective capacitance for the TSS-1 satellite is determined by a thin layer of paint which has a finite conductivity. The estimated capacitance of the satellite paint is 2 microfarads. The time dependent vehicle potential has been modeled for the TSS-1 mission taking into account the effective capacitance of the Shuttle and the TSS-1 satellite, and the results agree with measurements from the Vehicle Charging and Potential (VCAP) experiment on the OSS-l/STS-3 and Spacelab-2 missions.

Yamamoto, K. and T. Okuda, On the floating probe method for the measurement of ionized gas, J. Phys. Soc. Japan, 11, 57-68, 1956. (9 refs)

Abstract: Several problems with the floating probe method are discussed and solved. Analysing the variation of ion current due to sheath expansion, the authors obtain the criterion for applicability of the usual double probe method and

also the general expression available for the estimation of electron temperature. From the relation between probe characteristic and electron energy distribution, a floating triple method is newly proposed. This method is useful for the measuring of energy distribution in electrodeless or h.f. discharges. By special use of the floating double or triple probe, the measurement of potential distribution in such a quasi-plasma as ion sheath and the transition region between plasma and ion sheath can be attained. As one example, the floating potential becomes measurable. The floating, or insulating, potential is discussed in detail and expression is given showing good agreement with measured values.

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